

ELITE IGCSE ACADEMY

Student Worked Solutions

Elite IGCSE Classified Unit 2 Expertise Answers (Q20+)

133 Questions ■ Question page then worked-solution page

PREPARED & CLASSIFIED BY

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Student worked-solution book - Q20+ expertise.
Each question is followed by its worked solution on the next page.

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Dr. Eslam Ahmed

Topic 1

Direct & Inverse Proportion

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jan 2022

Q#: 20

Marks: 4

TOPIC: **Direct & Inverse Proportion**

Source: Jan 2022 P2HR

- 20 y is inversely proportional to $\sqrt{x}$
 x is directly proportional to T^3

Given that $y = 8$ when $T = 25$

find the exact value of T when $y = 27$

$T = \dots\dots\dots$

(Total for Question 20 is 4 marks)

WORKED SOLUTION

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Jan 2022 P2HR | Question 20

Direct & Inverse Proportion

Topic check. Direct & Inverse Proportion is correct.**Method**

$$y \propto \frac{1}{\sqrt{x}}$$

and

$$x \propto T^3$$

So

$$\sqrt{x} \propto T^{3/2}$$

Therefore

$$y \propto \frac{1}{T^{3/2}}$$

So $yT^{3/2}$ is constant.When $y = 8$ and $T = 25$,

$$8(25)^{3/2} = 8(125) = 1000$$

When $y = 27$,

$$27T^{3/2} = 1000$$

$$T^{3/2} = \frac{1000}{27} = \left(\frac{10}{3}\right)^3$$

$$T = \left(\frac{10}{3}\right)^2 = \frac{100}{9}$$

FINAL ANSWER

$$\frac{100}{9}$$

Algebraic Fractions

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Specimen

Q#: 21

Marks: 3

TOPIC: **Algebraic Fractions**

Source: Specimen 4WM2H

21 $a = \frac{14}{3x-7}$ $x = \frac{1}{4y-3}$

Express a in the form $\frac{py+q}{ry+s}$ where p, q, r and s are integers.

Give your answer in its simplest form.

$$a = \dots\dots\dots$$

(Total for Question 21 is 3 marks)

WORKED SOLUTION*Dr. Eslam Ahmed***Specimen 4WM2H | Question 21**

Algebraic Fractions

Topic check. Algebraic Fractions. The tag is correct because the task is substitution and simplification of algebraic fractions.

Solution

$$a = \frac{14}{3x - 7}, \quad x = \frac{7}{4y - 3}$$

Substitute for x :

$$a = \frac{14}{3\left(\frac{7}{4y-3}\right) - 7}$$

Simplify the denominator:

$$\begin{aligned} 3\left(\frac{7}{4y-3}\right) - 7 &= \frac{21}{4y-3} - 7 \\ &= \frac{21 - 7(4y-3)}{4y-3} = \frac{42 - 28y}{4y-3} \end{aligned}$$

So

$$\begin{aligned} a &= \frac{14}{\frac{42-28y}{4y-3}} = \frac{14(4y-3)}{42-28y} \\ a &= \frac{4y-3}{3-2y} \end{aligned}$$

FINAL ANSWER

$$a = \frac{4y-3}{3-2y}.$$

Algebraic Proof

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: May2025

Q#: 20

Marks: 3

TOPIC: **Algebraic Proof**

Source: May 2025 4WM2H

20 | a, b and c are three consecutive even numbers where $a < b < c$

Prove algebraically that $b^2 = ac + 4$

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2H | Question 20

Algebraic Proof

Solution

Let

$$a = 2n, \quad b = 2n + 2, \quad c = 2n + 4$$

Then

$$ac + 4 = (2n)(2n + 4) + 4$$

$$= 4n^2 + 8n + 4$$

$$= (2n + 2)^2 = b^2$$

So $b^2 = ac + 4$.

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jan 2020

Q#: 25

Marks: 3

TOPIC: **Algebraic Proof**

Source: Jan 2020 P2HR

25 N is a multiple of 5

$$A = N + 1$$

$$B = N - 1$$

Prove, using algebra, that $A^2 - B^2$ is always a multiple of 20

(Total for Question 25 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2HR | Question 25

Algebraic Proof

Topic check. Corrected from Right-Angled Triangles to Algebraic Proof.**Method**Since N is a multiple of 5, let

$$N = 5k$$

where k is an integer.

Given

$$A = N + 1, \quad B = N - 1$$

So

$$A^2 - B^2 = (N + 1)^2 - (N - 1)^2$$

Expand:

$$(N + 1)^2 = N^2 + 2N + 1$$

$$(N - 1)^2 = N^2 - 2N + 1$$

Therefore

$$A^2 - B^2 = (N^2 + 2N + 1) - (N^2 - 2N + 1)$$

$$= 4N$$

Substitute $N = 5k$:

$$4N = 4(5k) = 20k$$

Since k is an integer, $20k$ is a multiple of 20.**FINAL ANSWER** $A^2 - B^2$ is always a multiple of 20.

CLASSIFIED PROBLEM**Paper: P2H****Session: Nov 2024****Q#: 23****Marks: 4**TOPIC: **Algebraic Proof**

Source: Nov 2024 P2H

23 Show that $\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where n is an integer to be found.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2024 P2H | Question 23

Algebraic Proof

Topic check. Correct.**Method**

$$\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x)$$

Factorise:

$$16x^2 - 36 = 4(2x - 3)(2x + 3)$$

$$2x^2 + 7x + 6 = (2x + 3)(x + 2)$$

$$x^2 - 5x - 14 = (x - 7)(x + 2)$$

So

$$\begin{aligned} & \frac{4(2x - 3)(2x + 3)}{x - 7} \div \frac{(2x + 3)(x + 2)}{(x - 7)(x + 2)} - (7 + 8x) \\ &= \frac{4(2x - 3)(2x + 3)}{x - 7} \times \frac{(x - 7)(x + 2)}{(2x + 3)(x + 2)} - (7 + 8x) \\ &= 4(2x - 3) - (7 + 8x) \\ &= 8x - 12 - 7 - 8x \\ &= -19 \end{aligned}$$

So $n = -19$, which is an integer.**FINAL ANSWER**

$$n = -19$$

Solving Inequalities

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: 4WM2HR

Session: May2025

Q#: 20

Marks: 3

TOPIC: **Solving Inequalities**

Source: May 2025 4WM2HR

20 Solve the inequality $2x^2 - 7x - 15 > 0$

Show clear algebraic working.

.....
(Total for Question 20 is 3 marks)

WORKED SOLUTION*Dr. Eslam Ahmed***May 2025 4WM2HR | Question 20**

Solving Inequalities

Required:Solve $2x^2 - 7x - 15 > 0$. Show clear algebraic working.**Step:**

Factorise the quadratic:

$$2x^2 - 7x - 15 = (2x+3)(x-5)$$

$$\text{Check: } (2x+3)(x-5) = 2x^2 - 10x + 3x - 15 = 2x^2 - 7x - 15$$

Step:

Roots: set each factor to zero:

$$2x + 3 = 0 \implies x = -\frac{3}{2}; \quad x - 5 = 0 \implies x = 5$$

Step:

Since the coefficient of x^2 is $+2 > 0$, the parabola opens **upward**. The expression is > 0 *outside* the roots:

FINAL ANSWER

$$x < -\frac{3}{2} \text{ or } x > 5$$

Simultaneous Equations

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CLASSIFIED PROBLEM

Paper: 4WM2HR

Session: May2025

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

Source: May 2025 4WM2HR

22 Solve the simultaneous equations

$$y^2 + 5y + x^2 = 12$$

$$x = y + 5$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

WORKED SOLUTION

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May 2025 4WM2HR | **Question 22**
Simultaneous Equations

Required:

$$\text{Solve } y^2 + 5y + x^2 = 12 \text{ and } x = y + 5.$$

Step:

Substitute $x = y + 5$ into the first equation :

$$y^2 + 5y + (y + 5)^2 = 12$$

Step:

$$\text{Expand } (y + 5)^2 = y^2 + 10y + 25 :$$

$$y^2 + 5y + y^2 + 10y + 25 = 12 \implies 2y^2 + 15y + 13 = 0$$

Step:

Factorise:

$$(2y + 13)(y + 1) = 0 \implies y = -\frac{13}{2} \text{ or } y = -1$$

Step:

Find corresponding x values using $x = y + 5$:

FINAL ANSWER

$$(x, y) = (4, -1) \text{ and } \left(-\frac{3}{2}, -\frac{13}{2}\right)$$

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Nov2025

Q#: 20

Marks: 5

TOPIC: **Simultaneous Equations**

Source: Nov 2025 4WM2H

20 Solve the simultaneous equations

$$\begin{aligned}x^2 + y^2 &= 3xy - 11 \\ y &= x + 2\end{aligned}$$

Show clear algebraic working.

(Total for Question 20 is 5 marks)

WORKED SOLUTION

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Nov 2025 4WM2H | Question 20

Simultaneous Equations

Step 1: Substitute $y = (x+2)$ into the quadratic equation.

$$x^2 + (x+2)^2 = 3x(x+2) - 11$$

Step 2: Expand all brackets.

$$x^2 + (x^2+4x+4) = 3x^2 + 6x - 11$$

$$2x^2 + 4x + 4 = 3x^2 + 6x - 11$$

Step 3: Move all terms to one side to set the equation to zero.

$$0 = 3x^2 - 2x^2 + 6x - 4x - 11 - 4$$

$$x^2 + 2x - 15 = 0$$

Step 4: Factorise the quadratic equation.

$$(x+5)(x-3) = 0$$

So, $x = -5$ or $x = 3$.**Step 5: Find corresponding y values using $y = x + 2$.**

$$\text{If } x = -5 \implies y = -5 + 2 = -3$$

$$\text{If } x = 3 \implies y = 3 + 2 = 5$$

FINAL ANSWER

(-5,-3) and (3,5)

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Specimen

Q#: 20

Marks: 5

TOPIC: **Simultaneous Equations**

Source: Specimen 4WM2H

20 Solve the simultaneous equations

$$x - 2y = 3$$

$$x^2 - y^2 + 2x = 10$$

Show clear algebraic working.

(Total for Question 20 is 5 marks)

WORKED SOLUTION

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Specimen 4WM2H | Question 20

Simultaneous Equations

Topic check. Simultaneous Equations. The tag is correct; this is a linear and quadratic simultaneous equation.

Solution

From

$$x - 2y = 3$$

$$x = 2y + 3$$

Substitute into $x^2 - y^2 + 2x = 10$:

$$(2y + 3)^2 - y^2 + 2(2y + 3) = 10$$

$$4y^2 + 12y + 9 - y^2 + 4y + 6 = 10$$

$$3y^2 + 16y + 5 = 0$$

$$(3y + 1)(y + 5) = 0$$

So

$$y = -\frac{1}{3} \quad \text{or} \quad y = -5$$

If $y = -\frac{1}{3}$,

$$x = 2\left(-\frac{1}{3}\right) + 3 = \frac{7}{3}$$

If $y = -5$,

$$x = 2(-5) + 3 = -7$$

FINAL ANSWER

$\left(\frac{7}{3}, -\frac{1}{3}\right)$ and $(-7, -5)$.

Solving Inequalities

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CLASSIFIED PROBLEM

Paper: P2H

Session: Jan 2022

Q#: 22

Marks: 5

TOPIC: **Solving Inequalities**

Source: Jan 2022 P2H

22 Here is a rectangle.

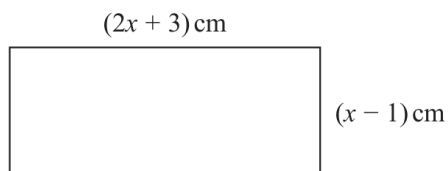


Diagram **NOT**
accurately drawn

Given that the area of the rectangle is less than 75 cm^2

find the range of possible values of x

(Total for Question 22 is 5 marks)

WORKED SOLUTION*Dr. Eslam Ahmed*

Jan 2022 P2H | Question 22
Solving Inequalities

Topic check. Corrected from Functions to Solving Inequalities.

Method

The rectangle has side lengths

$$2x + 3$$

and

$$x - 1$$

For the side lengths to be possible,

$$x > 1$$

The area is less than 75:

$$(2x + 3)(x - 1) < 75$$

$$2x^2 + x - 3 < 75$$

$$2x^2 + x - 78 < 0$$

Factorise:

$$2x^2 + x - 78 = (2x + 13)(x - 6)$$

So

$$-\frac{13}{2} < x < 6$$

Combining with $x > 1$:

$$1 < x < 6$$

FINAL ANSWER

$$1 < x < 6$$

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2024

Q#: 20

Marks: 4

TOPIC: **Solving Inequalities**

Source: Jun 2024 P2HR

- 20** Solve $6x^2 - 7x - 20 > 0$
Show clear algebraic working.

(Total for Question 20 is 4 marks)

WORKED SOLUTION*Dr. Eslam Ahmed*

Jun 2024 P2HR | Question 20
Solving Inequalities

Topic check. Corrected from Solving Quadratic Equations to Solving Inequalities.

Method

$$6x^2 - 7x - 20 > 0$$

Factorise:

$$6x^2 - 7x - 20 = (3x + 4)(2x - 5)$$

The critical values are

$$x = -\frac{4}{3} \quad \text{and} \quad x = \frac{5}{2}$$

Since the coefficient of x^2 is positive, the quadratic is greater than 0 outside the roots.

FINAL ANSWER

$$x < -\frac{4}{3} \quad \text{or} \quad x > \frac{5}{2}$$

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2025

Q#: 21

Marks: 3

TOPIC: **Solving Inequalities**

Source: Jun 2025 P2HR

21 Solve the inequality $2x^2 - 7x - 15 > 0$

Show clear algebraic working.

.....
(Total for Question 21 is 3 marks)

WORKED SOLUTION*Dr. Eslam Ahmed*

Jun 2025 P2HR | Question 21
Solving Inequalities

Topic check. Correct.

Method

$$2x^2 - 7x - 15 > 0$$

Factorise:

$$2x^2 - 7x - 15 = (2x + 3)(x - 5)$$

The critical values are

$$x = -\frac{3}{2} \quad \text{and} \quad x = 5$$

The quadratic is greater than 0 outside the roots.

FINAL ANSWER

$$x < -\frac{3}{2} \quad \text{or} \quad x > 5$$

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2024

Q#: 20

Marks: 3

TOPIC: **Solving Inequalities**

Source: Nov 2024 P2H

- 20** Solve the inequality $10x^2 + 11x - 21 < 0$
Show clear algebraic working.

.....
(Total for Question 20 is 3 marks)

WORKED SOLUTION*Dr. Eslam Ahmed***Nov 2024 P2H | Question 20**

Solving Inequalities

Topic check. Correct.**Method**

$$10x^2 + 11x - 21 < 0$$

Factorise:

$$10x^2 + 11x - 21 = (10x + 21)(x - 1)$$

The critical values are

$$x = -\frac{21}{10} \quad \text{and} \quad x = 1$$

The quadratic is less than 0 between the roots.

FINAL ANSWER

$$-\frac{21}{10} < x < 1$$

CLASSIFIED PROBLEM

Paper: P1H

Session: Nov 2025

Q#: 25

Marks: 3

TOPIC: **Solving Inequalities**

Source: Nov 2025 P1H

- 25** Solve the inequality $2x^2 + x - 28 > 0$
Show clear algebraic working.

(Total for Question 25 is 3 marks)

WORKED SOLUTION*Dr. Eslam Ahmed***Nov 2025 P1H | Question 25**

Solving Inequalities

Topic check. Correct.**Method**

$$2x^2 + x - 28 > 0$$

Factorise:

$$2x^2 + x - 28 = (2x - 7)(x + 4)$$

The critical values are

$$x = \frac{7}{2} \quad \text{and} \quad x = -4$$

The quadratic is greater than 0 outside the roots.

FINAL ANSWER

$$x < -4 \text{ or } x > \frac{7}{2}$$

Sequences

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: May2025

Q#: 24

Marks: 4

TOPIC: Sequences

Source: May 2025 4WM2H

- 24** The 21st term of an arithmetic series is 109
The sum of the first 52 terms of the series is 4381
Work out the 5th term of the series.
Show clear algebraic working.

.....
(Total for Question 24 is 4 marks)

WORKED SOLUTION

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May 2025 4WM2H | Question 24
Sequences

Solution

For an arithmetic series,

$$a + 20d = 109$$

and

$$\frac{52}{2}(2a + 51d) = 4381$$

$$26(2a + 51d) = 4381$$

Solving gives

$$d = -4.5, \quad a = 199$$

The fifth term is

$$a + 4d = 199 + 4(-4.5) = 181$$

FINAL ANSWER

181.

CLASSIFIED PROBLEM

Paper: 4WM2HR

Session: May2025

Q#: 24

Marks: 6

TOPIC: **Sequences**

Source: May 2025 4WM2HR

- 24** An arithmetic series P has first term a and common difference d
- The sum of the first 40 terms of P is 2070
- An arithmetic series Q has first term $2a$ and common difference $3d$
- The 25th term of Q is 186
- Work out the sum of the first 40 terms of Q
- Show clear algebraic working.

.....

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2HR | Question 24
Sequences

Required:

Series P : first term a , c.d. d ; $S_{40}(P) = 2070$. Series Q : first term $2a$, c.d. $3d$; 25th term of $Q = 186$. Find $S_{40}(Q)$.

Step:

Form equation from $S_{40}(P) = 2070$:

$$S_{40} = \frac{40}{2}[2a + 39d] = 20(2a + 39d) = 2070 \implies 2a + 39d = 103.5 \quad (1)$$

Step:

Form equation from 25th term of $Q = 186$:

$$T_{25}(Q) = 2a + 24(3d) = 2a + 72d = 186$$

Step:

Solve simultaneously – subtract (1) from (2):

$$33d = 82.5 \implies d = 2.5$$

Step:

Substituted $d = 2.5$ into (1) :

$$2a + 39(2.5) = 103.5 \implies 2a = 6 \implies a = 3$$

Step:

Compute $S_{40}(Q)$, with first term $= 2a = 6$ and c.d. $= 3d = 7.5$:

$$S_{40}(Q) = \frac{40}{2}[2(6) + 39(7.5)] = 20[12 + 292.5] = 20 \times 304.5 = 6090$$

FINAL ANSWER

$$S_{40}(Q) = 6090$$

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Nov2025

Q#: 26

Marks: 5

TOPIC: Sequences

Source: Nov 2025 4WM2H

- 26** The first term of an arithmetic series is 10
The 20th term of the series is 86

The sum of the first N terms of the series is 5194

Work out the value of N
Show your working clearly.

 $N = \dots\dots\dots$ **(Total for Question 26 is 5 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM2H | Question 26

Sequences

Step 1: Find the common difference d . The n^{th} term is given by $u_n = a + (n-1)d$.

$$u_{20} = 10 + 19d = 86$$

$$19d = 76$$

$$d = \frac{76}{19} = 4$$

Step 2: Use the sum formula to form an equation in terms of N .

$$S_N = \frac{N}{2} [2a + (N-1)d]$$

Substitute the known values $S_N = 5194$, $a = 10$, and $d = 4$:

$$5194 = \frac{N}{2} [2(10) + (N-1)(4)]$$

Step 3: Simplify and solve the quadratic equation. Multiply by 2 to clear the fraction:

$$10388 = N[20 + 4N - 4]$$

$$10388 = N(4N + 16)$$

$$4N^2 + 16N - 10388 = 0$$

Divide the entire equation by 4 to make it simpler:

$$N^2 + 4N - 2597 = 0$$

Step 4: Use the quadratic formula to find N .

$$N = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$N = \frac{-4 \pm \sqrt{16 - 4(1)(-2597)}}{2}$$

$$N = \frac{-4 \pm \sqrt{16 + 10388}}{2}$$

$$N = \frac{-4 \pm \sqrt{10404}}{2}$$

$$N = \frac{-4 \pm 102}{2}$$

Since N must be a positive integer, we discard the negative root:

$$N = \frac{-4 + 102}{2} = \frac{98}{2} = 49$$

FINAL ANSWER

$$N = 49$$

Always double-check that your N (number of terms) evaluates to a positive whole integer. If it ends up as a decimal or fraction, recheck your arithmetic progression formulas.

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Specimen

Q#: 24

Marks: 5

TOPIC: **Sequences**

Source: Specimen 4WM2H

- 24** The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.
The 8th term of this series is 45
- Find the first term of this series.
Show clear algebraic working.

(Total for Question 24 is 5 marks)

WORKED SOLUTION

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Specimen 4WM2H | Question 24
Sequences**Topic check.** Sequences. The tag is correct.**Solution**Let the first term be a and the common difference be d .

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

So

$$S_{10} = 5(2a + 9d)$$

and

$$S_5 = \frac{5}{2}(2a + 4d)$$

Given $S_{10} = 4S_5$:

$$5(2a + 9d) = 4\left(\frac{5}{2}(2a + 4d)\right)$$

$$10a + 45d = 20a + 40d$$

$$5d = 10a$$

$$d = 2a$$

The 8th term is 45:

$$a + 7d = 45$$

Substitute $d = 2a$:

$$a + 14a = 45$$

$$15a = 45$$

$$a = 3$$

FINAL ANSWER

the first term is 3.

Simultaneous Equations

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2022

Q#: 21

Marks: 5

TOPIC: **Simultaneous Equations**

Source: Jun 2022 P2HR

21 Solve the simultaneous equations

$$\begin{aligned}x - 2y &= 3 \\ x^2 - y^2 + 2x &= 10\end{aligned}$$

Show clear algebraic working.

(Total for Question 21 is 5 marks)

WORKED SOLUTION*Dr. Eslam Ahmed*

Jun 2022 P2HR | Question 21
Simultaneous Equations

Topic check. Correct.

Method

$$x - 2y = 3$$

So

$$x = 2y + 3$$

Substitute into

$$x^2 - y^2 + 2x = 10$$

$$(2y + 3)^2 - y^2 + 2(2y + 3) = 10$$

$$4y^2 + 12y + 9 - y^2 + 4y + 6 = 10$$

$$3y^2 + 16y + 5 = 0$$

$$(3y + 1)(y + 5) = 0$$

So

$$y = -\frac{1}{3} \quad \text{or} \quad y = -5$$

Find x :

$$y = -\frac{1}{3} \Rightarrow x = \frac{7}{3}$$

$$y = -5 \Rightarrow x = -7$$

FINAL ANSWER

$$(x, y) = \left(\frac{7}{3}, -\frac{1}{3}\right) \text{ or } (-7, -5)$$

CLASSIFIED PROBLEM

Paper: P2H

Session: Jun 2023

Q#: 21

Marks: 5

TOPIC: **Simultaneous Equations**

Source: Jun 2023 P2H

21 Solve the simultaneous equations

$$\begin{aligned}2x^2 + 3y^2 &= 11 \\ x &= 3y - 1\end{aligned}$$

Show clear algebraic working.

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2H | Question 21

Simultaneous Equations

Topic check. Correct.

Method

$$2x^2 + 3y^2 = 11$$

$$x = 3y - 1$$

Substitute:

$$2(3y - 1)^2 + 3y^2 = 11$$

$$18y^2 - 12y + 2 + 3y^2 = 11$$

$$21y^2 - 12y - 9 = 0$$

$$7y^2 - 4y - 3 = 0$$

$$(7y + 3)(y - 1) = 0$$

So

$$y = -\frac{3}{7} \quad \text{or} \quad y = 1$$

Find x :

$$y = -\frac{3}{7} \Rightarrow x = -\frac{16}{7}$$

$$y = 1 \Rightarrow x = 2$$

FINAL ANSWER

$$(x, y) = \left(-\frac{16}{7}, -\frac{3}{7}\right) \text{ or } (2, 1)$$

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2024

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

Source: Jun 2024 P2HR

22 Solve the simultaneous equations

$$x^2 + y^2 = y + 11$$

$$y = 3x - 1$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

WORKED SOLUTION*Dr. Eslam Ahmed***Jun 2024 P2HR | Question 22**

Simultaneous Equations

Topic check. Correct.**Method**

$$x^2 + y^2 = y + 11$$

$$y = 3x - 1$$

Substitute:

$$x^2 + (3x - 1)^2 = 3x - 1 + 11$$

$$x^2 + 9x^2 - 6x + 1 = 3x + 10$$

$$10x^2 - 9x - 9 = 0$$

$$(5x + 3)(2x - 3) = 0$$

So

$$x = -\frac{3}{5} \quad \text{or} \quad x = \frac{3}{2}$$

Find y :

$$x = -\frac{3}{5} \Rightarrow y = -\frac{14}{5}$$

$$x = \frac{3}{2} \Rightarrow y = \frac{7}{2}$$

FINAL ANSWER

$$(x, y) = \left(-\frac{3}{5}, -\frac{14}{5}\right) \text{ or } \left(\frac{3}{2}, \frac{7}{2}\right)$$

CLASSIFIED PROBLEM

Paper: P1H

Session: Jun 2025

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

Source: Jun 2025 P1H

22 Solve the simultaneous equations

$$\begin{aligned}x^2 + 3y + y^2 &= 7 \\ y &= x + 2\end{aligned}$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

WORKED SOLUTION*Dr. Eslam Ahmed***Jun 2025 P1H | Question 22**

Simultaneous Equations

Topic check. Correct.**Method**

$$x^2 + y^2 + 3y = 7$$

$$y = x + 2$$

Substitute:

$$x^2 + (x + 2)^2 + 3(x + 2) = 7$$

$$x^2 + x^2 + 4x + 4 + 3x + 6 = 7$$

$$2x^2 + 7x + 3 = 0$$

$$(2x + 1)(x + 3) = 0$$

So

$$x = -\frac{1}{2} \quad \text{or} \quad x = -3$$

Find y :

$$x = -\frac{1}{2} \Rightarrow y = \frac{3}{2}$$

$$x = -3 \Rightarrow y = -1$$

FINAL ANSWER

$$(x, y) = \left(-\frac{1}{2}, \frac{3}{2}\right) \text{ or } (-3, -1)$$

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jun 2025

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

Source: Jun 2025 P1HR

22 Solve the simultaneous equations

$$x^2 + y^2 = 41$$

$$2x + y = 3$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

WORKED SOLUTION*Dr. Eslam Ahmed*

Jun 2025 P1HR | Question 22
Simultaneous Equations

Topic check. Correct.

Method

$$x^2 + y^2 = 41$$

$$2x + y = 3$$

So

$$y = 3 - 2x$$

Substitute:

$$x^2 + (3 - 2x)^2 = 41$$

$$x^2 + 9 - 12x + 4x^2 = 41$$

$$5x^2 - 12x - 32 = 0$$

$$(5x + 8)(x - 4) = 0$$

So

$$x = -\frac{8}{5} \quad \text{or} \quad x = 4$$

Find y :

$$x = -\frac{8}{5} \Rightarrow y = \frac{31}{5}$$

$$x = 4 \Rightarrow y = -5$$

FINAL ANSWER

$$(x, y) = \left(-\frac{8}{5}, \frac{31}{5}\right) \text{ or } (4, -5)$$

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2024

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

Source: Nov 2024 P2H

22 Solve the simultaneous equations

$$x^2 + y^2 + y = 3$$

$$x + 2 = y$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2024 P2H | Question 22

Simultaneous Equations

Topic check. Correct.

Method

$$x^2 + y^2 + y = 3$$

$$y = x + 2$$

Substitute:

$$x^2 + (x + 2)^2 + (x + 2) = 3$$

$$x^2 + x^2 + 4x + 4 + x + 2 = 3$$

$$2x^2 + 5x + 3 = 0$$

$$(2x + 3)(x + 1) = 0$$

So

$$x = -\frac{3}{2} \quad \text{or} \quad x = -1$$

Find y :

$$x = -\frac{3}{2} \Rightarrow y = \frac{1}{2}$$

$$x = -1 \Rightarrow y = 1$$

FINAL ANSWER

$$(x, y) = \left(-\frac{3}{2}, \frac{1}{2}\right) \text{ or } (-1, 1)$$

Topic 9

Forming & Solving Equations

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2025

Q#: 20

Marks: 3

TOPIC: **Forming & Solving Equations**

Source: Nov 2025 P2H

20 Here is a quadratic equation.

$$ax^2 + 4x + c = 0$$

The solutions of this equation are given by $x = \frac{-4 \pm 2\sqrt{39}}{10}$

Find the value of a and the value of c
Show your working clearly.

 $a = \dots\dots\dots$ $c = \dots\dots\dots$

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P2H | Question 20

Forming & Solving Equations

Topic check. Correct.**Method**

For

$$ax^2 + 4x + c = 0$$

the quadratic formula gives

$$x = \frac{-4 \pm \sqrt{4^2 - 4ac}}{2a}$$

The given solutions are

$$x = \frac{-4 \pm 2\sqrt{39}}{10}$$

Compare denominators:

$$2a = 10$$

$$a = 5$$

Compare the square root parts:

$$\sqrt{16 - 4ac} = 2\sqrt{39}$$

Square both sides:

$$16 - 4ac = 156$$

Use $a = 5$:

$$16 - 20c = 156$$

$$-20c = 140$$

$$c = -7$$

FINAL ANSWER

$$a = 5, c = -7$$

Functions

Dr. Eslam Ahmed

CLASSIFIED PROBLEM**Paper: 4WM2H****Session: May2025****Q#: 21****Marks: 7****TOPIC: Functions**

Source: May 2025 4WM2H

- 21 The function g is defined as

$$g(x) = \frac{7x+20}{2x} \quad x \neq 0$$

- (a) Solve $gg(x) = 6$
Show clear algebraic working.

$$x = \dots\dots\dots (3)$$

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The function h is defined as

$$h(x) = 5x^2 + 30x - 7 \quad x \geq -3$$

- (b) Express the inverse function h^{-1} in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots (4)$$

(Total for Question 21 is 7 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2H | Question 21

Functions

Solution(a) If $g(g(x)) = 6$, then

$$\frac{7g(x) + 20}{2g(x)} = 6$$

$$7g(x) + 20 = 12g(x)$$

$$g(x) = 4$$

So

$$\frac{7x + 20}{2x} = 4$$

$$7x + 20 = 8x$$

$$x = 20$$

(b)

$$h(x) = 5x^2 + 30x - 7 = 5(x + 3)^2 - 52$$

Let $y = 5(x + 3)^2 - 52$. Since $x \geq -3$, use the positive square root:

$$x + 3 = \sqrt{\frac{y + 52}{5}}$$

$$h^{-1}(x) = \sqrt{\frac{x + 52}{5}} - 3$$

FINAL ANSWER

$$x = 20; h^{-1}(x) = \sqrt{\frac{x + 52}{5}} - 3.$$

CLASSIFIED PROBLEM**Paper: 4WM2H****Session: Nov2025****Q#: 21****Marks: 7****TOPIC: Functions**

Source: Nov 2025 4WM2H

21 The functions f and g are defined as

$$f(x) = 5x - 3$$

$$g(x) = 4x^2 + 16x - 9 \quad \text{where } x \geq -2$$

(a) Find $g(-1)$

(1)

(b) Find $fg(x)$

Give your answer in its simplest form.

$$fg(x) = \dots$$

(2)

$$g(x) = 4x^2 + 16x - 9 \quad \text{where } x \geq -2$$

(c) Express the inverse function g^{-1} in the form $g^{-1}(x) = \dots$

$$g^{-1}(x) = \dots$$

(4)

(Total for Question 21 is 7 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM2H | Question 21

Functions

(a) Find $g(-1)$:

$$g(-1) = 4(-1)^2 + 16(-1) - 9 = 4 - 16 - 9 = -21$$

FINAL ANSWER

$$a - 21$$

(b) Find $fg(x)$:

$$\begin{aligned} fg(x) &= f(g(x)) = 5(4x^2 + 16x - 9) - 3 \\ &= 20x^2 + 80x - 45 - 3 = 20x^2 + 80x - 48 \end{aligned}$$

FINAL ANSWER

$$b \ 20x^2 + 80x - 48$$

Let $y = 4x^2 + 16x - 9$. We must rearrange to make x the subject.

Step 1: Complete the square for the x terms.

$$y = 4(x^2 + 4x) - 9$$

$$y = 4((x+2)^2 - 4) - 9$$

$$y = 4(x+2)^2 - 16 - 9$$

$$y = 4(x+2)^2 - 25$$

Step 2: Rearrange for x .

$$y + 25 = 4(x + 2)^2$$

$$\frac{y + 25}{4} = (x + 2)^2$$

$$x + 2 = \pm \sqrt{\frac{y + 25}{4}}$$

Since $x \geq -2$, $x + 2$ is positive, so we take the positive root.

$$x + 2 = \frac{\sqrt{y + 25}}{2}$$

$$x = \frac{\sqrt{y + 25}}{2} - 2$$

Step 3: Swap y for x to write the inverse function.**FINAL ANSWER**

$$g^{-1}(x) = \frac{\sqrt{x+25}}{2} - 2$$

When finding the inverse of a quadratic, always complete the square. The restricted domain ($x \geq -2$) tells you whether to take the positive or negative square root!

Sequences

Dr. Eslam Ahmed

CLASSIFIED PROBLEM**Paper: P2H****Session: Jan 2020****Q#: 25****Marks: 3**TOPIC: **Sequences**

Source: Jan 2020 P2H

25 Mario is going to save \$50 in the year 2021

He is going to continue to save, up to and including the year 2070, by increasing the amount he saves each year by \$ k

Mario will save a total of \$33 125 from 2021 to 2070

Work out the value of k .

$k = \dots\dots\dots$

(Total for Question 25 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2H | Question 25
Sequences

Topic check. Corrected from Right-Angled Triangles to Sequences.

Method

Mario saves from 2021 up to and including 2070.

Number of years:

$$2070 - 2021 + 1 = 50$$

This is an arithmetic sequence.

First term:

$$50$$

Common difference:

$$k$$

Last term:

$$50 + 49k$$

Use the arithmetic series formula:

$$S_n = \frac{n}{2}(a + l)$$

$$33125 = \frac{50}{2} (50 + (50 + 49k))$$

$$33125 = 25(100 + 49k)$$

$$1325 = 100 + 49k$$

$$49k = 1225$$

$$k = 25$$

FINAL ANSWER

$$k = 25$$

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jan 2020

Q#: 22

Marks: 4

TOPIC: Sequences

Source: Jan 2020 P2HR

- 22** The first term of an arithmetic series S is -6
The sum of the first 30 terms of S is 2865
Find the 9th term of S .

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2HR | Question 22
Sequences

Topic check. Correct.

Method

First term:

$$a = -6$$

$$S_{30} = 2865$$

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$2865 = \frac{30}{2}(2(-6) + 29d)$$

$$2865 = 15(-12 + 29d)$$

$$191 = -12 + 29d$$

$$29d = 203$$

$$d = 7$$

The 9th term is

$$u_9 = a + 8d$$

$$u_9 = -6 + 8(7) = 50$$

FINAL ANSWER

50

CLASSIFIED PROBLEM**Paper: P1H****Session: Jan 2021****Q#: 24****Marks: 4**TOPIC: **Sequences**

Source: Jan 2021 P1H

24 An arithmetic series has first term a and common difference d .

The sum of the first $2n$ terms of the series is four times the sum of the first n terms of the series.

Find an expression for a in terms of d .
Show your working clearly.

$a = \dots\dots\dots$

(Total for Question 24 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1H | Question 24
Sequences

Topic check. Correct.

Method

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_{2n} = n(2a + (2n-1)d)$$

Given

$$S_{2n} = 4S_n$$

$$n(2a + (2n-1)d) = 4 \cdot \frac{n}{2}(2a + (n-1)d)$$

Divide by n :

$$2a + (2n-1)d = 2(2a + (n-1)d)$$

$$2a + 2nd - d = 4a + 2nd - 2d$$

$$d = 2a$$

$$a = \frac{d}{2}$$

FINAL ANSWER

$$a = \frac{d}{2}$$

CLASSIFIED PROBLEM**Paper: P1HR****Session: Jan 2021****Q#: 21****Marks: 6**TOPIC: **Sequences**

Source: Jan 2021 P1HR

- 21 The n th term of an arithmetic series is u_n where $u_n > 0$ for all n
 The sum to n terms of the series is S_n
 Given that $u_4 = 6$ and that $S_{11} = (u_6)^2 + 18$
 find the value of u_{20}

(Total for Question 21 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1HR | Question 21
Sequences

Topic check. Correct.

Method

Let $u_6 = v$.

For an arithmetic series,

$$S_{11} = \frac{11}{2}(u_1 + u_{11})$$

The average of u_1 and u_{11} is u_6 , so

$$S_{11} = 11u_6 = 11v$$

Given

$$S_{11} = (u_6)^2 + 18$$

$$11v = v^2 + 18$$

$$v^2 - 11v + 18 = 0$$

$$(v - 2)(v - 9) = 0$$

So

$$u_6 = 2 \quad \text{or} \quad u_6 = 9$$

But $u_4 = 6$. If $u_6 = 2$, then the common difference is negative and later terms would not all be positive. Therefore,

$$u_6 = 9$$

$$d = \frac{u_6 - u_4}{2} = \frac{9 - 6}{2} = \frac{3}{2}$$

$$u_{20} = u_6 + 14d$$

$$u_{20} = 9 + 14\left(\frac{3}{2}\right) = 30$$

FINAL ANSWER

$$u_{20} = 30$$

CLASSIFIED PROBLEM

Paper: P2H

Session: Jan 2022

Q#: 26

Marks: 5

TOPIC: Sequences

Source: Jan 2022 P2H

- 26** An arithmetic series has first term a and common difference d , where d is a prime number.

The sum of the first n terms of the series is S_n and

$$S_m = 39$$

$$S_{2m} = 320$$

Find the value of d and the value of m
Show clear algebraic working.

$$d = \dots\dots\dots$$

$$m = \dots\dots\dots$$

(Total for Question 26 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2H | Question 26
Sequences

Topic check. Correct.

Method

$$S_m = 39$$

$$S_{2m} = 320$$

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

Then

$$2S_m = m(2a + (m-1)d) = 78$$

and

$$S_{2m} = m(2a + (2m-1)d) = 320$$

Subtract:

$$S_{2m} - 2S_m = m[(2m-1)d - (m-1)d]$$

$$320 - 78 = m(md)$$

$$242 = m^2d$$

$$242 = 2 \times 11^2$$

Since d is prime and m^2 is a square factor,

$$d = 2, \quad m^2 = 121$$

$$m = 11$$

FINAL ANSWER

$$d = 2, \quad m = 11$$

CLASSIFIED PROBLEM**Paper: P1HR****Session: Jun 2022****Q#: 25****Marks: 5**TOPIC: **Sequences**

Source: Jun 2022 P1HR

- 25** The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.

The 8th term of this series is 45

Find the first term of this series.
Show clear algebraic working.

.....
(Total for Question 25 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1HR | Question 25
Sequences

Topic check. Correct.

Method

$$S_{10} = 4S_5$$

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_{10} = 5(2a + 9d)$$

$$S_5 = \frac{5}{2}(2a + 4d) = 5(a + 2d)$$

So

$$5(2a + 9d) = 4 \cdot 5(a + 2d)$$

$$2a + 9d = 4a + 8d$$

$$d = 2a$$

The 8th term is 45:

$$u_8 = a + 7d = 45$$

$$a + 7(2a) = 45$$

$$15a = 45$$

$$a = 3$$

FINAL ANSWER

The first term is 3

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2023

Q#: 23

Marks: 5

TOPIC: Sequences

Source: Jun 2023 P2HR

23 Here are the first three terms of an arithmetic sequence.

$$8p \qquad 7p - 3 \qquad 4p + 2$$

The sum of the first n terms of the sequence is -1914

Work out the value of n
Show your working clearly.

$$n = \dots\dots\dots$$

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2HR | Question 23
Sequences

Topic check. Correct.

Method

The first three terms are

$$8p, \quad 7p - 3, \quad 4p + 2$$

For an arithmetic sequence, consecutive differences are equal:

$$(7p - 3) - 8p = (4p + 2) - (7p - 3)$$

$$-p - 3 = -3p + 5$$

$$2p = 8$$

$$p = 4$$

So the first term is

$$8p = 32$$

and the common difference is

$$(7p - 3) - 8p = 25 - 32 = -7$$

The sum of the first n terms is

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

$$-1914 = \frac{n}{2}(64 - 7(n - 1))$$

$$-1914 = \frac{n}{2}(71 - 7n)$$

$$-3828 = n(71 - 7n)$$

$$7n^2 - 71n - 3828 = 0$$

$$n = \frac{71 \pm 335}{14}$$

Only the positive value is possible:

$$n = 29$$

FINAL ANSWER

$$n = 29$$

CLASSIFIED PROBLEM**Paper: P1H****Session: Jun 2025****Q#: 24****Marks: 5**TOPIC: **Sequences**

Source: Jun 2025 P1H

24 An arithmetic series has 30 terms.The first term is a The common difference is d

The 20th term is 123

The sum of the 30 terms is 2880

Work out the value of a and the value of d

Show clear algebraic working.

 $a = \dots\dots\dots$ $d = \dots\dots\dots$ **(Total for Question 24 is 5 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1H | Question 24
Sequences

Topic check. Correct.

Method

$$u_{20} = a + 19d = 123$$

$$S_{30} = \frac{30}{2}(2a + 29d) = 2880$$

$$2a + 29d = 192$$

Double the first equation:

$$2a + 38d = 246$$

Subtract:

$$9d = 54$$

$$d = 6$$

$$a + 19(6) = 123$$

$$a = 9$$

FINAL ANSWER

$$a = 9, d = 6$$

CLASSIFIED PROBLEM**Paper: P1HR****Session: Jun 2025****Q#: 24****Marks: 6****TOPIC: Sequences**

Source: Jun 2025 P1HR

24 The first 3 terms of an arithmetic series are

$$(2x + 5) \quad (3y - 4) \quad (4x - 2)$$

where x and y are constants.

The sum of the first 9 terms of the series is 216

Find the value of x and the value of y

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1HR | Question 24
Sequences

Topic check. Correct.

Method

The first 3 terms are

$$2x + 5, \quad 3y - 4, \quad 4x - 2$$

Equal differences give

$$(3y - 4) - (2x + 5) = (4x - 2) - (3y - 4)$$

$$6y - 6x = 11$$

For an arithmetic series,

$$S_9 = 9(a + 4d)$$

Here

$$a = 2x + 5$$

$$d = (3y - 4) - (2x + 5) = 3y - 2x - 9$$

Given $S_9 = 216$:

$$9[(2x + 5) + 4(3y - 2x - 9)] = 216$$

$$12y - 6x = 55$$

Now solve

$$6y - 6x = 11$$

$$12y - 6x = 55$$

Subtracting gives

$$6y = 44$$

$$y = \frac{22}{3}$$

Then

$$6\left(\frac{22}{3}\right) - 6x = 11$$

$$x = \frac{11}{2}$$

FINAL ANSWER

$$x = \frac{11}{2}, y = \frac{22}{3}$$

Dr. Eslam Ahmed

CLASSIFIED PROBLEM**Paper: P1H****Session: May 2021****Q#: 23****Marks: 5**TOPIC: **Sequences**

Source: May 2021 P1H

- 23** The sum of the first N terms of an arithmetic series, S , is 292
The 2nd term of S is 8.5
The 5th term of S is 13

Find the value of N .
Show clear algebraic working.

 $N = \dots\dots\dots$

(Total for Question 23 is 5 marks)

WORKED SOLUTION*Dr. Eslam Ahmed*

May 2021 P1H | **Question 23**
Sequences

Topic check. Correct.

Method

The 2nd term is

$$a + d = 8.5$$

The 5th term is

$$a + 4d = 13$$

Subtract:

$$3d = 4.5$$

$$d = 1.5$$

$$a = 8.5 - 1.5 = 7$$

Use

$$S_N = \frac{N}{2}(2a + (N - 1)d)$$

$$292 = \frac{N}{2}(14 + 1.5(N - 1))$$

$$292 = \frac{N}{2}(1.5N + 12.5)$$

$$3N^2 + 25N - 1168 = 0$$

$$N = \frac{-25 + 121}{6} = 16$$

FINAL ANSWER

$$N = 16$$

CLASSIFIED PROBLEM**Paper: P1H****Session: May 2023****Q#: 20****Marks: 6**TOPIC: **Sequences**

Source: May 2023 P1H

20 The sum of the first 80 terms of an arithmetic series, S , is 470

The 75th term of S is 14.5

The sum of the first X terms of S is 171

Work out the value of X

Show your working clearly.

$X = \dots\dots\dots$

(Total for Question 20 is 6 marks)

WORKED SOLUTION*Dr. Eslam Ahmed*

May 2023 P1H | Question 20
Sequences

Topic check. Correct.

Method

$$S_{80} = 470$$

$$40(2a + 79d) = 470$$

$$2a + 79d = \frac{47}{4}$$

The 75th term is

$$a + 74d = 14.5 = \frac{29}{2}$$

Double it:

$$2a + 148d = 29$$

Subtract:

$$69d = 29 - \frac{47}{4} = \frac{69}{4}$$

$$d = \frac{1}{4}$$

$$a + 74\left(\frac{1}{4}\right) = \frac{29}{2}$$

$$a = -4$$

Now

$$S_X = 171$$

$$171 = \frac{X}{2} \left(2(-4) + (X-1)\frac{1}{4} \right)$$

$$171 = \frac{X(X-33)}{8}$$

$$X^2 - 33X - 1368 = 0$$

$$X = \frac{33 + 81}{2} = 57$$

FINAL ANSWER

$$X = 57$$

CLASSIFIED PROBLEM**Paper: P1H****Session: May 2024****Q#: 24****Marks: 6**TOPIC: **Sequences**

Source: May 2024 P1H

24 A polygon has n sides, where $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is 84°

The common difference of the sequence is 4°

Work out the sum of the interior angles of the polygon.

Show clear algebraic working.

(Total for Question 24 is 6 marks)

WORKED SOLUTION*Dr. Eslam Ahmed*

May 2024 P1H | **Question 24**
Sequences

Topic check. Correct.

Method

The interior angles form an arithmetic sequence with first term 84 and common difference 4.
So the sum of the angles is

$$\begin{aligned} & \frac{n}{2}(2(84) + (n-1)4) \\ &= \frac{n}{2}(168 + 4n - 4) \\ &= 2n^2 + 82n \end{aligned}$$

The sum of the interior angles of an n -sided polygon is

$$180(n-2)$$

So

$$2n^2 + 82n = 180(n-2)$$

$$2n^2 - 98n + 360 = 0$$

$$n^2 - 49n + 180 = 0$$

$$(n-45)(n-4) = 0$$

Since $n > 5$,

$$n = 45$$

Therefore the sum of the interior angles is

$$180(45-2) = 7740$$

FINAL ANSWER

7740 degrees

CLASSIFIED PROBLEM

Paper: P1HR

Session: May 2024

Q#: 23

Marks: 4

TOPIC: **Sequences**

Source: May 2024 P1HR

23 Here are the first three terms of an arithmetic sequence.

$$(4x-14) \quad , \quad (x+2) \quad , \quad (7x-9)$$

Find, as an integer, the sum of the first 40 terms of the sequence.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

WORKED SOLUTION*Dr. Eslam Ahmed***May 2024 P1HR | Question 23**

Sequences

Topic check. Correct.**Method**

The first three terms are

$$4x - 14, \quad x + 2, \quad 7x - 9$$

Equal differences give

$$(x + 2) - (4x - 14) = (7x - 9) - (x + 2)$$

$$-3x + 16 = 6x - 11$$

$$9x = 27$$

$$x = 3$$

So the sequence is

$$-2, 5, 12, \dots$$

with common difference 7.

$$S_{40} = \frac{40}{2}(2(-2) + 39(7))$$

$$S_{40} = 20(-4 + 273) = 5380$$

FINAL ANSWER

5380

CLASSIFIED PROBLEM**Paper: P1H****Session: MayNov 2020****Q#: 24****Marks: 4**TOPIC: **Sequences**

Source: May-Nov 2020 P1H

24 Here are the first five terms of an arithmetic sequence.

8 15 22 29 36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

.....
(Total for Question 24 is 4 marks)

WORKED SOLUTION*Dr. Eslam Ahmed*

May-Nov 2020 P1H | **Question 24**
Sequences

Topic check. Correct.

Method

The arithmetic sequence is

$$8, 15, 22, 29, 36, \dots$$

So

$$a = 8, \quad d = 7$$

The n th term is

$$u_n = 8 + 7(n - 1) = 7n + 1$$

$$u_{50} = 7(50) + 1 = 351$$

$$u_{100} = 7(100) + 1 = 701$$

There are

$$100 - 50 + 1 = 51$$

terms from the 50th to the 100th inclusive.

$$\begin{aligned} \text{sum} &= \frac{51}{2}(351 + 701) \\ &= 26826 \end{aligned}$$

FINAL ANSWER

26826

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2021

Q#: 20

Marks: 5

TOPIC: Sequences

Source: Nov 2021 P2H

20 Here are the first four terms of an arithmetic series.

$$k \quad \frac{3k}{4} \quad \frac{k}{2} \quad \frac{k}{4}$$

Given that the 15th term of the series is $(90 + 2k)$,

calculate the sum of the first 30 terms of the series.

(Total for Question 20 is 5 marks)

WORKED SOLUTION*Dr. Eslam Ahmed*

Nov 2021 P2H | Question 20
Sequences

Topic check. Correct.

Method

The first four terms are

$$k, \quad \frac{3k}{4}, \quad \frac{k}{2}, \quad \frac{k}{4}$$

The common difference is

$$d = -\frac{k}{4}$$

The 15th term is

$$u_{15} = k + 14\left(-\frac{k}{4}\right)$$

$$u_{15} = k - \frac{7k}{2} = -\frac{5k}{2}$$

Given

$$-\frac{5k}{2} = 90 + 2k$$

$$-5k = 180 + 4k$$

$$k = -20$$

So

$$a = -20, \quad d = 5$$

$$S_{30} = \frac{30}{2}(2(-20) + 29(5))$$

$$S_{30} = 15(105) = 1575$$

FINAL ANSWER

1575

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2025

Q#: 25

Marks: 5

TOPIC: Sequences

Source: Nov 2025 P2H

- 25** The first term of an arithmetic series is 10
The 20th term of the series is 86
- The sum of the first N terms of the series is 5194
- Work out the value of N
Show your working clearly.

 $N = \dots\dots\dots$

(Total for Question 25 is 5 marks)

WORKED SOLUTION*Dr. Eslam Ahmed*

Nov 2025 P2H | Question 25
Sequences

Topic check. Correct.

Method

$$a = 10$$

$$u_{20} = 86$$

$$10 + 19d = 86$$

$$d = 4$$

Use

$$S_N = \frac{N}{2}(2a + (N - 1)d)$$

$$5194 = \frac{N}{2}(20 + 4(N - 1))$$

$$5194 = \frac{N}{2}(4N + 16)$$

$$5194 = 2N(N + 4)$$

$$N^2 + 4N - 2597 = 0$$

$$N = \frac{-4 + 102}{2} = 49$$

FINAL ANSWER

$$N = 49$$

Functions

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: P1H

Session: Jan 2020

Q#: 21

Marks: 5

TOPIC: **Functions**

Source: Jan 2020 P1H

21 The functions f and g are such that

$$f(x) = x^2 - 2x \qquad g(x) = x + 3$$

The function h is such that $h(x) = fg(x)$ for $x \geq -2$

Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots$$

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P1H | Question 21
Functions

Topic check. Correct.

Method

$$f(x) = x^2 - 2x$$

$$g(x) = x + 3$$

$$h(x) = fg(x) = f(g(x))$$

$$h(x) = (x + 3)^2 - 2(x + 3)$$

$$h(x) = x^2 + 4x + 3$$

$$h(x) = (x + 2)^2 - 1$$

Let

$$y = (x + 2)^2 - 1$$

Since the domain is $x \geq -2$, take the positive square root:

$$x + 2 = \sqrt{y + 1}$$

$$x = \sqrt{y + 1} - 2$$

FINAL ANSWER

$$h^{-1}(x) = \sqrt{x + 1} - 2$$

CLASSIFIED PROBLEM

Paper: P1H

Session: Jan 2021

Q#: 22

Marks: 3

TOPIC: **Functions**

Source: Jan 2021 P1H

22 The function f is such that $f(x) = x^2 - 8x + 5$ where $x \leq 4$

Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$$f^{-1}(x) = \dots\dots\dots$$

(Total for Question 22 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1H | Question 22

Functions

Topic check. Correct.

Method

$$f(x) = x^2 - 8x + 5$$

$$f(x) = (x - 4)^2 - 11$$

Let

$$y = (x - 4)^2 - 11$$

$$y + 11 = (x - 4)^2$$

Since the domain is $x < 4$, take the negative square root:

$$x - 4 = -\sqrt{y + 11}$$

$$x = 4 - \sqrt{y + 11}$$

FINAL ANSWER

$$f^{-1}(x) = 4 - \sqrt{x + 11}$$

CLASSIFIED PROBLEM**Paper: P2H****Session: Jan 2022****Q#: 25****Marks: 5**TOPIC: **Functions**

Source: Jan 2022 P2H

25 The function g is defined as

$$g : x \mapsto 5 + 6x - x^2 \quad \text{with domain } \{x : x \geq 3\}$$

(a) Express the inverse function g^{-1} in the form $g^{-1} : x \mapsto \dots$

$$g^{-1} : x \mapsto \dots \quad (4)$$

(b) State the domain of g^{-1}

$$\dots \quad (1)$$

(Total for Question 25 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2H | Question 25
Functions

Topic check. Correct.

Method

$$g(x) = 5 + 6x - x^2$$

Complete the square:

$$g(x) = 14 - (x - 3)^2$$

Let

$$y = 14 - (x - 3)^2$$

$$(x - 3)^2 = 14 - y$$

The domain of g is $x \geq 3$, so take the positive square root:

$$x - 3 = \sqrt{14 - y}$$

$$x = 3 + \sqrt{14 - y}$$

Therefore

$$g^{-1} : x \mapsto 3 + \sqrt{14 - x}$$

The range of g is $x \leq 14$, so this is the domain of g^{-1} .

FINAL ANSWER

(a) $g^{-1} : x \mapsto 3 + \sqrt{14 - x}$, (b) domain $x \leq 14$

CLASSIFIED PROBLEM**Paper: P2HR****Session: Jan 2022****Q#: 23****Marks: 4****TOPIC: Functions**

Source: Jan 2022 P2HR

23 The functions f and g are such that

$$f(x) = x + 25 \qquad g(x) = x^2 - 12x$$

The function h is such that $h(x) = fg(x)$ The domain of h is $\{x : x \leq 6\}$ Express the inverse function h^{-1} in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots$$

(Total for Question 23 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2HR | Question 23
Functions

Topic check. Correct.

Method

$$f(x) = x + 25$$

$$g(x) = x^2 - 12x$$

$$h(x) = fg(x) = f(g(x))$$

$$h(x) = x^2 - 12x + 25$$

$$h(x) = (x - 6)^2 - 11$$

Let

$$y = (x - 6)^2 - 11$$

$$y + 11 = (x - 6)^2$$

Since the domain of h is $x < 6$, take the negative square root:

$$x - 6 = -\sqrt{y + 11}$$

$$x = 6 - \sqrt{y + 11}$$

FINAL ANSWER

$$h^{-1}(x) = 6 - \sqrt{x + 11}$$

CLASSIFIED PROBLEM**Paper: P1H****Session: May 2021****Q#: 24****Marks: 6**TOPIC: **Functions**

Source: May 2021 P1H

24 The functions f and g are defined as

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(a) Find $fg(2)$

(2)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$$f^{-1}(x) = \dots$$

(4)

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P1H | Question 24
Functions

Topic check. Correct.

Method

(a)

$$g(2) = 7(2) - 6 = 8$$

$$fg(2) = f(8)$$

$$f(8) = 5(8)^2 - 10(8) + 7$$

$$f(8) = 247$$

(b)

$$f(x) = 5x^2 - 10x + 7$$

$$f(x) = 5(x - 1)^2 + 2$$

Let

$$y = 5(x - 1)^2 + 2$$

$$(x - 1)^2 = \frac{y - 2}{5}$$

Since the domain is $x \geq 1$, take the positive square root:

$$x = 1 + \sqrt{\frac{y - 2}{5}}$$

FINAL ANSWER

$$(a) 247, (b) f^{-1}(x) = 1 + \sqrt{\frac{x - 2}{5}}$$

CLASSIFIED PROBLEM**Paper: P1H****Session: Nov 2024****Q#: 25****Marks: 4****TOPIC: Functions**

Source: Nov 2024 P1H

25 The function f is such that $f(x) = 2x^2 - 24x + 7$ where $x \geq 6$

Find the inverse function $f^{-1}(x)$

$f^{-1}(x) = \dots\dots\dots$

(Total for Question 25 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2024 P1H | **Question 25**
Functions

Topic check. Correct.

Method

$$f(x) = 2x^2 - 24x + 7$$

Complete the square:

$$f(x) = 2(x^2 - 12x) + 7$$

$$f(x) = 2[(x - 6)^2 - 36] + 7$$

$$f(x) = 2(x - 6)^2 - 65$$

Let $y = f(x)$.

$$y = 2(x - 6)^2 - 65$$

$$y + 65 = 2(x - 6)^2$$

$$\frac{y + 65}{2} = (x - 6)^2$$

Since $x \geq 6$, take the positive square root:

$$x - 6 = \sqrt{\frac{y + 65}{2}}$$

$$x = 6 + \sqrt{\frac{y + 65}{2}}$$

Replace y by x .

FINAL ANSWER

$$f^{-1}(x) = 6 + \sqrt{\frac{x + 65}{2}}$$

Graphs of Functions

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: 4WM2HR

Session: May2025

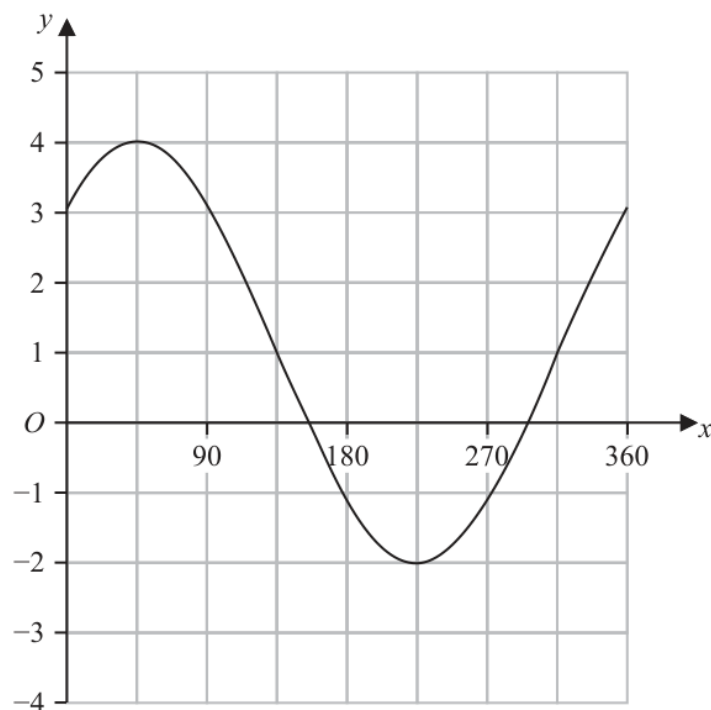
Q#: 23

Marks: 3

TOPIC: **Graphs of Functions**

Source: May 2025 4WM2HR

23 The graph of $y = a\sin(x + b)^\circ + c$ for $0 \leq x \leq 360$ is drawn on the grid below.



Find a suitable value for a , for b and for c

$a = \dots\dots\dots$

$b = \dots\dots\dots$

$c = \dots\dots\dots$

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2HR | Question 23

Graphs of Functions

Required:

The graph of $y = a \sin (x+b)^{\circ} + c$ for $0 \leq x \leq 360$ is shown. Find suitable values for a , b , and c .

Step:

Read the maximum and minimum values from the graph to find the **amplitude** a and **vertical shift** c :

$$a = \frac{\max - \min}{2} = \frac{4 - (-2)}{2} = 3 \quad c = \frac{\max + \min}{2} = \frac{4 + (-2)}{2} = 1$$

Step:

Find the **phase shift** b . The standard sin has its peak at $x = 90^{\circ}$. From the graph, the maximum occurs at $x \approx 45^{\circ}$:

$$x + b = 90^{\circ} \implies 45^{\circ} + b = 90^{\circ} \implies b = 45^{\circ}$$

Step:

Verify at $x = 0$: $y = 4 \sin(0 + 45)^{\circ} = 4 \times \frac{\sqrt{2}}{2} \approx 2.83$ – consistent with graph

FINAL ANSWER

$$a = 3, b = 45, c = 1$$

Note:

The period of the graph is still 360° (unchanged), so no horizontal stretch is applied.

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Nov2025

Q#: 25

Marks: 3

TOPIC: **Graphs of Functions**

Source: Nov 2025 4WM2H

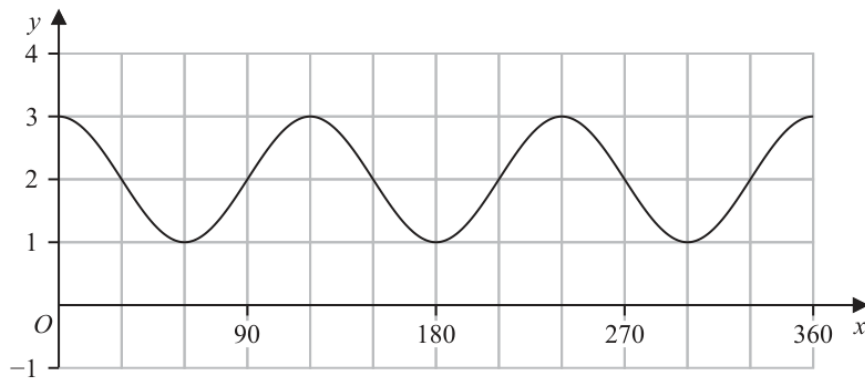
25 A curve has equation $y = f(x)$

There is one minimum point on this curve.

The coordinates of this minimum point are $(8, 1)$

- (a) Write down the coordinates of the minimum point on the curve with equation $y = f(4x)$

(.....,)
(1)

The diagram shows the graph of $y = \cos(ax) + b$ for $0 \leq x \leq 360$ 

- (b) (i) Find the value of a

$a = \dots\dots\dots$
(1)

- (ii) Find the value of b

$b = \dots\dots\dots$
(1)

(Total for Question 25 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM2H | Question 25

Graphs of Functions

The transformation $y = f(4x)$ represents a horizontal stretch by a scale factor of $\frac{1}{4}$. The x-coordinate is multiplied by $\frac{1}{4}$ while the y-coordinate stays the same.

$$\text{New minimum} = \left(8 \times \frac{1}{4}, 1\right) = (2, 1)$$

FINAL ANSWER

(2,1)

Step 1: Find the vertical shift *b*. The standard $y = \cos(x)$ graph fluctuates between -1 and 1, centered at $y = 0$. The given graph fluctuates between a minimum of $y = 1$ and a maximum of $y = 3$. The center line is $\frac{3+1}{2} = 2$. Therefore, $b = 2$.

Step 2: Find the frequency multiplier *a*. The period of the given graph completes one full cycle between $x = 0$ and $x = 90^\circ$ (it starts at 3, dips to 1, and returns to 3 at 90°). The standard cosine period is 360° .

$$\text{Period} = \frac{360^\circ}{a}$$

$$90^\circ = \frac{360^\circ}{a} \implies a = 4$$

FINAL ANSWER

 $a = 4, b = 2$

Transformations of Graphs

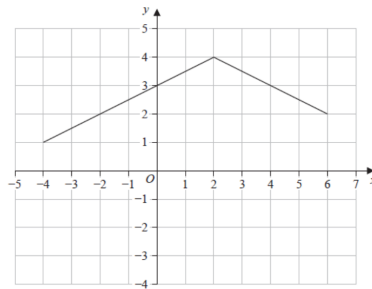
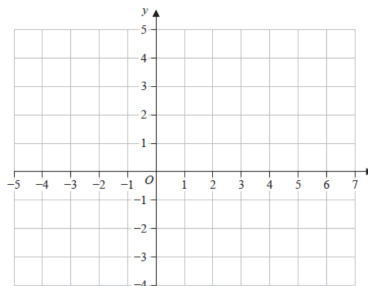
Dr. Eslam Ahmed

CLASSIFIED PROBLEM**Paper: 4WM2H****Session: May2025****Q#: 27****Marks: 4****TOPIC: Transformations of Graphs**

Source: May 2025 4WM2H

27 A curve has equation $y = f(x)$ There is one maximum point on this curve.
The coordinates of this maximum point are $(6, -5)$

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x - 4)$ (.....,)
(1)(ii) $y = f(3x)$ (.....,)
(1)P81641A
©2005 Pearson Education LtdHere is the graph of $y = g(x)$ (b) On the grid below, draw the graph of $y = g(2x) - 3$ 

(2)

(Total for Question 27 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2H | Question 27

Transformations of Graphs

Solution

(a)

For $y = f(x - 4)$, move the graph 4 units right:

$$(6, -5) \rightarrow (10, -5)$$

For $y = f(3x)$, divide the x -coordinate by 3:

$$(6, -5) \rightarrow (2, -5)$$

(b) For $y = g(2x) - 3$, halve each x -coordinate and subtract 3 from each y -coordinate.
Useful transformed points are

$$(-4, 1) \rightarrow (-2, -2), \quad (2, 4) \rightarrow (1, 1), \quad (6, 2) \rightarrow (3, -1)$$

Join these points with straight line segments.

FINAL ANSWER $(10, -5)$, $(2, -5)$, graph through $(-2, -2)$, $(1, 1)$, $(3, -1)$.

Differentiation

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: May2025

Q#: 22

Marks: 4

TOPIC: **Differentiation**

Source: May 2025 4WM2H

- 22 Work out the range of values of x for which the graph of $y = x^3 + 2x^2 - 10x + 5$ has a positive gradient.
Show clear algebraic working.

.....
(Total for Question 22 is 4 marks)

WORKED SOLUTION*Dr. Eslam Ahmed***May 2025 4WM2H | Question 22**

Differentiation

Solution

$$\frac{dy}{dx} = 3x^2 + 2x - 10$$

For a positive gradient,

$$3x^2 + 2x - 10 > 0$$

The roots are

$$x = \frac{-1 \pm \sqrt{31}}{3}$$

Since the quadratic opens upwards,

$$x < \frac{-1 - \sqrt{31}}{3} \quad \text{or} \quad x > \frac{-1 + \sqrt{31}}{3}$$

FINAL ANSWER

$$x < \frac{-1 - \sqrt{31}}{3} \quad \text{or} \quad x > \frac{-1 + \sqrt{31}}{3}.$$

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Nov2025

Q#: 22

Marks: 6

TOPIC: **Differentiation**

Source: Nov 2025 4WM2H

22 Curve **C** has equation $y = x^3 - 16x + 7$

At two points on **C**, the gradient is 11

The tangents to **C** at these two points have equations of the form $y = ax + b$

Work out the two possible values of b

Show clear algebraic working.

.....
(Total for Question 22 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM2H | Question 22

Differentiation

Step 1: Differentiate the curve to find the gradient expression.

$$\frac{dy}{dx} = 3x^2 - 16$$

Step 2: Set the gradient to 11 to find the x-coordinates.

$$3x^2 - 16 = 11$$

$$3x^2 = 27$$

$$x^2 = 9 \implies x = 3 \text{ or } x = -3$$

Step 3: Find the corresponding y-coordinates on the curve.

$$\text{For } x = 3: y = (3)^3 - 16(3) + 7 = 27 - 48 + 7 = -14$$

Point 1: (3,-14)

$$\text{For } x = -3: y = (-3)^3 - 16(-3) + 7 = -27 + 48 + 7 = 28$$

Point 2: (-3,28)**Step 4: Find the y-intercept (b) for each tangent ($y = 11x + b$).**

Using (3,-14):

$$-14 = 11(3) + b \implies -14 = 33 + b \implies b = -47$$

Using (-3,28):

$$28 = 11(-3) + b \implies 28 = -33 + b \implies b = 61$$

FINAL ANSWER

$$b = -47 \text{ and } 61$$

Transformations of Graphs

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jan 2020

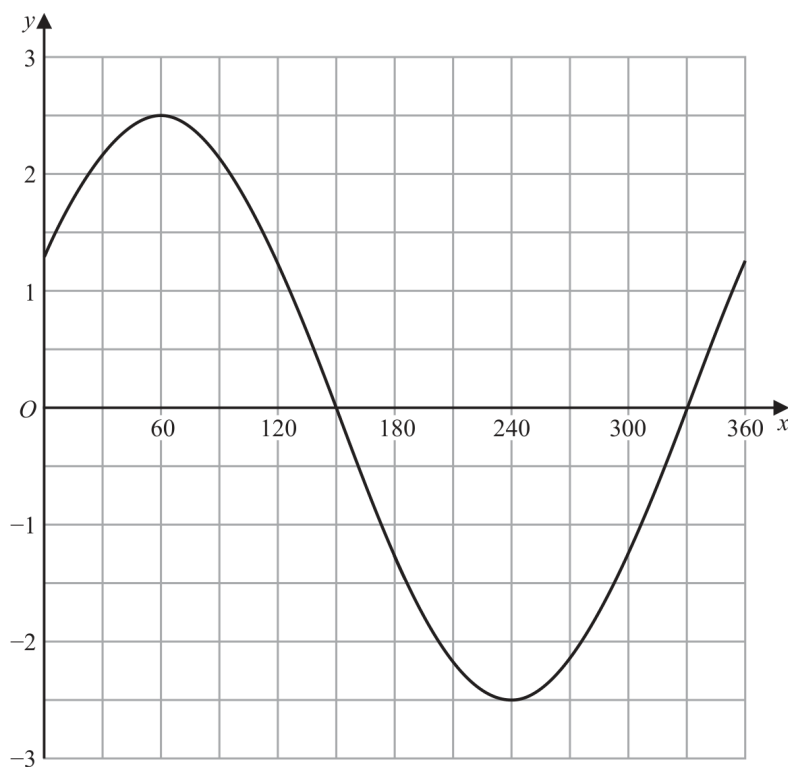
Q#: 22

Marks: 4

TOPIC: Transformations of Graphs

Source: Jan 2020 P1HR

22 The graph of $y = a \cos(x + b)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid.



(a) Find the value of a and the value of b .

$a = \dots\dots\dots$

$b = \dots\dots\dots$

(2)

Another curve C has equation $y = f(x)$

The coordinates of the minimum point of C are $(4, 5)$

(b) Write down the coordinates of the minimum point of the curve with equation

(i) $y = f(2x)$

(.....,)

(ii) $y = f(x) - 7$

(.....,)

(2)

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P1HR | Question 22

Transformations of Graphs

Topic check. Correct.**Method**

For part (a), the graph has maximum 2.5 and minimum -2.5 , so

$$a = 2.5$$

The maximum occurs at $x = 60^\circ$. For $y = a \cos(x + b)^\circ$, a maximum occurs when

$$x + b = 360$$

So

$$60 + b = 360$$

$$b = 300$$

For part (b), the minimum point of $y = f(x)$ is $(4, 5)$.

For

$$y = f(2x)$$

the x -coordinate is divided by 2:

$$(4, 5) \rightarrow (2, 5)$$

For

$$y = f(x) - 7$$

the y -coordinate decreases by 7:

$$(4, 5) \rightarrow (4, -2)$$

FINAL ANSWER

$a = 2.5$, $b = 300$; $(2, 5)$ and $(4, -2)$

CLASSIFIED PROBLEM

Paper: P2H

Session: Jan 2020

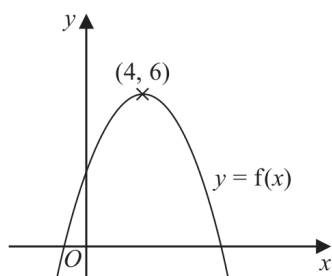
Q#: 21

Marks: 4

TOPIC: Transformations of Graphs

Source: Jan 2020 P2H

- 21 The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.
The coordinates of this maximum point are (4, 6)

- (a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 4)$

(.....,)

(ii) $y = f(2x)$

(.....,)

(2)

The equation of a curve **C** is $y = x^2 + 3x + 4$

The curve **C** is transformed to curve **S** under the translation $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$

- (b) Find an equation of curve **S**.

You do not need to simplify the equation.

(2)

(Total for Question 21 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2H | Question 21

Transformations of Graphs

Topic check. Correct.**Method**The maximum point of $y = f(x)$ is $(4, 6)$.

For

$$y = f(x + 4)$$

the graph moves 4 units left:

$$(4, 6) \rightarrow (0, 6)$$

For

$$y = f(2x)$$

the x -coordinate is divided by 2:

$$(4, 6) \rightarrow (2, 6)$$

For part (b),

$$y = x^2 + 3x + 4$$

The translation vector is

$$\begin{pmatrix} 4 \\ 6 \end{pmatrix}$$

So the graph moves 4 units right and 6 units up.

Replace x by $x - 4$, then add 6:

$$y = (x - 4)^2 + 3(x - 4) + 4 + 6$$

$$y = (x - 4)^2 + 3(x - 4) + 10$$

FINAL ANSWER $(0, 6)$, $(2, 6)$, and $y = (x - 4)^2 + 3(x - 4) + 10$

CLASSIFIED PROBLEM**Paper: P1H****Session: Jan 2021****Q#: 21****Marks: 4****TOPIC: Transformations of Graphs**

Source: Jan 2021 P1H

21 A curve has equation $y = f(x)$ The coordinates of the minimum point on this curve are $(-9, 15)$

(a) Write down the coordinates of the minimum point on the curve with equation

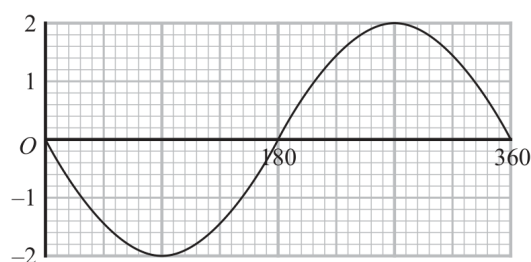
(i) $y = f(x + 3)$

(.....,)

(ii) $y = \frac{1}{3}f(x)$

(.....,)

(2)

The graph of $y = a \cos(x + b)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid below.Given that $a > 0$ and that $0 < b < 360$ (b) find the value of a and the value of b .

$a = \dots\dots\dots$

$b = \dots\dots\dots$

(2)

(Total for Question 21 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1H | Question 21

Transformations of Graphs

Topic check. Correct.**Method**The minimum point of $y = f(x)$ is $(-9, 15)$.

For

$$y = f(x + 3)$$

the graph moves 3 units left:

$$(-9, 15) \rightarrow (-12, 15)$$

For

$$y = \frac{1}{3}f(x)$$

the y -coordinate is divided by 3:

$$(-9, 15) \rightarrow (-9, 5)$$

For part (b), the cosine graph has maximum 2 and minimum -2 , so

$$a = 2$$

The graph has a minimum at $x = 90^\circ$. For $a \cos(x + b)^\circ$, a minimum occurs when

$$x + b = 180$$

So

$$90 + b = 180$$

$$b = 90$$

FINAL ANSWER

$$(-12, 15), (-9, 5), a = 2, b = 90$$

CLASSIFIED PROBLEM

Paper: P1H

Session: Jan 2022

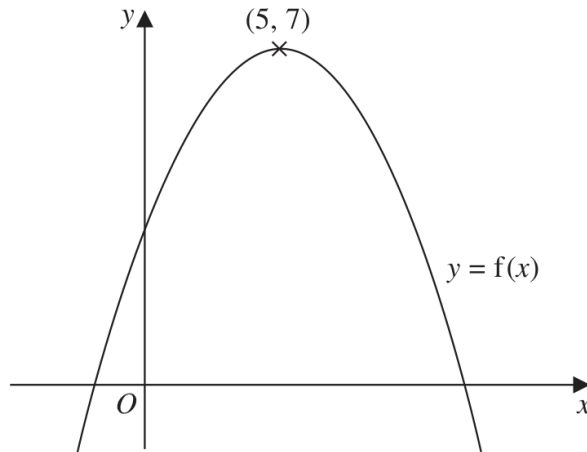
Q#: 23

Marks: 2

TOPIC: Transformations of Graphs

Source: Jan 2022 P1H

23 The diagram shows a sketch of the curve with equation $y = f(x)$



There is only one maximum point on the curve.
The coordinates of this maximum point are $(5, 7)$

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 9)$

(.....,)

(ii) $y = f(x) + 3$

(.....,)

(Total for Question 23 is 2 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1H | Question 23

Transformations of Graphs

Topic check. Correct.**Method**The maximum point of $y = f(x)$ is $(5, 7)$.

For

$$y = f(x + 9)$$

the graph moves 9 units left:

$$(5, 7) \rightarrow (-4, 7)$$

For

$$y = f(x) + 3$$

the graph moves 3 units up:

$$(5, 7) \rightarrow (5, 10)$$

FINAL ANSWER $(-4, 7)$ and $(5, 10)$

CLASSIFIED PROBLEM**Paper: P1HR****Session: Jan 2022****Q#: 20****Marks: 6****TOPIC: Transformations of Graphs**

Source: Jan 2022 P1HR

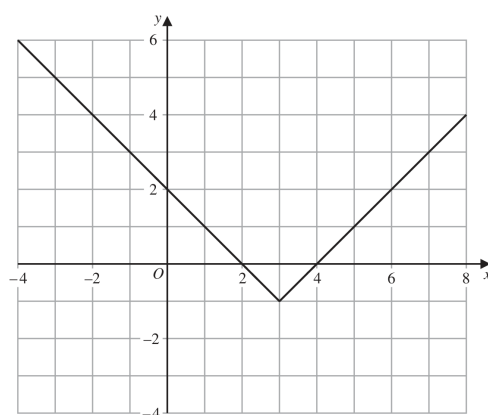
20 The curve with equation $y = f(x)$ has one turning point.The coordinates of this turning point are $(-6, -4)$

(a) Write down the coordinates of the turning point on the curve with equation

(i) $y = f(x) + 5$

(.....,) (1)

(ii) $y = f(3x)$

(.....,)
(2)The graph of $y = g(x)$ is shown on the grid below.(b) On the grid, sketch the graph of $y = 2g(x)$ for $-1 \leq x \leq 7$

(2)

The graph of $y = h(x)$ intersects the x -axis at two points.The coordinates of the two points are $(-1, 0)$ and $(6, 0)$ The graph of $y = h(x + a)$ passes through the point with coordinates $(2, 0)$, where a is a constant.(c) Find the two possible values of a,
(2)

(Total for Question 20 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1HR | Question 20

Transformations of Graphs

Topic check. Correct.**Method**The turning point of $y = f(x)$ is $(-6, -4)$.

For

$$y = f(x) + 5$$

the graph moves 5 units up:

$$(-6, -4) \rightarrow (-6, 1)$$

For

$$y = f(3x)$$

the x -coordinate is divided by 3:

$$(-6, -4) \rightarrow (-2, -4)$$

For the sketch of $y = 2g(x)$, keep every x -coordinate the same and double every y -coordinate.

Useful points are

$$(-1, 3) \rightarrow (-1, 6)$$

$$(2, 0) \rightarrow (2, 0)$$

$$(3, -1) \rightarrow (3, -2)$$

$$(4, 0) \rightarrow (4, 0)$$

$$(7, 3) \rightarrow (7, 6)$$

Draw straight line segments through these points for $-1 \leq x \leq 7$.For part (c), $h(x)$ has roots at $x = -1$ and $x = 6$.For $h(x + a)$ to pass through $(2, 0)$,

$$2 + a = -1 \quad \text{or} \quad 2 + a = 6$$

$$a = -3 \quad \text{or} \quad a = 4$$

FINAL ANSWER $(-6, 1)$, $(-2, -4)$, sketch through the doubled points, and $a = -3$ or $a = 4$

CLASSIFIED PROBLEM

Paper: P1H

Session: Jun 2022

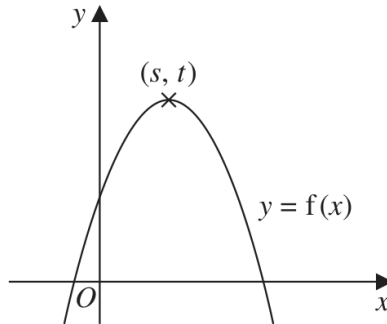
Q#: 20

Marks: 2

TOPIC: Transformations of Graphs

Source: Jun 2022 P1H

20 The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.

The coordinates of this maximum point are (s, t)

Find, in terms of s and t , the coordinates of the maximum point on the curve with equation

(i) $y = f(x - 2)$

(..... ,)
(1)

(ii) $y = 3f(x)$

(..... ,)
(1)

(Total for Question 20 is 2 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1H | Question 20

Transformations of Graphs

Topic check. Correct.**Method**The maximum point of $y = f(x)$ is (s, t) .

For

$$y = f(x - 2)$$

the graph moves 2 units right:

$$(s, t) \rightarrow (s + 2, t)$$

For

$$y = 3f(x)$$

the y -coordinate is multiplied by 3:

$$(s, t) \rightarrow (s, 3t)$$

FINAL ANSWER $(s + 2, t)$ and $(s, 3t)$

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2022

Q#: 22

Marks: 4

TOPIC: Transformations of Graphs

Source: Jun 2022 P2HR

22 The point A with coordinates $(-3, 2)$ lies on the straight line with equation $y = f(x)$

(a) Find the coordinates of the image of the point A on the straight line with equation

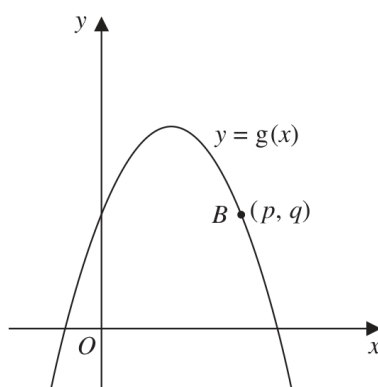
(i) $y = f(x) - 3$

(.....,)
(1)

(ii) $y = f\left(\frac{x}{2}\right)$

(.....,)
(1)

Here is a sketch of part of the curve with equation $y = g(x)$



The point B with coordinates (p, q) lies on the curve.

(b) Find the coordinates of the image of the point B on the curve with equation

$$y = -g(x - c)$$

where c is a constant.

(.....,)
(2)

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2HR | Question 22

Transformations of Graphs

Topic check. Correct.**Method**Point $A = (-3, 2)$ lies on $y = f(x)$.

For

$$y = f(x) - 3$$

the graph moves 3 units down:

$$(-3, 2) \rightarrow (-3, -1)$$

For

$$y = f\left(\frac{x}{2}\right)$$

the x -coordinate is multiplied by 2:

$$(-3, 2) \rightarrow (-6, 2)$$

For part (b), point $B = (p, q)$ lies on $y = g(x)$.

For

$$y = -g(x - c)$$

 $x - c$ moves the graph c units right, and the minus sign reflects the graph in the x -axis.

So

$$(p, q) \rightarrow (p + c, -q)$$

FINAL ANSWER $(-3, -1)$, $(-6, 2)$, and $(p + c, -q)$

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2024

Q#: 23

Marks: 2

TOPIC: Transformations of Graphs

Source: Jun 2024 P2HR

23 A curve has equation $y = f(x)$ The coordinates of the minimum point on this curve are $(6, -3)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x) + 10$

(.....,)
(1)

(ii) $y = f(3x)$

(.....,)
(1)

(Total for Question 23 is 2 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2HR | Question 23

Transformations of Graphs

Topic check. Correct.**Method**The minimum point of $y = f(x)$ is $(6, -3)$.

For

$$y = f(x) + 10$$

the graph moves 10 units up:

$$(6, -3) \rightarrow (6, 7)$$

For

$$y = f(3x)$$

the x -coordinate is divided by 3:

$$(6, -3) \rightarrow (2, -3)$$

FINAL ANSWER $(6, 7)$ and $(2, -3)$

CLASSIFIED PROBLEM**Paper: P1HR****Session: Jun 2025****Q#: 21****Marks: 3****TOPIC: Transformations of Graphs**

Source: Jun 2025 P1HR

- 21** The curve with equation $y = f(x)$ has one turning point.
The coordinates of this turning point are $(-6, 9)$

- (a) Write down the coordinates of the turning point on the curve
with equation $y = f(3x)$

(.....,)
(1)

The curve **C** with equation $y = g(x)$ is transformed to give the curve **S** with
equation $y = g(x + a) + b$

The point $(4, -5)$ on **C** is mapped to the point $(1, -16)$ on **S**

- (b) Write down the value of a and the value of b

$a = \dots\dots\dots$

$b = \dots\dots\dots$

(2)

(Total for Question 21 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1HR | Question 21

Transformations of Graphs

Topic check. Correct.**Method**The turning point of $y = f(x)$ is $(-6, 9)$.

For

$$y = f(3x)$$

the x -coordinate is divided by 3:

$$(-6, 9) \rightarrow (-2, 9)$$

For part (b), a point $(4, -5)$ on $y = g(x)$ maps to $(1, -16)$ on

$$y = g(x + a) + b$$

For the x -coordinate:

$$1 + a = 4$$

$$a = 3$$

For the y -coordinate:

$$-5 + b = -16$$

$$b = -11$$

FINAL ANSWER $(-2, 9), a = 3, b = -11$

CLASSIFIED PROBLEM**Paper: P1H****Session: May 2023****Q#: 21****Marks: 2****TOPIC: Transformations of Graphs**

Source: May 2023 P1H

21 A curve has equation $y = f(x)$

There is only one turning point on the curve.

The coordinates of this turning point are (6, 5)

Write down the coordinates of the turning point on the curve with equation

(a) $y = f(x - 4)$

(.....,)
(1)

(b) $y = f(3x)$

(.....,)
(1)**(Total for Question 21 is 2 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1H | Question 21

Transformations of Graphs

Topic check. Correct.**Method**The turning point of $y = f(x)$ is $(6, 5)$.

For

$$y = f(x - 4)$$

the graph moves 4 units right:

$$(6, 5) \rightarrow (10, 5)$$

For

$$y = f(3x)$$

the x -coordinate is divided by 3:

$$(6, 5) \rightarrow (2, 5)$$

FINAL ANSWER $(10, 5)$ and $(2, 5)$

CLASSIFIED PROBLEM**Paper: P1HR****Session: May 2023****Q#: 20****Marks: 7****TOPIC: Transformations of Graphs**

Source: May 2023 P1HR

20 Curve **C** has equation $y = f(x)$ The graph of curve **C** has one maximum point.

The coordinates of this maximum point are (3, 5)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = 2f(x)$

(.....,)
(1)

(ii) $y = f(x) - 7$

(.....,)
(1)

(iii) $y = f(-x)$

(.....,)
(1)Curve **L** has equation $y = x^2 + 7x + 20$ Curve **L** is transformed to curve **S** under the translation $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ (b) Find an equation for **S**Give your answer in the form $y = ax^2 + bx + c$ $y = \dots\dots\dots$
(4)**(Total for Question 20 is 7 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1HR | Question 20

Transformations of Graphs

Topic check. Correct.**Method**The maximum point of $y = f(x)$ is $(3, 5)$.

For

$$y = 2f(x)$$

the y -coordinate is multiplied by 2:

$$(3, 5) \rightarrow (3, 10)$$

For

$$y = f(x) - 7$$

the graph moves 7 units down:

$$(3, 5) \rightarrow (3, -2)$$

For

$$y = f(-x)$$

the graph reflects in the y -axis:

$$(3, 5) \rightarrow (-3, 5)$$

For part (b),

$$y = x^2 + 7x + 20$$

The translation vector is

$$\begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

So the graph moves 2 units right.

Replace x by $x - 2$:

$$y = (x - 2)^2 + 7(x - 2) + 20$$

$$y = x^2 + 3x + 10$$

FINAL ANSWER $(3, 10)$, $(3, -2)$, $(-3, 5)$, and $y = x^2 + 3x + 10$

CLASSIFIED PROBLEM**Paper: P1H****Session: May 2024****Q#: 21****Marks: 2****TOPIC: Transformations of Graphs**

Source: May 2024 P1H

21 A curve has equation $y = f(x)$

There is one minimum point on this curve.

The coordinates of this minimum point are $(5, -4)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 7)$

(.....,.....)
(1)

(ii) $y = f(x) - 6$

(.....,.....)
(1)

(Total for Question 21 is 2 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1H | Question 21

Transformations of Graphs

Topic check. Correct.**Method**The minimum point of $y = f(x)$ is $(5, -4)$.

For

$$y = f(x + 7)$$

the graph moves 7 units left:

$$(5, -4) \rightarrow (-2, -4)$$

For

$$y = f(x) - 6$$

the graph moves 6 units down:

$$(5, -4) \rightarrow (5, -10)$$

FINAL ANSWER $(-2, -4)$ and $(5, -10)$

CLASSIFIED PROBLEM

Paper: P1H

Session: MayNov 2020

Q#: 25

Marks: 3

TOPIC: Transformations of Graphs

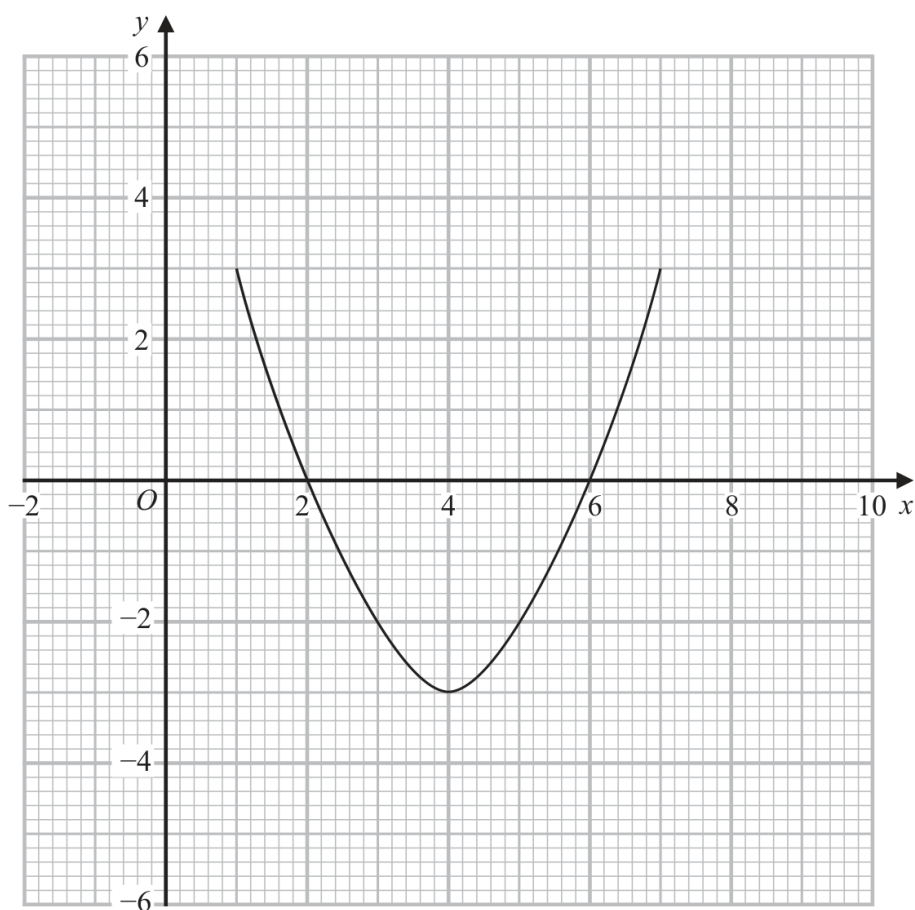
Source: May-Nov 2020 P1H

- 25** The curve with equation $y = g(x)$ is transformed to the curve with equation $y = -g(x)$ by the single transformation **T**.

(a) Describe fully the transformation **T**.

(1)

The diagram shows the graph of $y = f(x)$



(b) On the grid, draw the graph of $y = 2f(x - 1)$

(2)

(Total for Question 25 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1H | Question 25

Transformations of Graphs

Topic check. Correct.**Method**

For part (a),

$$y = g(x) \rightarrow y = -g(x)$$

The x -coordinates stay the same and the y -coordinates change sign.So the transformation is a reflection in the x -axis.

For part (b),

$$y = 2f(x - 1)$$

First, $x - 1$ moves the graph 1 unit right. Then the factor 2 stretches the graph vertically by scale factor 2.

Use points from the original graph:

$$(1, 3) \rightarrow (2, 6)$$

$$(2, 0) \rightarrow (3, 0)$$

$$(4, -3) \rightarrow (5, -6)$$

$$(6, 0) \rightarrow (7, 0)$$

$$(7, 3) \rightarrow (8, 6)$$

Draw the transformed parabola through these points.

FINAL ANSWERreflection in the x -axis; transformed graph through $(2, 6)$, $(3, 0)$, $(5, -6)$, $(7, 0)$, $(8, 6)$

CLASSIFIED PROBLEM

Paper: P1HR

Session: MayNov 2020

Q#: 20

Marks: 2

TOPIC: Transformations of Graphs

Source: May-Nov 2020 P1HR

20 A curve has equation $y = f(x)$

There is only one maximum point on the curve.

The coordinates of this maximum point are $(-3, 4)$

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x) - 6$

(..... ,)

(ii) $y = f(2x)$

(..... ,)

(Total for Question 20 is 2 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1HR | Question 20

Transformations of Graphs

Topic check. Correct.**Method**The maximum point of $y = f(x)$ is $(-3, 4)$.

For

$$y = f(x) - 6$$

the graph moves 6 units down:

$$(-3, 4) \rightarrow (-3, -2)$$

For

$$y = f(2x)$$

the x -coordinate is divided by 2:

$$(-3, 4) \rightarrow \left(-\frac{3}{2}, 4\right)$$

FINAL ANSWER $(-3, -2)$ and $\left(-\frac{3}{2}, 4\right)$

CLASSIFIED PROBLEM**Paper: P2H****Session: Nov 2021****Q#: 21****Marks: 6****TOPIC: Transformations of Graphs**

Source: Nov 2021 P2H

- 21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$.
The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(.....,)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

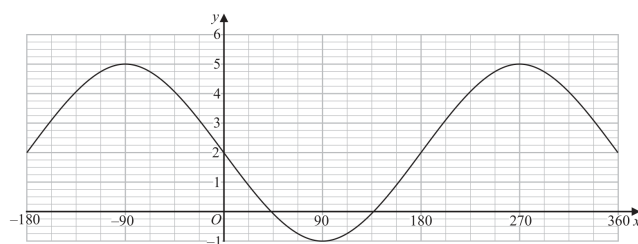
(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

(1)

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



(d) Find the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(3)

(Total for Question 21 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2021 P2H | Question 21

Transformations of Graphs

Topic check. Correct.**Method**

$$f(x) = 9 - 3(x + 2)^2$$

This is in completed square form, so the maximum point is

$$(-2, 9)$$

For the translation

$$\begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

the graph moves 4 units right. Replace x by $x - 4$:

$$y = 9 - 3((x - 4) + 2)^2$$

$$y = 9 - 3(x - 2)^2$$

The curve T has equation

$$y = 3(x + 2)^2 - 9$$

This is

$$y = -\left(9 - 3(x + 2)^2\right)$$

So C is reflected in the x -axis.For the cosine graph, the maximum is 5 and the minimum is -1 .

$$a = \frac{5 - (-1)}{2} = 3$$

$$c = \frac{5 + (-1)}{2} = 2$$

For

$$y = a \cos(x - b)^\circ + c$$

a maximum occurs when $x - b = 0$. The graph has a maximum at $x = 270^\circ$, so

$$b = 270$$

FINAL ANSWER $A = (-2, 9)$, $y = 9 - 3(x - 2)^2$, reflection in the x -axis, $a = 3$, $b = 270$, $c = 2$

CLASSIFIED PROBLEM

Paper: P1H

Session: Nov 2023

Q#: 24

Marks: 4

TOPIC: Transformations of Graphs

Source: Nov 2023 P1H

24 The curve with equation $f(x) = 5x^2 + 9x + 2$ is transformed to the curve with equation

$$g(x) = 5(x+4)^2 + 9(x+4) + 8 \text{ by the translation } \begin{pmatrix} a \\ b \end{pmatrix}$$

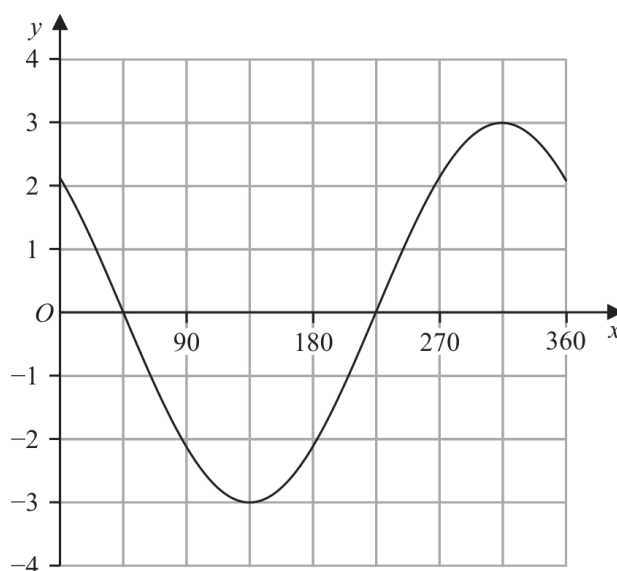
(a) Write down the value of a and the value of b

$a = \dots\dots\dots$

$b = \dots\dots\dots$

(2)

The graph of $y = p \cos(x+q)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid below.



Given that $p > 0$ and $0 < q < 360$

(b) find the value of p and the value of q

$p = \dots\dots\dots$

$q = \dots\dots\dots$

(2)

(Total for Question 24 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P1H | Question 24

Transformations of Graphs

Topic check. Moved from Functions to Transformations of Graphs.**Method**

For part (a),

$$f(x) = 5x^2 + 9x + 2$$

$$g(x) = 5(x + 4)^2 + 9(x + 4) + 8$$

Since

$$f(x + 4) = 5(x + 4)^2 + 9(x + 4) + 2$$

$$g(x) = f(x + 4) + 6$$

So the graph has moved 4 units left and 6 units up.

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -4 \\ 6 \end{pmatrix}$$

For part (b), the graph has maximum value 3 and minimum value -3 , so

$$p = 3$$

The minimum occurs when

$$x + q = 180$$

From the graph, the minimum is at $x = 135$, so

$$135 + q = 180$$

$$q = 45$$

FINAL ANSWER

$$a = -4, b = 6, p = 3, q = 45$$

CLASSIFIED PROBLEM**Paper: P1H****Session: Nov 2025****Q#: 24****Marks: 3****TOPIC: Transformations of Graphs**

Source: Nov 2025 P1H

24 A curve has equation $y = f(x)$

There is only one turning point on the curve.

The coordinates of this turning point are (4, 3)

Write down the coordinates of the turning point on the curve with equation

(i) $y = f(x + 5)$

(..... ,)
(1)

(ii) $y = f(x) + 7$

(..... ,)
(1)

(iii) $y = f(2x)$

(..... ,)
(1)**(Total for Question 24 is 3 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P1H | Question 24

Transformations of Graphs

Topic check. Correct.**Method**The turning point of $y = f(x)$ is $(4, 3)$.

For

$$y = f(x + 5)$$

the graph moves 5 units left:

$$(4, 3) \rightarrow (-1, 3)$$

For

$$y = f(x) + 7$$

the graph moves 7 units up:

$$(4, 3) \rightarrow (4, 10)$$

For

$$y = f(2x)$$

the x -coordinate is divided by 2:

$$(4, 3) \rightarrow (2, 3)$$

FINAL ANSWER $(-1, 3)$, $(4, 10)$, and $(2, 3)$

Differentiation

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jan 2020

Q#: 23

Marks: 6

TOPIC: **Differentiation**

Source: Jan 2020 P1HR

- 23**
- A particle moves along a straight line.

The fixed point O lies on this line.The displacement of the particle from O at time t seconds, $t \geq 0$, is s metres where

$$s = t^3 + 4t^2 - 5t + 7$$

At time T seconds the velocity of P is V m/s where $V \geq -5$ Find an expression for T in terms of V .Give your expression in the form $\frac{-4 + \sqrt{k + mV}}{3}$ where k and m are integers to be found. $T = \dots\dots\dots$ **(Total for Question 23 is 6 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P1HR | Question 23

Differentiation

Topic check. Correct.**Method**

$$s = t^3 + 4t^2 - 5t + 7$$

Velocity is the derivative of displacement:

$$v = \frac{ds}{dt} = 3t^2 + 8t - 5$$

At time T , the velocity is V , so

$$V = 3T^2 + 8T - 5$$

$$3T^2 + 8T - (V + 5) = 0$$

Use the quadratic formula:

$$T = \frac{-8 \pm \sqrt{8^2 - 4(3)(-(V+5))}}{2(3)}$$

$$T = \frac{-8 \pm \sqrt{64 + 12V + 60}}{6}$$

$$T = \frac{-8 \pm \sqrt{124 + 12V}}{6}$$

$$T = \frac{-4 \pm \sqrt{31 + 3V}}{3}$$

Since $T \geq 0$, take the positive root:

$$T = \frac{-4 + \sqrt{31 + 3V}}{3}$$

FINAL ANSWER

$$T = \frac{-4 + \sqrt{31 + 3V}}{3}$$

CLASSIFIED PROBLEM**Paper: P1HR****Session: Jan 2021****Q#: 20****Marks: 5****TOPIC: Differentiation**

Source: Jan 2021 P1HR

20 A particle P is moving along a straight line.

The fixed point O lies on the line.

At time t seconds ($t \geq 0$), the displacement of P from O is s metres where

$$s = t^3 - 9t^2 + 33t - 6$$

Find the minimum speed of P .

..... m/s

(Total for Question 20 is 5 marks)

WORKED SOLUTION*Dr. Eslam Ahmed***Jan 2021 P1HR | Question 20**

Differentiation

Topic check. Moved from Linear Graphs to Differentiation.**Method**

$$s = t^3 - 9t^2 + 33t - 6$$

Velocity is the rate of change of displacement:

$$v = \frac{ds}{dt}$$

$$v = 3t^2 - 18t + 33$$

Complete the square:

$$v = 3(t^2 - 6t + 11)$$

$$v = 3[(t - 3)^2 + 2]$$

$$v = 3(t - 3)^2 + 6$$

Since $(t - 3)^2 \geq 0$, the minimum value of v is

$$6$$

The velocity is always positive, so the minimum speed is also 6.

FINAL ANSWER

6 m/s

CLASSIFIED PROBLEM

Paper: P1H

Session: Jan 2022

Q#: 24

Marks: 6

TOPIC: **Differentiation**

Source: Jan 2022 P1H

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**

The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3

Show clear algebraic working.

$y = \dots\dots\dots$

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1H | Question 24
Differentiation

Topic check. Correct.

Method

$$y = ax^3 + bx^2 - 12x + 6$$

Point $A = (2, -6)$ lies on the curve:

$$-6 = 8a + 4b - 24 + 6$$

$$8a + 4b = 12$$

$$2a + b = 3$$

Differentiate:

$$\frac{dy}{dx} = 3ax^2 + 2bx - 12$$

The gradient at $x = 2$ is 16:

$$16 = 12a + 4b - 12$$

$$12a + 4b = 28$$

$$3a + b = 7$$

Now solve

$$2a + b = 3$$

$$3a + b = 7$$

Subtracting gives

$$a = 4$$

Then

$$2(4) + b = 3$$

$$b = -5$$

For $x = 3$,

$$y = 4(3)^3 - 5(3)^2 - 12(3) + 6$$

$$y = 108 - 45 - 36 + 6$$

$$y = 33$$

FINAL ANSWER

$$y = 33$$

Dr. Eslam Ahmed

CLASSIFIED PROBLEM**Paper: P1HR****Session: Jan 2022****Q#: 23****Marks: 6****TOPIC: Differentiation**

Source: Jan 2022 P1HR

- 23 Two particles, P and Q , move along a straight line.
The fixed point O lies on this line.

The displacement of P from O at time t seconds is s metres, where

$$s = t^3 - 4t^2 + 5t \quad \text{for } t > 1$$

The displacement of Q from O at time t seconds is x metres, where

$$x = t^2 - 4t + 4 \quad \text{for } t > 1$$

Find the range of values of t where $t > 1$ for which both particles are moving in the same direction along the straight line.

(Total for Question 23 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1HR | Question 23
Differentiation

Topic check. Corrected from Functions to Differentiation.

Method

Particle P :

$$s = t^3 - 4t^2 + 5t$$

$$\frac{ds}{dt} = 3t^2 - 8t + 5$$

$$\frac{ds}{dt} = (3t - 5)(t - 1)$$

Particle Q :

$$x = t^2 - 4t + 4$$

$$\frac{dx}{dt} = 2t - 4 = 2(t - 2)$$

For $t > 1$:

- P moves in the negative direction when $1 < t < \frac{5}{3}$.
- P moves in the positive direction when $t > \frac{5}{3}$.
- Q moves in the negative direction when $1 < t < 2$.
- Q moves in the positive direction when $t > 2$.

So both particles move in the same direction when both velocities have the same sign.

FINAL ANSWER

$$1 < t < \frac{5}{3} \text{ or } t > 2$$

CLASSIFIED PROBLEM

Paper: P2H

Session: Jan 2022

Q#: 24

Marks: 5

TOPIC: **Differentiation**

Source: Jan 2022 P2H

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geq 0$, is given by

$$x = 4t^3 - 27t + 8$$

The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is $a \text{ m/s}^2$

Find the value of a

$a = \dots\dots\dots$

(Total for Question 24 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2H | Question 24
Differentiation

Topic check. Corrected from Ratio Toolkit to Differentiation.

Method

$$x = 4t^3 - 27t + 8$$

Velocity is

$$\frac{dx}{dt} = 12t^2 - 27$$

The direction of motion reverses when velocity is zero:

$$12t^2 - 27 = 0$$

$$12t^2 = 27$$

$$t^2 = \frac{9}{4}$$

Since $t \geq 0$, $t = \frac{3}{2}$.
Acceleration is

$$\frac{d^2x}{dt^2} = 24t$$

At $t = \frac{3}{2}$,

$$a = 24 \left(\frac{3}{2} \right) = 36$$

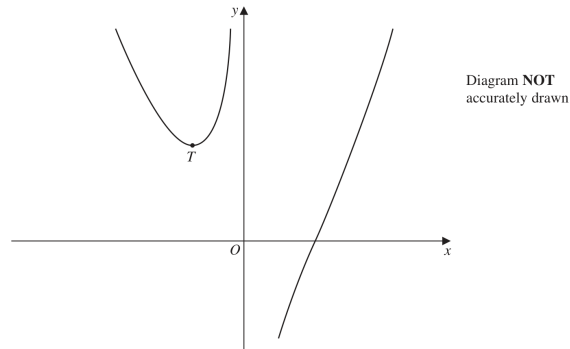
FINAL ANSWER

$$a = 36$$

CLASSIFIED PROBLEM**Paper: P1HR****Session: Jun 2022****Q#: 22****Marks: 7****TOPIC: Differentiation**

Source: Jun 2022 P1HR

- 22 The diagram shows a sketch of part of the curve with equation $y = x^2 - \frac{p}{x}$ where p is a positive constant.



For all values of p , the curve has exactly one turning point and this turning point is a minimum shown as the point T in the sketch.

For the curve where the x coordinate of T is -3

- (a) find the value of p

$$p = \dots\dots\dots (4)$$

The line with equation $y = k$ is a tangent to the curve with equation $y = x^2 - \frac{16}{x}$

- (b) Find the value of k

$$k = \dots\dots\dots (3)$$

(Total for Question 22 is 7 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1HR | Question 22

Differentiation

Topic check. Moved from Estimating Gradients to Differentiation.**Method**

For part (a),

$$y = x^2 - \frac{p}{x}$$

$$\frac{dy}{dx} = 2x + \frac{p}{x^2}$$

At the minimum point, $x = -3$ and the gradient is 0.

$$2(-3) + \frac{p}{(-3)^2} = 0$$

$$-6 + \frac{p}{9} = 0$$

$$p = 54$$

For part (b),

$$y = x^2 - \frac{16}{x}$$

$$\frac{dy}{dx} = 2x + \frac{16}{x^2}$$

At a horizontal tangent,

$$2x + \frac{16}{x^2} = 0$$

$$2x^3 + 16 = 0$$

$$x^3 = -8$$

$$x = -2$$

Now substitute $x = -2$ into the curve:

$$k = (-2)^2 - \frac{16}{-2}$$

$$k = 4 + 8 = 12$$

FINAL ANSWER

$$p = 54, k = 12$$

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2025

Q#: 22

Marks: 5

TOPIC: Differentiation

Source: Jun 2025 P2HR

22 The diagram shows a solid cuboid.

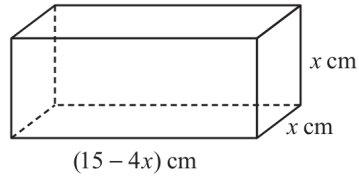


Diagram **NOT**
accurately drawn

The volume of the cuboid is $V \text{ cm}^3$

Find the maximum value of V

.....
(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2HR | Question 22
Differentiation

Topic check. Correct.

Method

The cuboid has dimensions

$$x, \quad x, \quad 15 - 4x$$

So

$$V = x^2(15 - 4x)$$

$$V = 15x^2 - 4x^3$$

Differentiate:

$$\frac{dV}{dx} = 30x - 12x^2$$

At a maximum,

$$30x - 12x^2 = 0$$

$$6x(5 - 2x) = 0$$

The useful value is

$$x = 2.5$$

Now substitute into the volume:

$$V = (2.5)^2(15 - 4(2.5))$$

$$V = 6.25 \times 5$$

$$V = 31.25$$

FINAL ANSWER

$$31.25 \text{ cm}^3$$

CLASSIFIED PROBLEM**Paper: P2H****Session: May 2021****Q#: 21****Marks: 5**TOPIC: **Differentiation**

Source: May 2021 P2H

- 21** The point A is the only stationary point on the curve with equation $y = kx^2 + \frac{16}{x}$ where k is a constant.

Given that the coordinates of A are $\left(\frac{2}{3}, a\right)$

find the value of a .

Show your working clearly.

$a = \dots\dots\dots$

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P2H | Question 21

Differentiation

Topic check. Correct.**Method**

$$y = kx^2 + \frac{16}{x}$$

Differentiate:

$$\frac{dy}{dx} = 2kx - \frac{16}{x^2}$$

At the stationary point, $x = \frac{2}{3}$, so the gradient is 0:

$$2k \left(\frac{2}{3} \right) - \frac{16}{\left(\frac{2}{3} \right)^2} = 0$$

$$\frac{4k}{3} - \frac{16}{\frac{4}{9}} = 0$$

$$\frac{4k}{3} - 36 = 0$$

$$\frac{4k}{3} = 36$$

$$k = 27$$

Now find a , the y -coordinate when $x = \frac{2}{3}$:

$$a = 27 \left(\frac{2}{3} \right)^2 + \frac{16}{\frac{2}{3}}$$

$$a = 27 \cdot \frac{4}{9} + 24$$

$$a = 12 + 24 = 36$$

FINAL ANSWER

$$a = 36$$

CLASSIFIED PROBLEM

Paper: P1HR

Session: May 2023

Q#: 21

Marks: 6

TOPIC: **Differentiation**

Source: May 2023 P1HR

21 The curve **T** has equation $y = x^3 - 2x^2 - 9x + 15$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots$$

(2)

(b) Find the range of values of x for which **T** has a positive gradient.
Give your values correct to 3 significant figures.
Show your working clearly.

(4)

(Total for Question 21 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1HR | Question 21

Differentiation

Topic check. Corrected from Functions to Differentiation.**Method**

(a)

$$y = x^3 - 2x^2 - 9x + 15$$

$$\frac{dy}{dx} = 3x^2 - 4x - 9$$

(b) The curve has a positive gradient when

$$3x^2 - 4x - 9 > 0$$

Solve

$$3x^2 - 4x - 9 = 0$$

$$x = \frac{4 \pm \sqrt{16 + 108}}{6}$$

$$x = \frac{2 \pm \sqrt{31}}{3}$$

To 3 significant figures, the roots are

$$x = -1.19, \quad x = 2.52$$

Since the quadratic opens upwards, it is positive outside the roots.

FINAL ANSWER(a) $\frac{dy}{dx} = 3x^2 - 4x - 9$, (b) $x < -1.19$ or $x > 2.52$

CLASSIFIED PROBLEM

Paper: P1H

Session: MayNov 2020

Q#: 23

Marks: 4

TOPIC: **Differentiation**

Source: May-Nov 2020 P1H

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x , in terms of p and m , for which the gradient of **C** is negative.

(Total for Question 23 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1H | Question 23

Differentiation

Topic check. Corrected from Functions to Differentiation.**Method**

$$y = px^3 - mx$$

Differentiate:

$$\frac{dy}{dx} = 3px^2 - m$$

The gradient is negative when

$$3px^2 - m < 0$$

$$3px^2 < m$$

$$x^2 < \frac{m}{3p}$$

So

$$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$$

FINAL ANSWER

$$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$$

CLASSIFIED PROBLEM**Paper: P1H****Session: Nov 2024****Q#: 20****Marks: 4****TOPIC: Differentiation**

Source: Nov 2024 P1H

20 The curve with equation $y = 2x^4 - 64x$ has a minimum point.

Find an equation of the tangent to the curve at the minimum point.
Show clear algebraic working.

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2024 P1H | Question 20
Differentiation

Topic check. Moved from Estimating Gradients to Differentiation.

Method

$$y = 2x^4 - 64x$$

Differentiate:

$$\frac{dy}{dx} = 8x^3 - 64$$

At the minimum point, the gradient is 0.

$$8x^3 - 64 = 0$$

$$8x^3 = 64$$

$$x^3 = 8$$

$$x = 2$$

Find the y -coordinate:

$$y = 2(2)^4 - 64(2)$$

$$y = 32 - 128 = -96$$

The tangent at a minimum point is horizontal, so its equation is

$$y = -96$$

FINAL ANSWER

$$y = -96$$

CLASSIFIED PROBLEM**Paper: P2H****Session: Nov 2025****Q#: 22****Marks: 6**TOPIC: **Differentiation**

Source: Nov 2025 P2H

22 Curve **C** has equation $y = x^3 - 16x + 7$

At two points on **C**, the gradient is 11

The tangents to **C** at these two points have equations of the form $y = ax + b$

Work out the two possible values of b

Show clear algebraic working.

(Total for Question 22 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P2H | Question 22
Differentiation

Topic check. Moved to Differentiation.

Method

$$y = x^3 - 16x + 7$$

Differentiate:

$$\frac{dy}{dx} = 3x^2 - 16$$

The gradient of the tangent is 11, so

$$3x^2 - 16 = 11$$

$$3x^2 = 27$$

$$x^2 = 9$$

$$x = 3 \quad \text{or} \quad x = -3$$

When $x = 3$:

$$y = 3^3 - 16(3) + 7 = 27 - 48 + 7 = -14$$

The tangent has equation $y = 11x + b$, so

$$-14 = 11(3) + b$$

$$b = -47$$

When $x = -3$:

$$y = (-3)^3 - 16(-3) + 7 = -27 + 48 + 7 = 28$$

$$28 = 11(-3) + b$$

$$b = 61$$

FINAL ANSWER

$$b = -47 \text{ or } b = 61$$

Angles in Polygons & Parallel Lines

Dr. Eslam Ahmed

CLASSIFIED PROBLEM**Paper: P1H****Session: Jun 2022****Q#: 23****Marks: 6****TOPIC: Angles in Polygons & Parallel Lines**

Source: Jun 2022 P1H

23 A polygon has n sides, where $n > 5$

When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence.

Here are the first five terms of this sequence.

177 175 173 171 169

Find the value of n

Show clear algebraic working.

Question 23 continued

$n =$

(Total for Question 23 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1H | Question 23

Angles in Polygons & Parallel Lines

Topic check. Correct.**Method**

The interior angles form an arithmetic sequence:

$$177, 175, 173, 171, 169, \dots$$

So

$$a = 177, \quad d = -2$$

For an n -sided polygon, the sum of the interior angles is

$$180(n - 2)$$

The sum of the arithmetic sequence is

$$\begin{aligned} & \frac{n}{2} (2(177) + (n - 1)(-2)) \\ &= \frac{n}{2} (354 - 2n + 2) \\ &= \frac{n}{2} (356 - 2n) \\ &= n(178 - n) \end{aligned}$$

Set this equal to the polygon angle sum:

$$n(178 - n) = 180(n - 2)$$

$$178n - n^2 = 180n - 360$$

$$n^2 + 2n - 360 = 0$$

Factorise:

$$(n + 20)(n - 18) = 0$$

Since $n > 5$,

$$n = 18$$

FINAL ANSWER

$$n = 18$$

Bearings, Scale Drawing & Constructions

Dr. Eslam Ahmed

CLASSIFIED PROBLEM**Paper: P2H****Session: Nov 2024****Q#: 24****Marks: 6****TOPIC: Bearings, Scale Drawing & Constructions**

Source: Nov 2024 P2H

24 A , B and C are three points on horizontal ground.

$AB = 8.4$ metres $BC = 9.2$ metres

B is on a bearing of 067° from A

C is on a bearing of 129° from B

Calculate the bearing of A from C

Give your answer correct to the nearest degree.

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2024 P2H | Question 24

Bearings, Scale Drawing & Constructions

Topic check. Correct.**Method**

Resolve each journey into east and north components.

From A to B :

$$\text{east} = 8.4 \sin 67^\circ = 7.7322 \dots$$

$$\text{north} = 8.4 \cos 67^\circ = 3.2821 \dots$$

From B to C :

$$\text{east} = 9.2 \sin 129^\circ = 7.1497 \dots$$

$$\text{north} = 9.2 \cos 129^\circ = -5.7897 \dots$$

So from A to C :

$$\text{east} = 14.8820 \dots$$

$$\text{north} = -2.5076 \dots$$

Therefore from C to A :

$$\text{east} = -14.8820 \dots, \quad \text{north} = 2.5076 \dots$$

This direction is west of north.

$$\theta = \tan^{-1} \left(\frac{14.8820 \dots}{2.5076 \dots} \right) = 80.4355 \dots^\circ$$

Bearing of A from C :

$$360^\circ - 80.4355 \dots^\circ = 279.5644 \dots^\circ$$

FINAL ANSWER**280°**

Circle Theorems

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2023

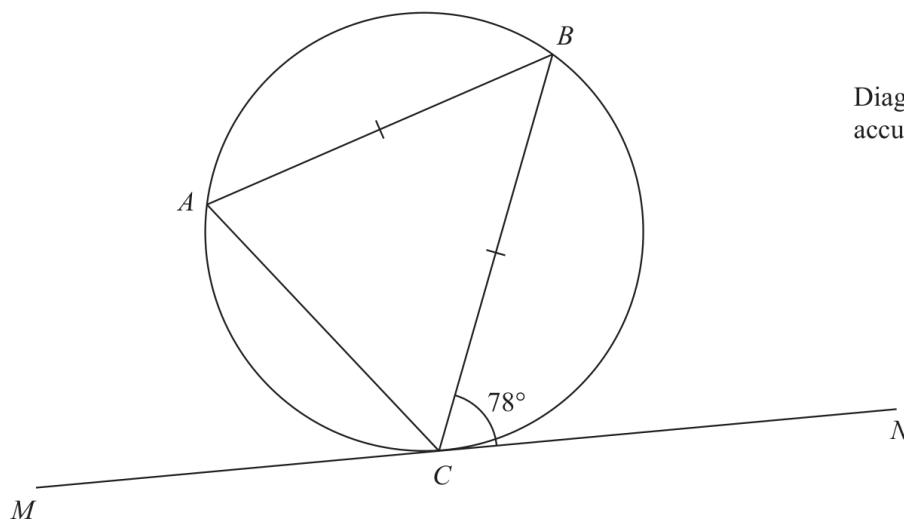
Q#: 20

Marks: 2

TOPIC: Circle Theorems

Source: Jun 2023 P2HR

20 A , B and C are points on a circle.



MN is the tangent to the circle at C

$AB = CB$

Angle $BCN = 78^\circ$

Find the size of angle ABC

(Total for Question 20 is 2 marks)

WORKED SOLUTION*Dr. Eslam Ahmed*

Jun 2023 P2HR | Question 20
 Circle Theorems

Topic check. Correct.

Method

The angle between the tangent CN and the chord CB is 78° .
 By the alternate segment theorem,

$$\angle BAC = 78^\circ$$

Also,

$$AB = CB$$

so triangle ABC is isosceles and the base angles are equal:

$$\angle BAC = \angle ACB = 78^\circ$$

Using the angle sum of a triangle:

$$\angle ABC = 180^\circ - 78^\circ - 78^\circ$$

$$\angle ABC = 24^\circ$$

FINAL ANSWER

24°

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jun 2025

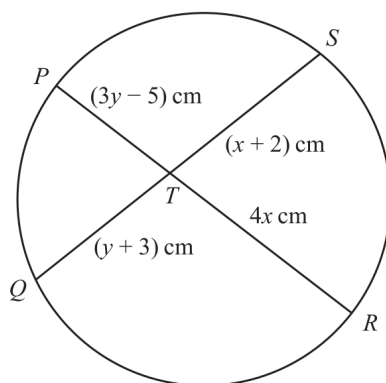
Q#: 23

Marks: 5

TOPIC: **Circle Theorems**

Source: Jun 2025 P1HR

23

Diagram **NOT**
accurately drawn $\overline{PTR}$ and $\overline{QTS}$ are chords of a circle.

$$PT = (3y - 5) \text{ cm} \quad QT = (y + 3) \text{ cm} \quad RT = 4x \text{ cm} \quad ST = (x + 2) \text{ cm}$$

Find an expression for y in terms of x

$$y = \dots\dots\dots$$

Total for Question 23 is 5 marks)

WORKED SOLUTION*Dr. Eslam Ahmed*

Jun 2025 P1HR | Question 23
 Circle Theorems

Topic check. Correct.

Method

For two intersecting chords in a circle,

$$PT \times TR = QT \times TS$$

Substitute the given lengths:

$$(3y - 5)(4x) = (y + 3)(x + 2)$$

Expand:

$$12xy - 20x = xy + 2y + 3x + 6$$

Collect the y terms on one side:

$$12xy - xy - 2y = 20x + 3x + 6$$

$$11xy - 2y = 23x + 6$$

Factorise:

$$y(11x - 2) = 23x + 6$$

$$y = \frac{23x + 6}{11x - 2}$$

FINAL ANSWER

$$y = \frac{23x + 6}{11x - 2}$$

Volume & Surface Area

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: May2025

Q#: 23

Marks: 3

TOPIC: **Volume & Surface Area**

Source: May 2025 4WM2H

- 23 The diagram shows a cuboid.

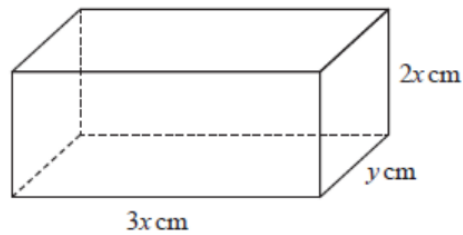


Diagram NOT
accurately drawn

The cuboid measures $3x$ cm by $2x$ cm by y cm

The volume of the cuboid is 1014 cm^3

The total surface area of the cuboid is $A \text{ cm}^2$

Show that $A = 12x^2 + \frac{1690}{x}$

You must show all the stages of your working.

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2H | Question 23

Volume & Surface Area

Solution

$$3x \times 2x \times y = 1014$$

$$6x^2y = 1014$$

$$y = \frac{169}{x^2}$$

Surface area:

$$A = 2(3x \cdot 2x + 3x \cdot y + 2x \cdot y)$$

$$= 12x^2 + 10xy$$

$$= 12x^2 + 10x \left(\frac{169}{x^2} \right)$$

$$= 12x^2 + \frac{1690}{x}$$

FINAL ANSWER

$$A = 12x^2 + \frac{1690}{x}.$$

CLASSIFIED PROBLEM

Paper: 4WM2HR

Session: May2025

Q#: 21

Marks: 5

TOPIC: **Volume & Surface Area**

Source: May 2025 4WM2HR

- 21 The diagram shows a solid cuboid.

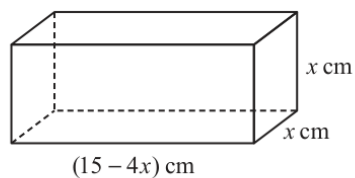


Diagram **NOT**
accurately drawn

The volume of the cuboid is $V \text{ cm}^3$

Find the maximum value of V

.....
(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2HR | Question 21

Volume & Surface Area

Required:

Cuboid has a square cross-section of side x cm and length $(15-4x)$ cm. Find the maximum value of the volume V .

Step:

Write V as a function of x :

$$V = x \cdot x \cdot (15-4x) = x^2(15-4x) = 15x^2 - 4x^3$$

Step:

Differentiate:

$$\frac{dV}{dx} = 30x - 12x^2$$

Step:

Set $\frac{dV}{dx} = 0$:

$$30x - 12x^2 = 0 \implies 6x(5 - 2x) = 0 \implies x = 0 \text{ or } x = \frac{5}{2} = 2.5$$

($x = 0$ gives zero volume, so the meaningful solution is $x = 2.5$.)

Step:

Second derivative test to confirm maximum:

$$\frac{d^2V}{dx^2} = 30 - 24x \Bigg|_{x=2.5} = 30 - 60 = -30 < 0 \quad \checkmark \text{ (maximum)}$$

Step:

Calculate $V_{\max}$:

$$V_{\max} = (2.5)^2 \times (15-4 \times 2.5) = 6.25 \times 5 = 31.25 \text{ cm}^3$$

FINAL ANSWER

$$\text{Maximum volume } V = 31.25 \text{ cm}^3$$

Note:

Calculus optimisation: always verify the nature of the stationary point using $\frac{d^2V}{dx^2}$. Also check $x > 0$ and $15 - 4x > 0$ (i.e. $x < 3.75$) to ensure a valid cuboid.

CLASSIFIED PROBLEM

TOPIC: **Volume & Surface Area**

Source: May 2025 4WM2HR

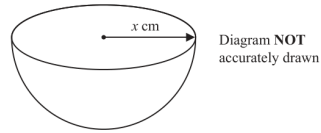
Paper: 4WM2HR

Session: May2025

Q#: 25

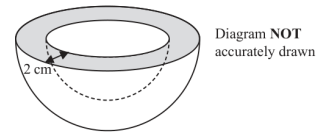
Marks: 5

25 The diagram shows a solid hemisphere, H



The radius of H is x cm
The volume of H is $6174\pi \text{ cm}^3$

A bowl is made by removing a solid hemisphere from H such that the uniform thickness of the bowl is 2 cm



Work out the **total** surface area of the bowl.
Give your answer in terms of π

..... cm^2

(Total for Question 25 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2HR | Question 25

Volume & Surface Area

Required:

Hemisphere H has radius x cm and volume 6174π cm³. A bowl is made by removing an inner hemisphere, leaving 2 cm thickness. Find total surface area in terms of π .

Step:

Find outer radius x . Volume of hemisphere = $\frac{2}{3}\pi r^3$:

$$\frac{2}{3}\pi x^3 = 6174\pi \implies x^3 = \frac{6174 \times 3}{2} = 9261 \implies x = \sqrt[3]{9261} = 21 \text{ cm}$$

Step:

Find inner radius:

$$r_{\text{inner}} = 21 - 2 = 19 \text{ cm}$$

Step:

Three surfaces to add:

1. Outer curved surface (half-sphere, radius 21):

$$A_1 = 2\pi(21)^2 = 2\pi \times 441 = 882\pi$$

2. Inner curved surface (half-sphere, radius 19):

$$A_2 = 2\pi(19)^2 = 2\pi \times 361 = 722\pi$$

3. Flat annular ring at the top rim:

$$A_3 = \pi(21)^2 - \pi(19)^2 = \pi(441 - 361) = 80\pi$$

Step:

Total surface area:

$$A_{\text{total}} = 882\pi + 722\pi + 80\pi = 1684\pi$$

FINAL ANSWER

$$\text{Total surface area} = 1684\pi \text{ cm}^2$$

Note:

Three separate surfaces! Missing the flat annular ring is the most common error on this type of question.

CLASSIFIED PROBLEM

Paper: 4WM2H

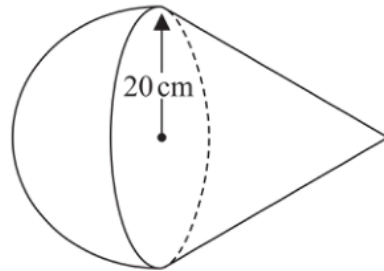
Session: Specimen

Q#: 22

Marks: 5

TOPIC: **Volume & Surface Area**

Source: Specimen 4WM2H

22 A solid is made from a cone and a hemisphere.Diagram **NOT**
accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
 The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm

The curved surface area of the cone is $580\pi\text{ cm}^2$

The volume of the solid is $k\pi\text{ cm}^3$

Work out the exact value of k

$k = \dots\dots\dots$

(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Specimen 4WM2H | Question 22

Volume & Surface Area

Topic check. Volume & Surface Area. The tag is correct.**Solution**

The curved surface area of a cone is

$$\pi r l$$

Here $r = 20$ and $\pi r l = 580\pi$, so

$$20l = 580$$

$$l = 29$$

Use Pythagoras to find the height of the cone:

$$h^2 = 29^2 - 20^2$$

$$h^2 = 841 - 400 = 441$$

$$h = 21$$

Volume of the cone:

$$\frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(20)^2(21) = 2800\pi$$

Volume of the hemisphere:

$$\frac{2}{3}\pi r^3 = \frac{2}{3}\pi(20)^3 = \frac{16000}{3}\pi$$

Total volume:

$$2800\pi + \frac{16000}{3}\pi = \frac{24400}{3}\pi$$

Since the volume is $k\pi$,

$$k = \frac{24400}{3}$$

FINAL ANSWER

$$k = \frac{24400}{3}.$$

Area & Volume of Similar Shapes

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: May2025

Q#: 25

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Source: May 2025 4WM2H

25 **A** and **B** are two similar solids.

The height of solid **A** is 31 cm

The height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**

..... cm³

(Total for Question 25 is 4 marks)

WORKED SOLUTION*Dr. Eslam Ahmed***May 2025 4WM2H | Question 25**

Area & Volume of Similar Shapes

SolutionLinear scale factor from solid A to solid B :

$$\frac{18.6}{31} = 0.6$$

So the volume scale factor is

$$0.6^3 = 0.216$$

Let the volume of A be V . Then

$$V - 0.216V = 735$$

$$0.784V = 735$$

$$V = 937.5$$

FINAL ANSWER937.5 cm³.

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Nov2025

Q#: 23

Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

Source: Nov 2025 4WM2H

23 Shape **P** is similar to shape **Q**

The table shows some information about shape **P** and shape **Q**

	Surface area (cm ²)	Volume (cm ³)
Shape P	200	672
Shape Q	450	

Work out the volume of shape **Q**

..... cm³

(Total for Question 23 is 3 marks)

WORKED SOLUTION*Dr. Eslam Ahmed***Nov 2025 4WM2H | Question 23**

Area & Volume of Similar Shapes

Step 1: Find the Area Scale Factor (ASF).

$$\text{ASF} = \frac{\text{Area of Q}}{\text{Area of P}} = \frac{450}{200} = \frac{9}{4} = 2.25$$

Step 2: Find the Length Scale Factor (LSF).

$$\text{LSF} = \sqrt{\text{ASF}} = \sqrt{\frac{9}{4}} = \frac{3}{2} = 1.5$$

Step 3: Find the Volume Scale Factor (VSF).

$$\text{VSF} = (\text{LSF})^3 = \left(\frac{3}{2}\right)^3 = \frac{27}{8} = 3.375$$

Step 4: Calculate the volume of Shape Q.

$$\begin{aligned} \text{Volume of Q} &= \text{Volume of P} \times \text{VSF} = 672 \times \frac{27}{8} \\ &= 84 \times 27 = 2268 \end{aligned}$$

FINAL ANSWER2268 cm³

Volume & Surface Area

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

TOPIC: **Volume & Surface Area**

Source: Jan 2020 P2H

Paper: P2H

Session: Jan 2020

Q#: 26

Marks: 6

26 Here is a sector, AOB , of a circle with centre O and angle $AOB = x^\circ$

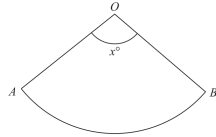


Diagram **NOT**
accurately drawn

The sector can form the curved surface of a cone by joining OA to OB .

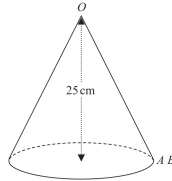


Diagram **NOT**
accurately drawn

The height of the cone is 25 cm.
The volume of the cone is 1600 cm^3

Work out the value of x .
Give your answer correct to the nearest whole number.

$x =$ _____

(Total for Question 26 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2H | Question 26

Volume & Surface Area

Topic check. Moved to Volume & Surface Area.**Method**Let the radius of the base of the cone be r , and let the slant height be l .

The volume of the cone is

$$\frac{1}{3}\pi r^2 h = 1600$$

The height is 25 cm, so

$$\frac{1}{3}\pi r^2 (25) = 1600$$

$$r^2 = \frac{4800}{25\pi}$$

$$r = 7.8176\dots$$

The slant height is found using Pythagoras:

$$l^2 = r^2 + 25^2$$

$$l = 26.1938\dots$$

When the sector is joined to form the cone, the arc length of the sector becomes the circumference of the cone base.

$$\frac{x}{360} \times 2\pi l = 2\pi r$$

Cancel 2π :

$$\frac{x}{360} l = r$$

$$x = \frac{360r}{l}$$

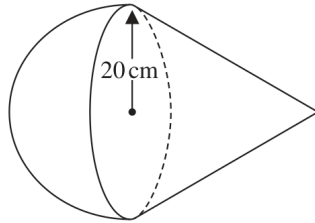
$$x = \frac{360(7.8176\dots)}{26.1938\dots} = 107.4433\dots$$

FINAL ANSWER

107°

CLASSIFIED PROBLEM**Paper: P1H****Session: Jun 2022****Q#: 22****Marks: 5**TOPIC: **Volume & Surface Area**

Source: Jun 2022 P1H

22 A solid is made from a cone and a hemisphere.Diagram **NOT**
accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
 The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm.

The curved surface area of the cone is $580\pi \text{ cm}^2$

The volume of the solid is $k\pi \text{ cm}^3$

Work out the exact value of k

 $k = \dots\dots\dots$ **(Total for Question 22 is 5 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1H | Question 22

Volume & Surface Area

Topic check. Correct.**Method**

The cone has radius 20 cm.

The curved surface area of the cone is

$$\pi r l$$

So

$$\pi(20)l = 580\pi$$

$$l = 29$$

Use Pythagoras to find the height of the cone:

$$h^2 + 20^2 = 29^2$$

$$h^2 = 841 - 400 = 441$$

$$h = 21$$

Volume of the cone:

$$\frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(20)^2(21) = 2800\pi$$

Volume of the hemisphere:

$$\frac{2}{3}\pi r^3 = \frac{2}{3}\pi(20)^3 = \frac{16000}{3}\pi$$

Total volume:

$$2800\pi + \frac{16000}{3}\pi = \frac{24400}{3}\pi$$

The volume is $k\pi$, so

$$k = \frac{24400}{3}$$

FINAL ANSWER

24400
3

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2023

Q#: 24

Marks: 5

TOPIC: **Volume & Surface Area**

Source: Jun 2023 P2HR

- 24** The surface area of sphere **A** is nine times the surface area of sphere **B**
The difference between the volume of sphere **A** and the volume of sphere **B** is $117\pi \text{ cm}^3$
Find the radius of the smaller sphere.
Show your working clearly.

..... cm

(Total for Question 24 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2HR | Question 24
Volume & Surface Area

Topic check. Correct.

Method

Let the radius of the smaller sphere be r .

Let the radius of the larger sphere be R .

Surface area of a sphere is

$$4\pi r^2$$

The surface area of sphere A is nine times the surface area of sphere B , so

$$4\pi R^2 = 9(4\pi r^2)$$

$$R^2 = 9r^2$$

$$R = 3r$$

The difference between the volumes is 117π :

$$\frac{4}{3}\pi(3r)^3 - \frac{4}{3}\pi r^3 = 117\pi$$

$$\frac{4}{3}\pi(27r^3 - r^3) = 117\pi$$

$$\frac{104}{3}\pi r^3 = 117\pi$$

Cancel π :

$$\frac{104}{3}r^3 = 117$$

$$r^3 = \frac{117 \times 3}{104}$$

$$r^3 = \frac{27}{8}$$

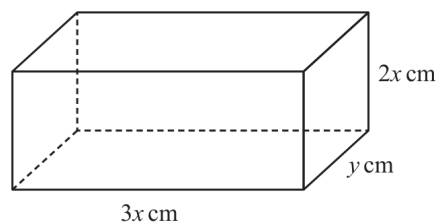
$$r = \frac{3}{2}$$

FINAL ANSWER

1.5 cm

CLASSIFIED PROBLEM**Paper: P2H****Session: Jun 2025****Q#: 21****Marks: 3**TOPIC: **Volume & Surface Area**

Source: Jun 2025 P2H

21 The diagram shows a cuboid.Diagram **NOT**
accurately drawnThe cuboid measures $3x$ cm by $2x$ cm by y cmThe volume of the cuboid is 1014 cm^3 The total surface area of the cuboid is $A\text{ cm}^2$ Show that $A = 12x^2 + \frac{1690}{x}$

You must show all the stages of your working.

(Total for Question 21 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2H | Question 21

Volume & Surface Area

Topic check. Correct.**Method**

The cuboid has dimensions

$$3x, \quad 2x, \quad y$$

Its volume is 1014 cm^3 , so

$$(3x)(2x)(y) = 1014$$

$$6x^2y = 1014$$

$$y = \frac{1014}{6x^2}$$

$$y = \frac{169}{x^2}$$

The total surface area is

$$A = 2(3x \cdot 2x + 3x \cdot y + 2x \cdot y)$$

$$A = 2(6x^2 + 5xy)$$

$$A = 12x^2 + 10xy$$

Substitute $y = \frac{169}{x^2}$:

$$A = 12x^2 + 10x \left(\frac{169}{x^2} \right)$$

$$A = 12x^2 + \frac{1690}{x}$$

FINAL ANSWER

$$A = 12x^2 + \frac{1690}{x}$$

CLASSIFIED PROBLEM

TOPIC: **Volume & Surface Area**

Source: Jun 2025 P2HR

Paper: P2HR

Session: Jun 2025

Q#: 26

Marks: 5

26 The diagram shows a solid hemisphere, H

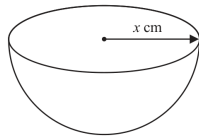


Diagram **NOT** accurately drawn

The radius of H is x cm
The volume of H is $6174\pi \text{ cm}^3$

A bowl is made by removing a solid hemisphere from H such that the uniform thickness of the bowl is 2 cm

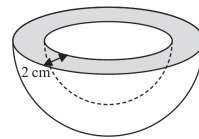


Diagram **NOT** accurately drawn

Work out the **total** surface area of the bowl.
Give your answer in terms of π

..... cm^2

(Total for Question 26 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2HR | Question 26
Volume & Surface Area

Topic check. Correct.

Method

The volume of a hemisphere is

$$\frac{2}{3}\pi r^3$$

For the original hemisphere H ,

$$\frac{2}{3}\pi x^3 = 6174\pi$$

Cancel π :

$$\frac{2}{3}x^3 = 6174$$

$$x^3 = 9261$$

$$x = 21$$

The outer radius of the bowl is 21 cm.

The thickness is 2 cm, so the inner radius is

$$21 - 2 = 19$$

Total surface area of the bowl is:

- outer curved hemisphere
- inner curved hemisphere
- flat annular rim

Outer curved surface area:

$$2\pi(21)^2 = 882\pi$$

Inner curved surface area:

$$2\pi(19)^2 = 722\pi$$

Flat rim area:

$$\pi(21^2 - 19^2) = 80\pi$$

Total:

$$882\pi + 722\pi + 80\pi = 1684\pi$$

FINAL ANSWER

$$1684\pi \text{ cm}^2$$

CLASSIFIED PROBLEM**Paper: P1H****Session: May 2024****Q#: 23****Marks: 5****TOPIC: Volume & Surface Area**

Source: May 2024 P1H

- 23 A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

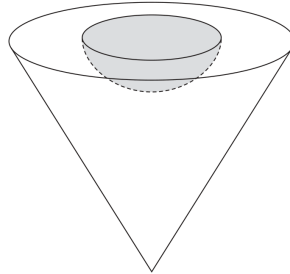


Diagram **NOT**
accurately drawn

The radius of the hemisphere is $2x$ cm
 The radius of the base of the cone is $5x$ cm
 The vertical height of the cone is $6x$ cm

The volume of the solid shape is 6948π cm³

Work out the **total** surface area of the solid hemisphere that has been removed from the cone.

Give your answer correct to the nearest integer.

..... cm²

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1H | Question 23

Volume & Surface Area

Topic check. Correct.**Method**The cone has radius $5x$ and height $6x$.

Volume of the cone:

$$\begin{aligned}\frac{1}{3}\pi(5x)^2(6x) \\ = 50\pi x^3\end{aligned}$$

The removed hemisphere has radius $2x$.

Volume of the hemisphere:

$$\begin{aligned}\frac{2}{3}\pi(2x)^3 \\ = \frac{16}{3}\pi x^3\end{aligned}$$

The remaining solid has volume 6948π , so

$$\begin{aligned}50\pi x^3 - \frac{16}{3}\pi x^3 &= 6948\pi \\ \frac{134}{3}\pi x^3 &= 6948\pi\end{aligned}$$

Cancel π :

$$\begin{aligned}\frac{134}{3}x^3 &= 6948 \\ x^3 &= \frac{10422}{67}\end{aligned}$$

$$x = 5.3781\dots$$

The solid hemisphere removed has radius $2x$.

Total surface area of a solid hemisphere is

$$3\pi r^2$$

So

$$3\pi(2x)^2 = 12\pi x^2$$

$$12\pi(5.3781\dots)^2 = 1090.3903\dots$$

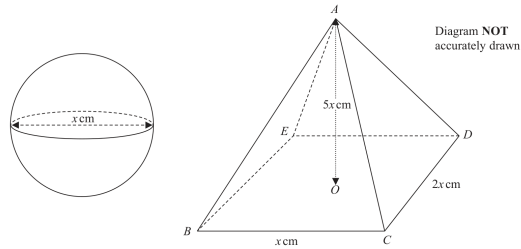
FINAL ANSWER

$$1090 \text{ cm}^2$$

CLASSIFIED PROBLEM**Paper: P2H****Session: Nov 2021****Q#: 22****Marks: 6****TOPIC: Volume & Surface Area**

Source: Nov 2021 P2H

- 22 The diagram shows a sphere of diameter x cm and a pyramid $ABCDE$ with a horizontal rectangular base $BCDE$.



The vertex A of the pyramid is vertically above the centre O of the base so that $AB = AC = AD = AE$.

$BC = x$ cm, $CD = 2x$ cm and $AO = 5x$ cm.

The volume of the sphere is 288π cm³.

Calculate the total surface area of the pyramid.
Give your answer correct to the nearest cm².

_____ cm²

(Total for Question 22 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2021 P2H | Question 22

Volume & Surface Area

Topic check. Corrected from Right-Angled Triangles to Volume & Surface Area.**Method**The sphere has diameter x , so its radius is

$$\frac{x}{2}$$

The volume of the sphere is $288\pi \text{ cm}^3$:

$$\frac{4}{3}\pi \left(\frac{x}{2}\right)^3 = 288\pi$$

$$\frac{\pi x^3}{6} = 288\pi$$

$$x^3 = 1728$$

$$x = 12$$

The rectangular base of the pyramid has dimensions x and $2x$, so its area is

$$x(2x) = 2x^2$$

For the two faces with base x , the slant height is

$$\sqrt{(5x)^2 + x^2} = x\sqrt{26}$$

Their total area is

$$2\left(\frac{1}{2}x \cdot x\sqrt{26}\right) = x^2\sqrt{26}$$

For the two faces with base $2x$, the slant height is

$$\sqrt{(5x)^2 + \left(\frac{x}{2}\right)^2} = \frac{x\sqrt{101}}{2}$$

Their total area is

$$2\left(\frac{1}{2}(2x) \cdot \frac{x\sqrt{101}}{2}\right) = x^2\sqrt{101}$$

Total surface area:

$$2x^2 + x^2\sqrt{26} + x^2\sqrt{101}$$

Substitute $x = 12$:

$$144(2 + \sqrt{26} + \sqrt{101}) = 2469.440\dots$$

Correct to the nearest cm^2 :

2469

FINAL ANSWER 2469 cm^2

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: P1H

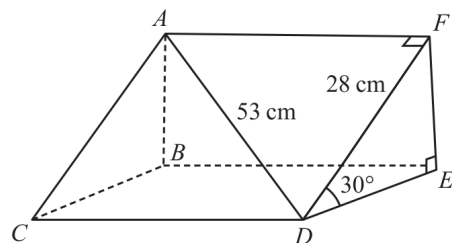
Session: Nov 2023

Q#: 21

Marks: 5

TOPIC: **Volume & Surface Area**

Source: Nov 2023 P1H

21 Here is a triangular prism $ABCDEF$ Diagram **NOT**
accurately drawn

$$AD = 53 \text{ cm}$$

$$DF = 28 \text{ cm}$$

$$\text{Angle } FDE = 30^\circ$$

Work out the volume of the triangular prism.

Give your answer correct to the nearest whole number.

..... cm^3

(Total for Question 21 is 5 marks)

WORKED SOLUTION*Dr. Eslam Ahmed***Nov 2023 P1H | Question 21**

Volume & Surface Area

Topic check. Correct.**Method**The triangular cross-section is triangle DEF .It is right-angled at E , with

$$DF = 28$$

and

$$\angle FDE = 30^\circ$$

So

$$FE = 28 \sin 30^\circ = 14$$

and

$$DE = 28 \cos 30^\circ = 14\sqrt{3}$$

Area of triangle DEF :

$$\frac{1}{2}(14)(14\sqrt{3}) = 98\sqrt{3}$$

The length of the prism is

$$AD = 53$$

Volume of the prism:

$$98\sqrt{3} \times 53$$

$$= 5194\sqrt{3}$$

$$= 8996.2719 \dots$$

FINAL ANSWER

$$8996 \text{ cm}^3$$

CLASSIFIED PROBLEM

Paper: P1H

Session: Nov 2024

Q#: 22

Marks: 3

TOPIC: **Volume & Surface Area**

Source: Nov 2024 P1H

- 22 The diagram shows a bowl in the shape of a hemisphere.
The bowl is made from metal.

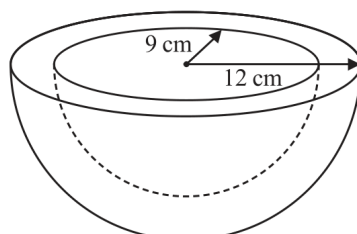


Diagram **NOT**
accurately drawn

The outer radius of the bowl is 12 cm

The inner radius of the bowl is 9 cm

The thickness of the bowl is uniform.

Work out the volume of the metal.

Give your answer correct to the nearest whole number.

..... cm³

(Total for Question 22 is 3 marks)

WORKED SOLUTION*Dr. Eslam Ahmed***Nov 2024 P1H | Question 22**

Volume & Surface Area

Topic check. Correct.**Method**

The bowl is a hemispherical shell.

The volume of the metal is:

outer hemisphere volume – inner hemisphere volume

Volume of a hemisphere:

$$\frac{2}{3}\pi r^3$$

So

$$\text{metal volume} = \frac{2}{3}\pi(12)^3 - \frac{2}{3}\pi(9)^3$$

$$= \frac{2}{3}\pi(12^3 - 9^3)$$

$$= \frac{2}{3}\pi(1728 - 729)$$

$$= 666\pi$$

$$= 2092.3007\dots$$

FINAL ANSWER

$$2092 \text{ cm}^3$$

Area & Volume of Similar Shapes

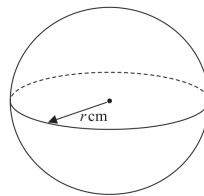
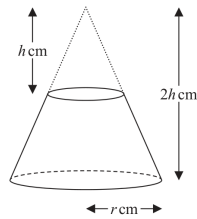
Dr. Eslam Ahmed

CLASSIFIED PROBLEM**Paper: P1HR****Session: Jan 2020****Q#: 20****Marks: 5****TOPIC: Area & Volume of Similar Shapes**

Source: Jan 2020 P1HR

20 The diagram shows a frustum of a cone and a sphere.

The frustum is made by removing a small cone from a large cone.
The cones are similar.

Diagram NOT
accurately drawn

The height of the small cone is h cm.
The height of the large cone is $2h$ cm.
The radius of the base of the large cone is r cm.

The radius of the sphere is r cm.

Given that the volume of the frustum is equal to the volume of the sphere,

find an expression for r in terms of h .
Give your expression in its simplest form.

 $r = \dots\dots\dots$

(Total for Question 20 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P1HR | Question 20

Area & Volume of Similar Shapes

Topic check. Correct.**Method**

The two cones are similar.

The large cone has height $2h$ and base radius r .The small cone has height h , so its linear scale factor compared with the large cone is

$$\frac{h}{2h} = \frac{1}{2}$$

So the small cone has radius

$$\frac{r}{2}$$

Volume of the large cone:

$$\frac{1}{3}\pi r^2(2h) = \frac{2}{3}\pi r^2h$$

Volume of the small cone:

$$\frac{1}{3}\pi \left(\frac{r}{2}\right)^2 h = \frac{1}{12}\pi r^2h$$

So the volume of the frustum is

$$\frac{2}{3}\pi r^2h - \frac{1}{12}\pi r^2h = \frac{7}{12}\pi r^2h$$

The sphere has radius r , so its volume is

$$\frac{4}{3}\pi r^3$$

The two volumes are equal:

$$\frac{7}{12}\pi r^2h = \frac{4}{3}\pi r^3$$

Divide by πr^2 :

$$\frac{7}{12}h = \frac{4}{3}r$$

$$7h = 16r$$

$$r = \frac{7h}{16}$$

FINAL ANSWER

$$r = \frac{7h}{16}$$

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jan 2020

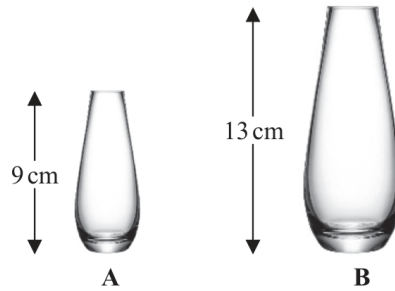
Q#: 20

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Source: Jan 2020 P2HR

20 The diagram shows two similar vases, **A** and **B**.



The height of vase **A** is 9 cm and the height of vase **B** is 13 cm.

Given that

$$\text{surface area of vase A} + \text{surface area of vase B} = 1800 \text{ cm}^2$$

calculate the surface area of vase **A**.

..... cm²

(Total for Question 20 is 4 marks)

WORKED SOLUTION*Dr. Eslam Ahmed***Jan 2020 P2HR | Question 20**

Area & Volume of Similar Shapes

Topic check. Correct.**Method**

The two vases are similar.

Their heights are in the ratio

$$9 : 13$$

So their surface areas are in the ratio

$$9^2 : 13^2 = 81 : 169$$

The total surface area is 1800 cm^2 , so there are

$$81 + 169 = 250$$

parts altogether.

Surface area of vase A:

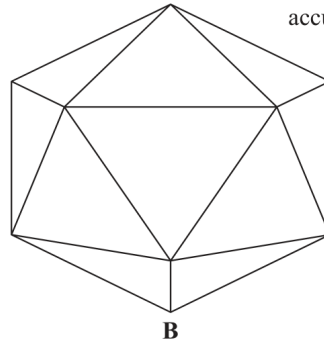
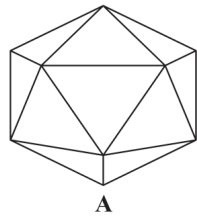
$$1800 \times \frac{81}{250} = 583.2$$

FINAL ANSWER

$$583.2 \text{ cm}^2$$

CLASSIFIED PROBLEM**Paper: P1H****Session: Jan 2021****Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

Source: Jan 2021 P1H

20 **A** and **B** are two similar solids.Diagram **NOT**
accurately drawn**A** has a volume of 1836 cm^3 **B** has a volume of 4352 cm^3 **B** has a total surface area of 1120 cm^2 Work out the total surface area of **A**...... cm^2 **(Total for Question 20 is 3 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1H | Question 20

Area & Volume of Similar Shapes

Topic check. Correct.**Method**

The volume ratio of solid A to solid B is

$$1836 : 4352$$

Simplify:

$$\frac{1836}{4352} = \frac{27}{64}$$

So the linear ratio is

$$\sqrt[3]{27} : \sqrt[3]{64} = 3 : 4$$

Surface area ratio:

$$3^2 : 4^2 = 9 : 16$$

Solid B has surface area 1120 cm^2 , so solid A has surface area

$$1120 \times \frac{9}{16} = 630$$

FINAL ANSWER

$$630 \text{ cm}^2$$

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jun 2022

Q#: 20

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Source: Jun 2022 P1HR

20 The diagram shows two similar metal statues.

**A****B**

Diagram **NOT**
accurately drawn

The volume of statue **B** is 20% less than the volume of statue **A**

The surface area of statue **B** is $k\%$ less than the surface area of statue **A**

Work out the value of k

Give your answer correct to 3 significant figures.

$k = \dots\dots\dots$

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1HR | Question 20

Area & Volume of Similar Shapes

Topic check. Corrected from Percentages to Area & Volume of Similar Shapes, because the key idea is the relationship between volume scale factor and surface area scale factor for similar solids.

Method

The volume of statue B is 20% less than statue A , so the volume scale factor is 0.8.

For similar solids, the linear scale factor is the cube root of the volume scale factor:

linear scale factor = $\sqrt[3]{0.8}$

Surface area scale factor is the square of the linear scale factor:

surface area scale factor = $(\sqrt[3]{0.8})^2 = 0.8^{2/3} = 0.861773\dots$

So statue B has 86.1773...% of the surface area of statue A .

$k = 100 - 86.1773\dots = 13.8226\dots$

FINAL ANSWER

$k = 13.8$ to 3 significant figures.

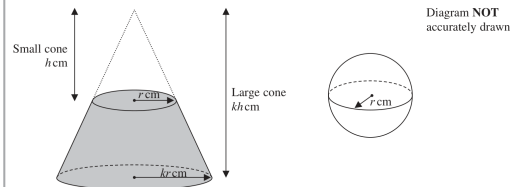
CLASSIFIED PROBLEM**Paper: P2HR****Session: Jun 2022****Q#: 24****Marks: 6****TOPIC: Area & Volume of Similar Shapes**

Source: Jun 2022 P2HR

24 The diagram shows a frustum of a cone, and a sphere.

The frustum, shown shaded in the diagram, is made by removing the small cone from the large cone.

The small cone and the large cone are similar.

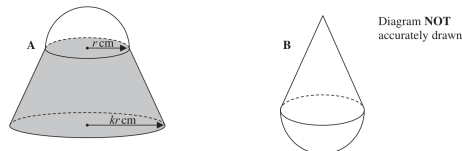


The height of the small cone is h cm and the radius of the base of the small cone is r cm. The height of the large cone is kh cm and the radius of the base of the large cone is kr cm. The radius of the sphere is r cm.

The sphere is divided into two hemispheres, each of radius r cm.

Solid A is formed by joining one of the hemispheres to the frustum. The plane face of the hemisphere coincides with the upper plane face of the frustum, as shown in the diagram below.

Solid B is formed by joining the other hemisphere to the small cone that was removed from the large cone. The plane face of the hemisphere coincides with the plane face of the base of the small cone, as shown in the diagram below.



The volume of solid A is 6 times the volume of solid B.

Given that $k > \sqrt[3]{7}$

find an expression for h in terms of k and r

$h =$ _____

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2HR | Question 24

Area & Volume of Similar Shapes

Topic check. Correct.**Method**

Small cone volume:

$$V_s = \frac{1}{3}\pi r^2 h$$

The larger similar cone has radius kr and height kh , so its volume is

$$V_L = \frac{1}{3}\pi(kr)^2(kh) = \frac{1}{3}\pi k^3 r^2 h$$

So the frustum volume is

$$V_L - V_s = \frac{1}{3}\pi r^2 h(k^3 - 1)$$

The volume of one hemisphere of radius r is

$$\frac{2}{3}\pi r^3$$

Solid A is the frustum plus one hemisphere:

$$V_A = \frac{1}{3}\pi r^2 h(k^3 - 1) + \frac{2}{3}\pi r^3$$

Solid B is the small cone plus one hemisphere:

$$V_B = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$$

Given $V_A = 6V_B$:

$$\frac{1}{3}\pi r^2 h(k^3 - 1) + \frac{2}{3}\pi r^3 = 6\left(\frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3\right)$$

Divide by $\frac{1}{3}\pi r^2$:

$$h(k^3 - 1) + 2r = 6h + 12r$$

$$h(k^3 - 7) = 10r$$

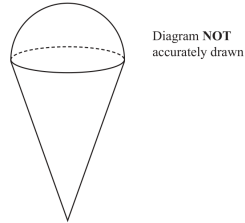
$$h = \frac{10r}{k^3 - 7}$$

FINAL ANSWER

$$h = \frac{10r}{k^3 - 7}$$

CLASSIFIED PROBLEM**Paper: P2HR****Session: Jun 2024****Q#: 24****Marks: 5****TOPIC: Area & Volume of Similar Shapes**

Source: Jun 2024 P2HR

24 The diagram shows a solid, S , made from a cone and a hemisphere.

The centre of the circular face of the cone coincides with the centre of the flat surface of the hemisphere.

The radius of the circular face of the cone, x cm, is equal to the radius of the hemisphere.

The total height of S is $4 \times$ the radius of the hemisphere.

A separate sphere has radius kx cm.

The volume of this sphere is $12.5 \times$ the volume of S

(a) Work out the value of k

$$k = \frac{\quad}{\quad} \quad (4)$$

A solid, T , is similar to solid S

The volume of T is $512 \times$ the volume of S

The total surface area of T is $d \times$ the total surface area of S

(b) Find the value of d

$$d = \frac{\quad}{\quad} \quad (1)$$

(Total for Question 24 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2HR | Question 24

Area & Volume of Similar Shapes

Topic check. Correct.**Method**The hemisphere has radius x , so its height is x .The total height of solid S is $4x$, so the cone height is

$$4x - x = 3x$$

Volume of the cone:

$$\frac{1}{3}\pi x^2(3x) = \pi x^3$$

Volume of the hemisphere:

$$\frac{2}{3}\pi x^3$$

So the volume of solid S is

$$\pi x^3 + \frac{2}{3}\pi x^3 = \frac{5}{3}\pi x^3$$

The sphere has radius kx , so its volume is

$$\frac{4}{3}\pi(kx)^3 = \frac{4}{3}\pi k^3 x^3$$

This is 12.5 times the volume of S:

$$\frac{4}{3}\pi k^3 x^3 = 12.5 \left(\frac{5}{3}\pi x^3 \right)$$

Cancel $\frac{1}{3}\pi x^3$:

$$4k^3 = 62.5$$

$$k^3 = 15.625 = \frac{125}{8}$$

$$k = 2.5$$

For solid T, the volume scale factor is 512.

$$512 = 8^3$$

So the linear scale factor is 8, and the surface area scale factor is

$$8^2 = 64$$

FINAL ANSWER

$$k = 2.5, d = 64$$

CLASSIFIED PROBLEM

Paper: P1HR

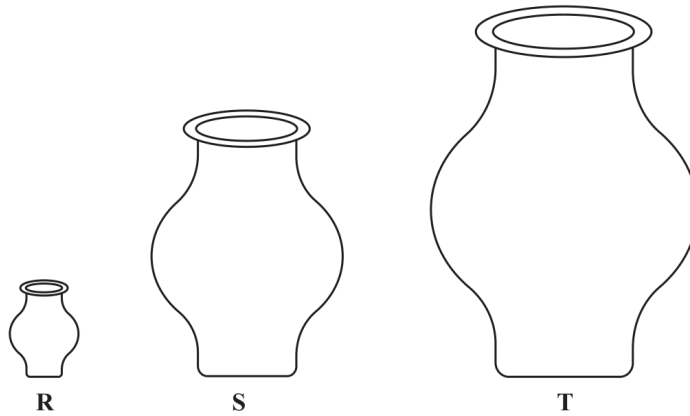
Session: Jun 2025

Q#: 26

Marks: 4

TOPIC: Area & Volume of Similar Shapes

Source: Jun 2025 P1HR

26 R, S and T are three similar vases.Diagram **NOT**
accurately drawnThe volume of vase **S** is 72.8% more than the volume of vase **R**The height of vase **R** is h cmThe height of vase **T** is $6h$ cmThe surface area of vase **S** is A cm²The surface area of vase **T** is kA cm²Work out the value of k $k = \dots\dots\dots$ **(Total for Question 26 is 4 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1HR | Question 26

Area & Volume of Similar Shapes

Topic check. Corrected from Percentages to Area & Volume of Similar Shapes, because the solution uses volume and surface area scale factors for similar shapes.

Method

The volume of vase S is 72.8% more than vase R , so

$$\frac{V_S}{V_R} = 1.728$$

For similar solids, volume scale factor is the cube of the linear scale factor:

linear scale factor from R to $S = \sqrt[3]{1.728} = 1.2$

The height of vase T is $6h$, so the linear scale factor from R to T is 6.

Therefore the linear scale factor from S to T is

$$\frac{6}{1.2} = 5$$

Surface area scale factor is the square of the linear scale factor:

$$k = 5^2 = 25$$

FINAL ANSWER

$$k = 25$$

CLASSIFIED PROBLEM**Paper: P2H****Session: Jun 2025****Q#: 26****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Source: Jun 2025 P2H

26 **A** and **B** are two similar solids.The height of solid **A** is 31 cmThe height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**..... cm³**(Total for Question 26 is 4 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2H | Question 26

Area & Volume of Similar Shapes

Topic check. Correct.**Method**

The heights of solids A and B are 31 cm and 18.6 cm.

$$31 : 18.6 = 5 : 3$$

So the volumes are in the ratio

$$5^3 : 3^3 = 125 : 27$$

The difference in volume is

$$125 - 27 = 98$$

parts.

This difference is 735 cm^3 , so one part is

$$\frac{735}{98} = 7.5$$

Volume of solid A:

$$125 \times 7.5 = 937.5$$

FINAL ANSWER

$$937.5 \text{ cm}^3$$

CLASSIFIED PROBLEM**Paper: P2H****Session: May 2021****Q#: 20****Marks: 3**TOPIC: **Area & Volume of Similar Shapes**

Source: May 2021 P2H

20 Mathematically similar wooden blocks are made in a workshop.

There are small blocks and there are large blocks.

The volume of each small block is 300 cm^3

Given that

the surface area of each small block : the surface area of each large block = $25 : 36$

work out the volume of each large block.

..... cm^3

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P2H | Question 20

Area & Volume of Similar Shapes

Topic check. Correct.**Method**

The surface area ratio of the small block to the large block is

$$25 : 36$$

So the linear ratio is

$$\sqrt{25} : \sqrt{36} = 5 : 6$$

Therefore the volume ratio is

$$5^3 : 6^3 = 125 : 216$$

The small block has volume 300 cm^3 .

Large block volume:

$$300 \times \frac{216}{125} = 518.4$$

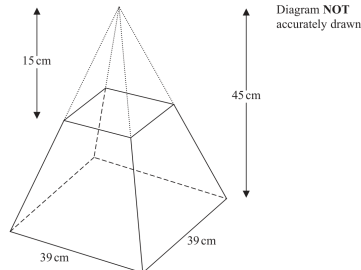
FINAL ANSWER

$$518.4 \text{ cm}^3$$

CLASSIFIED PROBLEM**Paper: P1H****Session: May 2023****Q#: 23****Marks: 5****TOPIC: Area & Volume of Similar Shapes**

Source: May 2023 P1H

- 23 A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram.



The height of the small pyramid is 15 cm.
 The height of the large pyramid is 45 cm.
 The square base of the large pyramid has side length 39 cm.

Work out the **total** surface area of the frustum.
 Give your answer correct to the nearest whole number.

_____ cm²

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1H | Question 23

Area & Volume of Similar Shapes

Topic check. Correct.**Method**

The small and large square-based pyramids are similar.

Their heights are

$$15 : 45 = 1 : 3$$

So the side length of the top square of the frustum is

$$\frac{39}{3} = 13 \text{ cm}$$

The vertical height of the frustum is

$$45 - 15 = 30 \text{ cm}$$

For one trapezium face, the difference between the half-side lengths is

$$\frac{39 - 13}{2} = 13 \text{ cm}$$

So the slant height of the trapezium face is

$$\sqrt{30^2 + 13^2} = \sqrt{1069}$$

Area of one trapezium face:

$$\frac{1}{2}(39 + 13)\sqrt{1069} = 26\sqrt{1069}$$

Area of four trapezium faces:

$$4(26\sqrt{1069}) = 104\sqrt{1069}$$

Add the top and bottom square faces:

$$13^2 + 39^2 = 169 + 1521 = 1690$$

Total surface area:

$$1690 + 104\sqrt{1069} = 5090.338 \dots$$

Correct to the nearest whole number:

$$5090$$

FINAL ANSWER

$$5090 \text{ cm}^2$$

CLASSIFIED PROBLEM**Paper: P1H****Session: MayNov 2020****Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

Source: May-Nov 2020 P1H

20 **R** and **S** are two similar solid shapes.

Shape **R** has surface area 108 cm^2 and volume 135 cm^3

Shape **S** has surface area 300 cm^2

Work out the volume of shape **S**.

..... cm^3

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1H | Question 20

Area & Volume of Similar Shapes

Topic check. Correct.**Method**Shape R has surface area 108 cm^2 and volume 135 cm^3 .Shape S has surface area 300 cm^2 .

Surface area ratio S:R:

$$300 : 108 = 25 : 9$$

So the linear ratio S:R is

$$5 : 3$$

Therefore the volume ratio S:R is

$$5^3 : 3^3 = 125 : 27$$

Volume of S:

$$135 \times \frac{125}{27} = 625$$

FINAL ANSWER

$$625 \text{ cm}^3$$

CLASSIFIED PROBLEM**Paper: P1H****Session: Nov 2023****Q#: 23****Marks: 5****TOPIC: Area & Volume of Similar Shapes**

Source: Nov 2023 P1H

23 Solid A is similar to solid B

Here is some information about solid A and solid B

	solid A	solid B
Height (cm)	3^x	
Area (cm^2)	7776	486
Volume (cm^3)	8^x	2^{x+4}

Work out the height of solid B
Give your answer as a decimal.

_____ cm

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P1H | Question 23

Area & Volume of Similar Shapes

Topic check. Correct.**Method**

The area ratio of solid A to solid B is

$$7776 : 486 = 16 : 1$$

So the linear ratio A:B is

$$\sqrt{16} : 1 = 4 : 1$$

Therefore the volume ratio A:B is

$$4^3 : 1^3 = 64 : 1$$

The volumes are 8^x and 2^{x+4} , so

$$\frac{8^x}{2^{x+4}} = 64$$

Write 8^x as a power of 2:

$$8^x = (2^3)^x = 2^{3x}$$

So

$$\frac{2^{3x}}{2^{x+4}} = 2^6$$

$$2^{3x-(x+4)} = 2^6$$

$$2x - 4 = 6$$

$$x = 5$$

The height of solid A is

$$3^x = 3^5 = 243$$

Since the linear ratio A:B is 4 : 1, the height of solid B is

$$\frac{243}{4} = 60.75$$

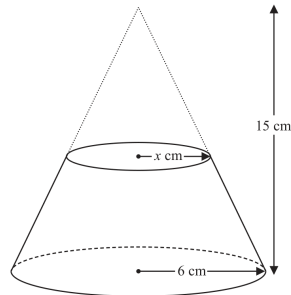
FINAL ANSWER

60.75 cm

CLASSIFIED PROBLEM**Paper: P2H****Session: Nov 2023****Q#: 23****Marks: 5****TOPIC: Area & Volume of Similar Shapes**

Source: Nov 2023 P2H

- 23 Here is a frustum of a cone.
The frustum is made by removing a small cone from a similar large cone.

Diagram NOT
accurately drawn

The height of the large cone is 15 cm.

The radius of the base of the large cone is 6 cm.

The radius of the base of the small cone is x cm.Given that the volume of the frustum is $\frac{4212}{25}\pi \text{ cm}^3$ work out the value of x

Show clear algebraic working.

 $x =$ _____

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P2H | Question 23

Area & Volume of Similar Shapes

Topic check. Correct.**Method**

The large cone has height 15 cm and radius 6 cm.

The small cone has radius x . Since the cones are similar,

$$\frac{\text{radius}}{\text{height}} = \frac{6}{15} = \frac{2}{5}$$

So for the small cone,

$$\frac{x}{h} = \frac{2}{5}$$

$$h = \frac{5}{2}x$$

Volume of the large cone:

$$\frac{1}{3}\pi(6)^2(15) = 180\pi$$

Volume of the small cone:

$$\frac{1}{3}\pi x^2 \left(\frac{5}{2}x\right) = \frac{5}{6}\pi x^3$$

The frustum volume is

$$\frac{4212}{25}\pi$$

So

$$180\pi - \frac{5}{6}\pi x^3 = \frac{4212}{25}\pi$$

Cancel π :

$$180 - \frac{5}{6}x^3 = \frac{4212}{25}$$

$$\frac{5}{6}x^3 = 180 - \frac{4212}{25}$$

$$\frac{5}{6}x^3 = \frac{288}{25}$$

$$x^3 = \frac{1728}{125}$$

$$x = \frac{12}{5} = 2.4$$

FINAL ANSWER

2.4 cm

CLASSIFIED PROBLEM**Paper: P2H****Session: Nov 2025****Q#: 23****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

Source: Nov 2025 P2H

23 Shape **P** is similar to shape **Q**The table shows some information about shape **P** and shape **Q**

	Surface area (cm ²)	Volume (cm ³)
Shape P	200	672
Shape Q	450	

Work out the volume of shape **Q**..... cm³**(Total for Question 23 is 3 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P2H | Question 23

Area & Volume of Similar Shapes

Topic check. Correct.**Method**

Surface area ratio Q:P:

$$450 : 200 = 9 : 4$$

So the linear ratio Q:P is

$$3 : 2$$

Therefore the volume ratio Q:P is

$$3^3 : 2^3 = 27 : 8$$

Shape P has volume 672 cm^3 .

Volume of Q:

$$672 \times \frac{27}{8} = 2268$$

FINAL ANSWER 2268 cm^3

Vectors

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: May2025

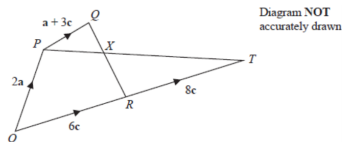
Q#: 26

Marks: 5

TOPIC: **Vectors**

Source: May 2025 4WM2H

26

The diagram shows a quadrilateral $OPQR$ in which

$$\vec{OP} = 2\mathbf{a} \quad \vec{OR} = 6\mathbf{c} \quad \vec{PQ} = \mathbf{a} + 3\mathbf{c}$$

(a) Find in terms of $\mathbf{a}$ and $\mathbf{c}$

$$\vec{QR} = \dots\dots\dots (1)$$

The point T is such that $\vec{RT} = 9\mathbf{c}$
 The point X lies on QR such that PXT is a straight line.

(b) Using a vector method, find the ratio $QX : XR$
 Give your answer in the form $m : n$ where m and n are integers.
 Show your working clearly.

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$$QX : XR = \dots\dots\dots (4)$$

(Total for Question 26 is 5 marks)

WORKED SOLUTION*Dr. Eslam Ahmed***May 2025 4WM2H | Question 26**

Vectors

Solution

(a)

$$\overrightarrow{QR} = \overrightarrow{OR} - \overrightarrow{OQ}$$

$$\overrightarrow{OQ} = 2a + (a + 3c) = 3a + 3c$$

$$\overrightarrow{QR} = 6c - (3a + 3c) = 3c - 3a$$

(b) Let X divide QR so that

$$\overrightarrow{QX} = \lambda \overrightarrow{QR}$$

Then

$$\overrightarrow{OX} = 3a + 3c + \lambda(3c - 3a)$$

Also X lies on PT , where $\overrightarrow{OT} = 14c$, so

$$\overrightarrow{OX} = 2a + \mu(14c - 2a)$$

Equating coefficients gives $\lambda = \frac{5}{9}$. Therefore

$$QX : XR = \frac{5}{9} : \frac{4}{9} = 5 : 4$$

FINAL ANSWER

$$\overrightarrow{QR} = 3c - 3a; QX : XR = 5 : 4.$$

CLASSIFIED PROBLEM

Paper: 4WM2H

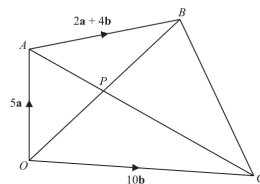
Session: Nov2025

Q#: 24

Marks: 6

TOPIC: **Vectors**

Source: Nov 2025 4WM2H

24 The diagram shows a quadrilateral $OABC$ Diagram NOT
accurately drawn

$$\vec{OA} = 5\mathbf{a} \quad \vec{OC} = 10\mathbf{b} \quad \vec{AB} = 2\mathbf{a} + 4\mathbf{b}$$

(a) Find $\vec{AC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

$$\vec{AC} = \underline{\hspace{2cm}} \quad (1)$$

(b) Find $\vec{OB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.

$$\vec{OB} = \underline{\hspace{2cm}} \quad (1)$$

The point P is such that APC and OPB are straight lines.(c) Using a vector method, find $\vec{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.
Show your working clearly.

$$\vec{OP} = \underline{\hspace{2cm}} \quad (4)$$

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM2H | Question 24

Vectors

(a) Find $\overrightarrow{AC}$:

$$\overrightarrow{AC} = \overrightarrow{AO} + \overrightarrow{OC} = -\overrightarrow{OA} + \overrightarrow{OC} = -5\mathbf{a} + 10\mathbf{b}$$

FINAL ANSWER

$$a - 5\mathbf{a} + 10\mathbf{b}$$

(b) Find $\overrightarrow{OB}$:

$$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = 5\mathbf{a} + (2\mathbf{a} + 4\mathbf{b}) = 7\mathbf{a} + 4\mathbf{b}$$

FINAL ANSWER

$$b \ 7\mathbf{a} + 4\mathbf{b}$$

**Step 1: Set up vector equations for $\overrightarrow{OP}$. Since OPB is a straight line, $\overrightarrow{OP}$ is a scalar multiple of $\overrightarrow{OB}$. Let this be λ .

$$\overrightarrow{OP} = \lambda(7\mathbf{a} + 4\mathbf{b}) = 7\lambda\mathbf{a} + 4\lambda\mathbf{b}$$

Since P lies on the straight line AC , $\overrightarrow{AP} = \mu\overrightarrow{AC}$ for some scalar μ .

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = 5\mathbf{a} + \mu(-5\mathbf{a} + 10\mathbf{b})$$

$$\overrightarrow{OP} = (5 - 5\mu)\mathbf{a} + 10\mu\mathbf{b}$$

Step 2: Equate the coefficients of $\mathbf{a}$ and $\mathbf{b}$.

$$7\lambda = 5 - 5\mu - (1)$$

$$4\lambda = 10\mu \implies \mu = \frac{4}{10}\lambda = 0.4\lambda \quad \text{--- (2)}$$

Step 3: Solve for λ . Substitute (2) into (1):

$$7\lambda = 5 - 5(0.4\lambda)$$

$$7\lambda = 5 - 2\lambda$$

$$9\lambda = 5 \implies \lambda = \frac{5}{9}$$

Step 4: Find $\overrightarrow{OP}$.

$$\overrightarrow{OP} = \frac{5}{9}(7\mathbf{a} + 4\mathbf{b}) = \frac{35}{9}\mathbf{a} + \frac{20}{9}\mathbf{b}$$

FINAL ANSWER

$$\frac{35}{9}\mathbf{a} + \frac{20}{9}\mathbf{b}$$

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Specimen

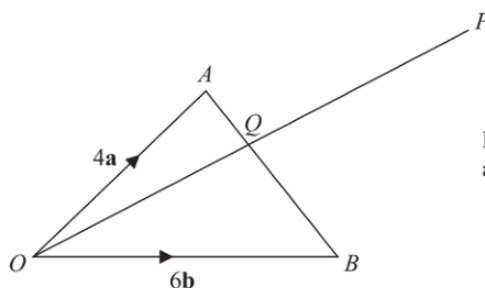
Q#: 23

Marks: 5

TOPIC: **Vectors**

Source: Specimen 4WM2H

23

Diagram **NOT**
accurately drawn OAB is a triangle. Q is the point on AB such that OQP is a straight line.

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AP} = 2\mathbf{a} + 8\mathbf{b}$$

Using a vector method, find the ratio $AQ : QB$

$$AQ : QB = \dots\dots\dots$$

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Specimen 4WM2H | Question 23

Vectors

Topic check. Vectors. The tag is correct.**Solution**

$$\overrightarrow{OA} = 4a, \quad \overrightarrow{OB} = 6b, \quad \overrightarrow{AP} = 2a + 8b$$

So

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}$$

$$\overrightarrow{OP} = 4a + (2a + 8b) = 6a + 8b$$

Since O, Q, P are collinear, let

$$\overrightarrow{OQ} = \lambda(6a + 8b)$$

Also Q lies on AB . Let

$$\overrightarrow{AQ} = \mu\overrightarrow{AB}$$

Then

$$\overrightarrow{AB} = 6b - 4a$$

$$\overrightarrow{OQ} = 4a + \mu(6b - 4a)$$

$$\overrightarrow{OQ} = (4 - 4\mu)a + 6\mu b$$

Compare coefficients with

$$\lambda(6a + 8b) = 6\lambda a + 8\lambda b$$

So

$$4 - 4\mu = 6\lambda, \quad 6\mu = 8\lambda$$

From $6\mu = 8\lambda$,

$$\lambda = \frac{3}{4}\mu$$

Substitute:

$$4 - 4\mu = 6\left(\frac{3}{4}\mu\right)$$

$$4 - 4\mu = \frac{9}{2}\mu$$

$$8 = 17\mu$$

$$\mu = \frac{8}{17}$$

Therefore

$$AQ : QB = \frac{8}{17} : \frac{9}{17} = 8 : 9$$

FINAL ANSWER

$$AQ : QB = 8 : 9.$$

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: P2H

Session: Jan 2020

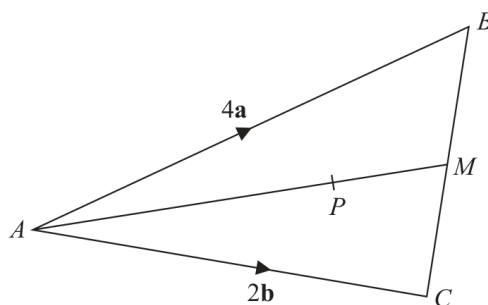
Q#: 23

Marks: 3

TOPIC: **Vectors**

Source: Jan 2020 P2H

23

Diagram **NOT**
accurately drawn

ABC is a triangle.
The midpoint of BC is M .
 P is a point on AM .

$$\vec{AB} = 4\mathbf{a}$$

$$\vec{AC} = 2\mathbf{b}$$

$$\vec{AP} = \frac{3}{2}\mathbf{a} + \frac{3}{4}\mathbf{b}$$

Find the ratio $AP:PM$

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2H | Question 23

Vectors

Topic check. Corrected from Ratio Toolkit to Vectors.**Method**Since M is the midpoint of BC ,

$$\overrightarrow{AM} = \frac{\overrightarrow{AB} + \overrightarrow{AC}}{2}$$

$$\overrightarrow{AM} = \frac{4a + 2b}{2} = 2a + b$$

Given

$$\overrightarrow{AP} = \frac{3}{2}a + \frac{3}{4}b$$

Factorise:

$$\overrightarrow{AP} = \frac{3}{4}(2a + b) = \frac{3}{4}\overrightarrow{AM}$$

So AP is $\frac{3}{4}$ of AM , leaving PM as $\frac{1}{4}$ of AM .**FINAL ANSWER**

$$AP : PM = 3 : 1$$

CLASSIFIED PROBLEM

Paper: P2HR

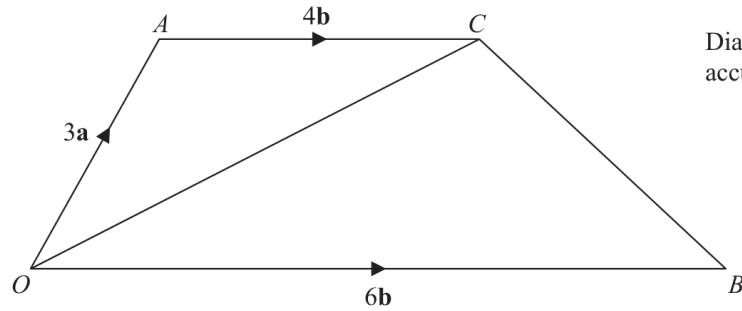
Session: Jan 2020

Q#: 26

Marks: 5

TOPIC: **Vectors**

Source: Jan 2020 P2HR

26 The diagram shows trapezium $OACB$.Diagram **NOT**
accurately drawn

$$\vec{OA} = 3\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AC} = 4\mathbf{b}$$

 N is the point on OC such that ANB is a straight line.Find $\vec{ON}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{b}$.

(Total for Question 26 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2HR | Question 26

Vectors

Topic check. Moved from Linear Graphs to Vectors.**Method**

$$\vec{OA} = 3\mathbf{a}, \quad \vec{OB} = 6\mathbf{b}, \quad \vec{AC} = 4\mathbf{b}$$

So

$$\vec{OC} = 3\mathbf{a} + 4\mathbf{b}$$

Point N lies on OC , so

$$\vec{ON} = t(3\mathbf{a} + 4\mathbf{b})$$

Point N also lies on AB , so

$$\vec{ON} = 3\mathbf{a} + s(6\mathbf{b} - 3\mathbf{a})$$

$$\vec{ON} = (3 - 3s)\mathbf{a} + 6s\mathbf{b}$$

Equate coefficients:

$$3t = 3 - 3s$$

$$4t = 6s$$

So

$$t = 1 - s$$

and

$$t = \frac{3}{2}s$$

$$\frac{3}{2}s = 1 - s$$

$$5s = 2$$

$$s = \frac{2}{5}$$

Then

$$t = \frac{3}{5}$$

Therefore

$$\vec{ON} = \frac{3}{5}(3\mathbf{a} + 4\mathbf{b})$$

$$O\vec{N} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$$

FINAL ANSWER

$$O\vec{N} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$$

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: P1H

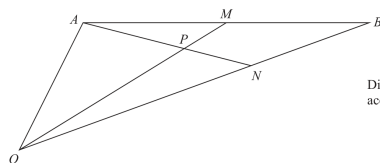
Session: Jan 2021

Q#: 23

Marks: 6

TOPIC: **Vectors**

Source: Jan 2021 P1H

23 OAB is a triangle.Diagram NOT
accurately drawn

$$\vec{OA} = 2\mathbf{a} \text{ and } \vec{OB} = 2\mathbf{b}$$

 M is the midpoint of AB . N is the point on OB such that $ON:NB = 2:1$ P is the point on AN such that OPM is a straight line.Use a vector method to find $OP:PM$
Show your working clearly.

(Total for Question 23 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1H | Question 23

Vectors

Topic check. Moved from Coordinate Geometry to Vectors.**Method**

$$\vec{OA} = 2\mathbf{a}, \quad \vec{OB} = 2\mathbf{b}$$

Since M is the midpoint of AB ,

$$\vec{OM} = \frac{2\mathbf{a} + 2\mathbf{b}}{2}$$

$$\vec{OM} = \mathbf{a} + \mathbf{b}$$

Since $ON : NB = 2 : 1$,

$$\vec{ON} = \frac{2}{3}\vec{OB}$$

$$\vec{ON} = \frac{2}{3}(2\mathbf{b}) = \frac{4}{3}\mathbf{b}$$

Point P lies on OM , so

$$\vec{OP} = t(\mathbf{a} + \mathbf{b})$$

Point P also lies on AN , so

$$\vec{OP} = 2\mathbf{a} + s\left(\frac{4}{3}\mathbf{b} - 2\mathbf{a}\right)$$

$$\vec{OP} = (2 - 2s)\mathbf{a} + \frac{4s}{3}\mathbf{b}$$

Equate coefficients of $\mathbf{a}$ and $\mathbf{b}$:

$$t = 2 - 2s$$

$$t = \frac{4s}{3}$$

So

$$\frac{4s}{3} = 2 - 2s$$

$$4s = 6 - 6s$$

$$10s = 6$$

$$s = \frac{3}{5}$$

Then

$$t = \frac{4}{5}$$

So P is $\frac{4}{5}$ of the way from O to M .

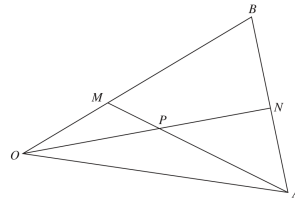
FINAL ANSWER

$$OP : PM = 4 : 1$$

Dr. Eslam Ahmed

CLASSIFIED PROBLEM**Paper: P1H****Session: Jan 2022****Q#: 22****Marks: 5**TOPIC: **Vectors**

Source: Jan 2022 P1H

22 The diagram shows triangle OAB Diagram **NOT**
accurately drawn

$$\vec{OA} = 8\mathbf{a} \quad \vec{OB} = 6\mathbf{b}$$

 M is the point on OB such that $OM:MB = 1:2$ N is the midpoint of AB P is the point of intersection of ON and AM Using a vector method, find $\vec{OP}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{b}$
Show your working clearly.

$$\vec{OP} =$$

(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1H | Question 22
Vectors

Topic check. Moved from Coordinate Geometry to Vectors.

Method

$$\vec{OA} = 8\mathbf{a}, \quad \vec{OB} = 6\mathbf{b}$$

Since $OM : MB = 1 : 2$,

$$\vec{OM} = \frac{1}{3}\vec{OB}$$

$$\vec{OM} = 2\mathbf{b}$$

Since N is the midpoint of AB ,

$$\vec{ON} = \frac{8\mathbf{a} + 6\mathbf{b}}{2}$$

$$\vec{ON} = 4\mathbf{a} + 3\mathbf{b}$$

Point P lies on ON , so

$$\vec{OP} = \lambda(4\mathbf{a} + 3\mathbf{b})$$

Point P also lies on AM , so

$$\vec{OP} = 8\mathbf{a} + \mu(2\mathbf{b} - 8\mathbf{a})$$

$$\vec{OP} = (8 - 8\mu)\mathbf{a} + 2\mu\mathbf{b}$$

Equate coefficients:

$$4\lambda = 8 - 8\mu$$

$$3\lambda = 2\mu$$

From $3\lambda = 2\mu$,

$$\mu = \frac{3}{2}\lambda$$

Substitute:

$$4\lambda = 8 - 8\left(\frac{3}{2}\lambda\right)$$

$$4\lambda = 8 - 12\lambda$$

$$16\lambda = 8$$

$$\lambda = \frac{1}{2}$$

Therefore

$$\vec{OP} = \frac{1}{2}(4\mathbf{a} + 3\mathbf{b})$$

$$\vec{OP} = 2\mathbf{a} + \frac{3}{2}\mathbf{b}$$

FINAL ANSWER

$$\vec{OP} = 2\mathbf{a} + \frac{3}{2}\mathbf{b}$$

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: P1HR

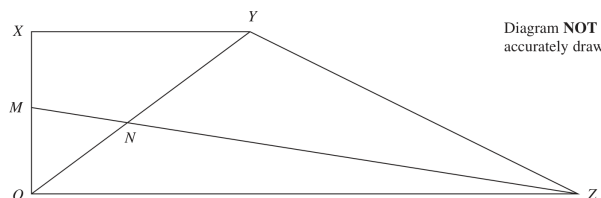
Session: Jan 2022

Q#: 24

Marks: 5

TOPIC: **Vectors**

Source: Jan 2022 P1HR

24 $OXYZ$ is a trapezium.Diagram **NOT**
accurately drawn

$$\overrightarrow{OX} = \mathbf{a}$$

$$\overrightarrow{XY} = \mathbf{b}$$

$$\overrightarrow{OZ} = 3\mathbf{b}$$

 M is the midpoint of OX N is the point such that MNZ and ONY are straight lines.Given that $ON : OY = \lambda : 1$ use a vector method to find the value of λ

$$\lambda = \dots\dots\dots$$

(Total for Question 24 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1HR | Question 24

Vectors

Topic check. Moved from Coordinate Geometry to Vectors.**Method**Take O as the origin.

$$\vec{OX} = \mathbf{a}, \quad \vec{XY} = \mathbf{b}, \quad \vec{OZ} = 3\mathbf{b}$$

So

$$\vec{OY} = \mathbf{a} + \mathbf{b}$$

Since M is the midpoint of OX ,

$$\vec{OM} = \frac{1}{2}\mathbf{a}$$

Given $ON : OY = \lambda : 1$,

$$\vec{ON} = \lambda(\mathbf{a} + \mathbf{b})$$

Point N also lies on MZ , so

$$\vec{ON} = \frac{1}{2}\mathbf{a} + t\left(3\mathbf{b} - \frac{1}{2}\mathbf{a}\right)$$

$$\vec{ON} = \left(\frac{1}{2} - \frac{t}{2}\right)\mathbf{a} + 3t\mathbf{b}$$

Equate coefficients with $\lambda\mathbf{a} + \lambda\mathbf{b}$:

$$\lambda = \frac{1}{2} - \frac{t}{2}$$

$$\lambda = 3t$$

So

$$t = \frac{\lambda}{3}$$

Substitute:

$$\lambda = \frac{1}{2} - \frac{1}{2}\left(\frac{\lambda}{3}\right)$$

$$\lambda = \frac{1}{2} - \frac{\lambda}{6}$$

$$\frac{7}{6}\lambda = \frac{1}{2}$$

$$\lambda = \frac{3}{7}$$

FINAL ANSWER

$$\lambda = \frac{3}{7}$$

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: P2H

Session: Jun 2022

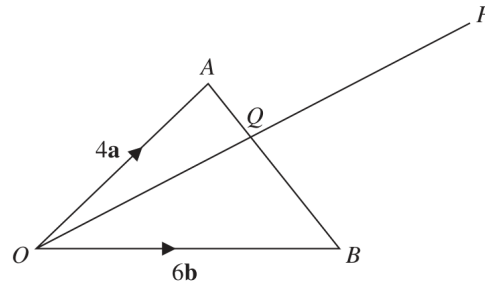
Q#: 21

Marks: 5

TOPIC: **Vectors**

Source: Jun 2022 P2H

21

Diagram **NOT**
accurately drawn OAB is a triangle. Q is the point on AB such that OQP is a straight line.

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AP} = 2\mathbf{a} + 8\mathbf{b}$$

Using a vector method, find the ratio $AQ:QB$

$$AQ:QB = \dots\dots\dots$$

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2H | Question 21

Vectors

Topic check. Corrected from Ratio Toolkit to Vectors.**Method**

$$\overrightarrow{OA} = 4a, \quad \overrightarrow{OB} = 6b, \quad \overrightarrow{AP} = 2a + 8b$$

So

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = 4a + (2a + 8b) = 6a + 8b$$

Since Q lies on OP , let

$$\overrightarrow{OQ} = \lambda(6a + 8b)$$

Since Q lies on AB , let

$$\overrightarrow{OQ} = \overrightarrow{OA} + \mu\overrightarrow{AB}$$

$$\overrightarrow{AB} = 6b - 4a$$

So

$$\overrightarrow{OQ} = 4a + \mu(6b - 4a) = (4 - 4\mu)a + 6\mu b$$

Equating coefficients:

$$6\lambda = 4 - 4\mu, \quad 8\lambda = 6\mu$$

From $8\lambda = 6\mu$, $\mu = \frac{4}{3}\lambda$.

$$6\lambda = 4 - \frac{16}{3}\lambda$$

$$34\lambda = 12, \quad \lambda = \frac{6}{17}$$

$$\mu = \frac{8}{17}$$

So $AQ : QB = \frac{8}{17} : \frac{9}{17}$.**FINAL ANSWER**

$$AQ : QB = 8 : 9$$

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2022

Q#: 25

Marks: 5

TOPIC: **Vectors**

Source: Jun 2022 P2HR

25 $ABCD$ is a parallelogram and ADM is a straight line.

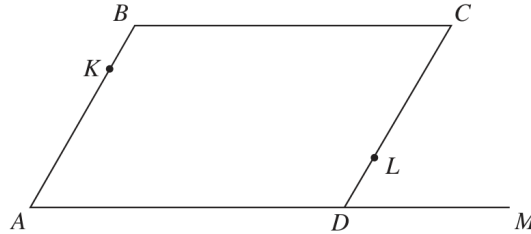


Diagram **NOT**
accurately drawn

$$\vec{AB} = \mathbf{a} \quad \vec{BC} = \mathbf{b} \quad \vec{DM} = \frac{1}{2}\mathbf{b}$$

K is the point on AB such that $AK:AB = \lambda:1$

L is the point on CD such that $CL:CD = \mu:1$

KLM is a straight line.

Given that $\lambda:\mu = 1:2$

use a vector method to find the value of λ and the value of μ

$$\lambda = \dots\dots\dots$$

$$\mu = \dots\dots\dots$$

(Total for Question 25 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2HR | Question 25
Vectors

Topic check. Moved from Linear Graphs to Vectors.

Method

Let A be the origin.

$$\vec{AB} = \mathbf{a}, \quad \vec{BC} = \mathbf{b}$$

So

$$B = \mathbf{a}, \quad D = \mathbf{b}, \quad C = \mathbf{a} + \mathbf{b}$$

Also

$$D\vec{M} = \frac{1}{2}\mathbf{b}$$

So

$$M = \frac{3}{2}\mathbf{b}$$

Point K is on AB :

$$K = \lambda\mathbf{a}$$

Point L is on CD , and $C\vec{D} = -\mathbf{a}$:

$$L = C + \mu(D - C)$$

$$L = \mathbf{a} + \mathbf{b} + \mu(-\mathbf{a})$$

$$L = (1 - \mu)\mathbf{a} + \mathbf{b}$$

Given

$$\lambda : \mu = 1 : 2$$

$$\mu = 2\lambda$$

Since K, L, M are collinear, $K\vec{M}$ is a multiple of $K\vec{L}$.

$$K\vec{M} = -\lambda\mathbf{a} + \frac{3}{2}\mathbf{b}$$

$$K\vec{L} = (1 - \mu - \lambda)\mathbf{a} + \mathbf{b}$$

$$K\vec{L} = (1 - 3\lambda)\mathbf{a} + \mathbf{b}$$

Compare coefficients using

$$K\vec{M} = \frac{3}{2}K\vec{L}$$

$$-\lambda = \frac{3}{2}(1 - 3\lambda)$$

$$-2\lambda = 3 - 9\lambda$$

$$7\lambda = 3$$

$$\lambda = \frac{3}{7}$$

Then

$$\mu = 2\lambda = \frac{6}{7}$$

FINAL ANSWER

$$\lambda = \frac{3}{7}, \quad \mu = \frac{6}{7}$$

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: P2H

Session: Jun 2024

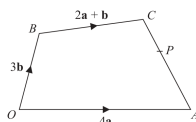
Q#: 24

Marks: 6

TOPIC: **Vectors**

Source: Jun 2024 P2H

24

Diagram NOT
accurately drawnThe diagram shows a quadrilateral $OACB$ in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

- (a) Find $\vec{AC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.

$$\vec{AC} = \underline{\hspace{2cm}} \quad (2)$$

The point P lies on AC such that $AP:PC = 3:2$ The point Q is such that OPQ and BCQ are straight lines.

- (b) Using a vector method, find $\vec{OQ}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.
Show your working clearly.

$$\vec{OQ} = \underline{\hspace{2cm}} \quad (4)$$

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2H | Question 24
Vectors

Topic check. Moved from Linear Graphs to Vectors.

Method

$$\vec{OA} = 4\mathbf{a}, \quad \vec{OB} = 3\mathbf{b}, \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

So

$$\vec{OC} = \vec{OB} + \vec{BC}$$

$$\vec{OC} = 2\mathbf{a} + 4\mathbf{b}$$

For part (a),

$$\vec{AC} = \vec{OC} - \vec{OA}$$

$$\vec{AC} = (2\mathbf{a} + 4\mathbf{b}) - 4\mathbf{a}$$

$$\vec{AC} = 4\mathbf{b} - 2\mathbf{a}$$

For part (b), $AP : PC = 3 : 2$, so

$$\vec{OP} = \vec{OA} + \frac{3}{5}\vec{AC}$$

$$\vec{OP} = 4\mathbf{a} + \frac{3}{5}(4\mathbf{b} - 2\mathbf{a})$$

$$\vec{OP} = \frac{14}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$$

Since O, P, Q are collinear,

$$\vec{OQ} = t\vec{OP}$$

$$\vec{OQ} = t\left(\frac{14}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}\right)$$

Since B, C, Q are collinear,

$$\vec{OQ} = \vec{OB} + s\vec{BC}$$

$$\vec{OQ} = 3\mathbf{b} + s(2\mathbf{a} + \mathbf{b})$$

$$\vec{OQ} = 2s\mathbf{a} + (3 + s)\mathbf{b}$$

Equate coefficients:

$$\frac{14}{5}t = 2s$$

$$\frac{12}{5}t = 3 + s$$

From the first equation,

$$s = \frac{7}{5}t$$

Substitute into the second:

$$\frac{12}{5}t = 3 + \frac{7}{5}t$$

$$t = 3$$

Therefore

$$\vec{OQ} = 3\vec{OP}$$

$$\vec{OQ} = \frac{42}{5}\mathbf{a} + \frac{36}{5}\mathbf{b}$$

FINAL ANSWER

$$\vec{AC} = 4\mathbf{b} - 2\mathbf{a}, \vec{OQ} = \frac{42}{5}\mathbf{a} + \frac{36}{5}\mathbf{b}$$

CLASSIFIED PROBLEM

Paper: P2HR

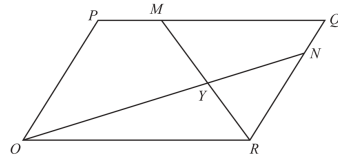
Session: Jun 2024

Q#: 25

Marks: 6

TOPIC: **Vectors**

Source: Jun 2024 P2HR

25 $OPQR$ is a parallelogram.Diagram **NOT**
accurately drawn

$$\vec{OP} = 2\mathbf{a} \quad \text{and} \quad \vec{OR} = 3\mathbf{b}$$

The point M lies on PQ such that $PM = \frac{1}{4}PQ$

The point N lies on RQ such that $RN = \frac{4}{5}RQ$

(a) Find, in terms of $\mathbf{a}$ and $\mathbf{b}$, giving your answers in simplest form

(i) $\vec{ON}$

(1)

(ii) $\vec{MR}$

(1)

MR and ON intersect at the point Y

Given that

$$OY = k \times ON$$

(b) use a vector method to find the value of k

$$k = \dots\dots\dots$$

(4)

(Total for Question 25 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2HR | Question 25

Vectors

Topic check. Correct.

Method

$$\overrightarrow{OP} = 2a, \quad \overrightarrow{OR} = 3b$$

So

$$\overrightarrow{OQ} = 2a + 3b$$

Since $PM : PQ = 1 : 4$,

$$\overrightarrow{PM} = \frac{1}{4}\overrightarrow{PQ}$$

$$\overrightarrow{PM} = \frac{1}{4}(3b) = \frac{3}{4}b$$

$$\overrightarrow{OM} = 2a + \frac{3}{4}b$$

Since $RN : RQ = 4 : 5$,

$$\overrightarrow{RN} = \frac{4}{5}\overrightarrow{RQ}$$

$$\overrightarrow{RN} = \frac{4}{5}(2a) = \frac{8}{5}a$$

$$\overrightarrow{ON} = 3b + \frac{8}{5}a$$

So

$$\overrightarrow{MR} = \overrightarrow{OR} - \overrightarrow{OM}$$

$$\overrightarrow{MR} = 3b - \left(2a + \frac{3}{4}b\right)$$

$$\overrightarrow{MR} = -2a + \frac{9}{4}b$$

Let Y be on ON , so

$$\overrightarrow{OY} = k\overrightarrow{ON} = k\left(\frac{8}{5}a + 3b\right)$$

Also Y is on MR , so

$$\overrightarrow{OY} = \overrightarrow{OM} + t\overrightarrow{MR}$$

$$\overrightarrow{OY} = 2a + \frac{3}{4}b + t\left(-2a + \frac{9}{4}b\right)$$

Compare coefficients of a and b :

$$\frac{8}{5}k = 2 - 2t$$

$$3k = \frac{3}{4} + \frac{9}{4}t$$

From the first equation,

$$t = 1 - \frac{4}{5}k$$

Substitute into the second equation:

$$3k = \frac{3}{4} + \frac{9}{4} \left(1 - \frac{4}{5}k \right)$$

$$3k = 3 - \frac{9}{5}k$$

$$\frac{24}{5}k = 3$$

$$k = \frac{5}{8}$$

FINAL ANSWER

$$\overrightarrow{ON} = \frac{8}{5}a + 3b, \overrightarrow{MR} = -2a + \frac{9}{4}b, \text{ and } k = \frac{5}{8}$$

CLASSIFIED PROBLEM

Paper: P1H

Session: Jun 2025

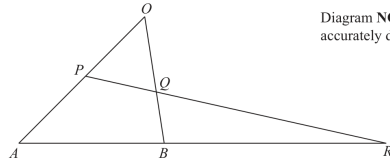
Q#: 23

Marks: 5

TOPIC: **Vectors**

Source: Jun 2025 P1H

23

Diagram NOT
accurately drawn

OAB is a triangle.
 P is the midpoint of OA
 Q is a point on OB

ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

(a) Express $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

(1)

$$AB:BR = 1:2 \quad \vec{OQ} = n\mathbf{b}$$

(b) Use a vector method to find the value of n

$n =$
 (4)

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1H | Question 23

Vectors

Topic check. Moved from Coordinate Geometry to Vectors.**Method**

For part (a),

$$\vec{OA} = 12\mathbf{a}, \quad \vec{OB} = 8\mathbf{b}$$

$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$\vec{AB} = 8\mathbf{b} - 12\mathbf{a}$$

For part (b), P is the midpoint of OA , so

$$\vec{OP} = 6\mathbf{a}$$

Given

$$AB : BR = 1 : 2$$

$$\vec{BR} = 2\vec{AB}$$

$$\vec{BR} = 16\mathbf{b} - 24\mathbf{a}$$

Therefore

$$\vec{OR} = \vec{OB} + \vec{BR}$$

$$\vec{OR} = 8\mathbf{b} + 16\mathbf{b} - 24\mathbf{a}$$

$$\vec{OR} = -24\mathbf{a} + 24\mathbf{b}$$

Since P, Q, R are collinear, write Q on PR :

$$\vec{OQ} = \vec{OP} + t(\vec{OR} - \vec{OP})$$

$$\vec{OQ} = 6\mathbf{a} + t(-30\mathbf{a} + 24\mathbf{b})$$

$$\vec{OQ} = (6 - 30t)\mathbf{a} + 24t\mathbf{b}$$

But Q lies on OB , so there is no $\mathbf{a}$ component:

$$6 - 30t = 0$$

$$t = \frac{1}{5}$$

Then

$$\vec{OQ} = 24 \left(\frac{1}{5} \right) \mathbf{b}$$

$$\vec{OQ} = \frac{24}{5} \mathbf{b}$$

Since $\vec{OQ} = n\mathbf{b}$,

$$n = \frac{24}{5}$$

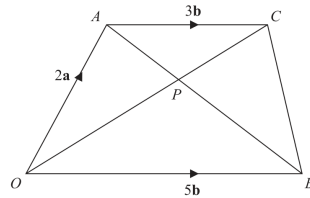
FINAL ANSWER

$$\vec{AB} = 8\mathbf{b} - 12\mathbf{a}, \quad n = \frac{24}{5}$$

Dr. Eslam Ahmed

CLASSIFIED PROBLEM**Paper: P1H****Session: May 2021****Q#: 26****Marks: 5**TOPIC: **Vectors**

Source: May 2021 P1H

26 $OACB$ is a trapezium.Diagram **NOT**
accurately drawn

$$\vec{OA} = 2\mathbf{a} \quad \vec{OB} = 5\mathbf{b} \quad \vec{AC} = 3\mathbf{b}$$

The diagonals, OC and AB , of the trapezium intersect at the point P .Find and simplify an expression, in terms of $\mathbf{a}$ and $\mathbf{b}$, for $\vec{OP}$.
Show your working clearly.

$$\vec{OP} = \dots\dots\dots$$

(Total for Question 26 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P1H | Question 26

Vectors

Topic check. Moved to Vectors.**Method**

$$\overrightarrow{OA} = 2\mathbf{a}, \quad \overrightarrow{OB} = 5\mathbf{b}, \quad \overrightarrow{AC} = 3\mathbf{b}$$

So

$$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC} = 2\mathbf{a} + 3\mathbf{b}$$

Since P lies on OC , let

$$\overrightarrow{OP} = t(2\mathbf{a} + 3\mathbf{b})$$

Since P also lies on AB , let

$$\overrightarrow{OP} = \overrightarrow{OA} + s\overrightarrow{AB}$$

Now

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 5\mathbf{b} - 2\mathbf{a}$$

So

$$\overrightarrow{OP} = 2\mathbf{a} + s(5\mathbf{b} - 2\mathbf{a})$$

$$\overrightarrow{OP} = (2 - 2s)\mathbf{a} + 5s\mathbf{b}$$

Compare this with

$$t(2\mathbf{a} + 3\mathbf{b}) = 2t\mathbf{a} + 3t\mathbf{b}$$

Equating coefficients:

$$2t = 2 - 2s$$

$$3t = 5s$$

From $t = 1 - s$,

$$3(1 - s) = 5s$$

$$3 = 8s$$

$$s = \frac{3}{8}$$

So

$$t = 1 - \frac{3}{8} = \frac{5}{8}$$

Therefore

$$\overrightarrow{OP} = \frac{5}{8}(2\mathbf{a} + 3\mathbf{b})$$

$$\overrightarrow{OP} = \frac{5}{4}\mathbf{a} + \frac{15}{8}\mathbf{b}$$

FINAL ANSWER

$$\frac{5}{4}\mathbf{a} + \frac{15}{8}\mathbf{b}$$

Dr. Eslam Ahmed

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2023

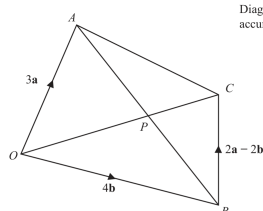
Q#: 22

Marks: 5

TOPIC: **Vectors**

Source: Nov 2023 P2H

22

Diagram **NOT** accurately drawn $OACB$ is a quadrilateral.

$$\vec{OA} = 3\mathbf{a} \quad \vec{OB} = 4\mathbf{b} \quad \vec{BC} = 2\mathbf{a} - 2\mathbf{b}$$

- (a) (i) Find the vector $\vec{OC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. Simplify your answer.

$$\vec{OC} = \text{_____}$$

(1)

- (ii) Find the vector $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

$$\vec{AB} = \text{_____}$$

(1)

The point P lies on AB and on OC

- (b) Using a vector method, find the ratio $AP : PB$. Show your working clearly.

(3)

(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P2H | Question 22

Vectors

Topic check. Corrected from Ratio Toolkit to Vectors.**Method**

$$\overrightarrow{OA} = 3a, \quad \overrightarrow{OB} = 4b, \quad \overrightarrow{BC} = 2a - 2b$$

Part (a):

$$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC} = 4b + (2a - 2b) = 2a + 2b$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 4b - 3a$$

For part (b), let P divide AB so that

$$\overrightarrow{AP} = \mu \overrightarrow{AB}$$

Then

$$\overrightarrow{OP} = 3a + \mu(4b - 3a) = (3 - 3\mu)a + 4\mu b$$

Since P lies on OC , also let

$$\overrightarrow{OP} = \lambda(2a + 2b)$$

Equating coefficients:

$$3 - 3\mu = 2\lambda, \quad 4\mu = 2\lambda$$

From $4\mu = 2\lambda$, $\lambda = 2\mu$.

$$3 - 3\mu = 4\mu$$

$$\mu = \frac{3}{7}$$

So $AP : PB = \frac{3}{7} : \frac{4}{7}$.**FINAL ANSWER** $\overrightarrow{OC} = 2a + 2b$, $\overrightarrow{AB} = 4b - 3a$, and $AP : PB = 3 : 4$

CLASSIFIED PROBLEM

Paper: P1H

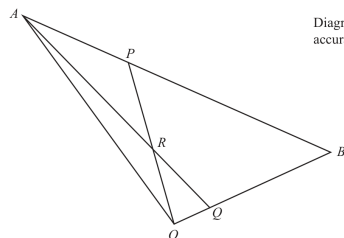
Session: Nov 2024

Q#: 24

Marks: 6

TOPIC: **Vectors**

Source: Nov 2024 P1H

24 OAB is a triangle.Diagram NOT
accurately drawn

$$\vec{OA} = 10\mathbf{a} \quad \vec{OB} = 10\mathbf{b}$$

 AQ and OP are straight lines.

$$\vec{AP} = \frac{1}{4} \vec{AB} \quad \text{and} \quad \vec{OQ} = \frac{1}{5} \vec{OB}$$

Write the following vectors in terms of $\mathbf{a}$ and $\mathbf{b}$.
Simplify your answers.

(i) $\vec{AQ}$

(1)

(ii) $\vec{OP}$

(1)

(iii) $\vec{OR}$

(4)

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2024 P1H | Question 24
Vectors

Topic check. Moved from Linear Graphs to Vectors.

Method

$$\vec{OA} = 10\mathbf{a}, \quad \vec{OB} = 10\mathbf{b}$$

Also,

$$\vec{AP} = \frac{1}{4}\vec{AB}$$

and

$$\vec{OQ} = \frac{1}{5}\vec{OB}$$

For part (i),

$$\vec{OQ} = 2\mathbf{b}$$

$$\vec{AQ} = \vec{OQ} - \vec{OA}$$

$$\vec{AQ} = 2\mathbf{b} - 10\mathbf{a}$$

For part (ii),

$$\vec{AB} = 10\mathbf{b} - 10\mathbf{a}$$

$$\vec{AP} = \frac{1}{4}(10\mathbf{b} - 10\mathbf{a})$$

$$\vec{AP} = \frac{5}{2}\mathbf{b} - \frac{5}{2}\mathbf{a}$$

$$\vec{OP} = \vec{OA} + \vec{AP}$$

$$\vec{OP} = 10\mathbf{a} - \frac{5}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$$

$$\vec{OP} = \frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$$

For part (iii), R lies on OP , so

$$\vec{OR} = t\vec{OP}$$

It also lies on AQ , so

$$\vec{OR} = \vec{OA} + s\vec{AQ}$$

$$\vec{OR} = 10\mathbf{a} + s(2\mathbf{b} - 10\mathbf{a})$$

$$\vec{OR} = (10 - 10s)\mathbf{a} + 2s\mathbf{b}$$

Equate with

$$t\left(\frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}\right)$$

$$\frac{15}{2}t = 10 - 10s$$

$$\frac{5}{2}t = 2s$$

From the second equation,

$$s = \frac{5}{4}t$$

Substitute:

$$\frac{15}{2}t = 10 - \frac{25}{2}t$$

$$20t = 10$$

$$t = \frac{1}{2}$$

So

$$\vec{OR} = \frac{1}{2}\vec{OP}$$

$$\vec{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$$

FINAL ANSWER

$$\vec{AQ} = 2\mathbf{b} - 10\mathbf{a}, \vec{OP} = \frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}, \vec{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$$

CLASSIFIED PROBLEM

Paper: P1H

Session: Nov 2025

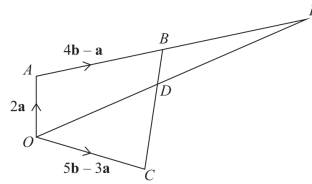
Q#: 27

Marks: 6

TOPIC: **Vectors**

Source: Nov 2025 P1H

27

Diagram NOT
accurately drawn

$OABC$ is a quadrilateral.
 ABP and ODP are straight lines.

$$\vec{OA} = 2\mathbf{a} \quad \vec{AB} = 4\mathbf{b} - \mathbf{a} \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

- (a) Find an expression in terms of $\mathbf{a}$ and $\mathbf{b}$ for the vector $\vec{BC}$.
 Simplify your answer.

(2)

The point D lies on BC such that $BD:DC = 1:3$

Given that $\vec{OP} = n\vec{OD}$

- (b) use a vector method to find the value of n

$$n = \dots \quad (4)$$

(Total for Question 27 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P1H | Question 27

Vectors

Topic check. Moved from Linear Graphs to Vectors.**Method**

$$\vec{OA} = 2\mathbf{a}, \quad \vec{AB} = 4\mathbf{b} - \mathbf{a}, \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

First find $\vec{OB}$:

$$\vec{OB} = \vec{OA} + \vec{AB}$$

$$\vec{OB} = 2\mathbf{a} + 4\mathbf{b} - \mathbf{a}$$

$$\vec{OB} = \mathbf{a} + 4\mathbf{b}$$

For part (a),

$$\vec{BC} = \vec{OC} - \vec{OB}$$

$$\vec{BC} = (5\mathbf{b} - 3\mathbf{a}) - (\mathbf{a} + 4\mathbf{b})$$

$$\vec{BC} = \mathbf{b} - 4\mathbf{a}$$

For part (b), $BD : DC = 1 : 3$, so

$$\vec{OD} = \vec{OB} + \frac{1}{4}\vec{BC}$$

$$\vec{OD} = \mathbf{a} + 4\mathbf{b} + \frac{1}{4}(\mathbf{b} - 4\mathbf{a})$$

$$\vec{OD} = \frac{17}{4}\mathbf{b}$$

Since A, B, P are collinear, let

$$\vec{OP} = \vec{OA} + s\vec{AB}$$

$$\vec{OP} = 2\mathbf{a} + s(4\mathbf{b} - \mathbf{a})$$

$$\vec{OP} = (2 - s)\mathbf{a} + 4s\mathbf{b}$$

But O, D, P are collinear, and $\vec{OD}$ has only a $\mathbf{b}$ component.
So

$$2 - s = 0$$

$$s = 2$$

Then

$$\vec{OP} = 8\mathbf{b}$$

Given

$$\vec{OP} = n\vec{OD}$$

$$8\mathbf{b} = n\left(\frac{17}{4}\mathbf{b}\right)$$

$$n = \frac{32}{17}$$

FINAL ANSWER

$$\vec{BC} = \mathbf{b} - 4\mathbf{a}, \quad n = \frac{32}{17}$$

Dr. Eslam Ahmed